Java Card ™ 2.1 Application Programming Interface



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Java Card API

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Java Card™ 2.1 Platform API Specification

Final Revision 1.1

This document is the specification for the Java Card 2.1 Application Programming Interface.

See:

Description

Packages	
java.lang	Provides classes that are fundamental to the design of the Java Card technology subset of the Java programming language.
javacard.framework	Provides framework of classes and interfaces for the core functionality of a Java Card applet.
javacard.security	Provides the classes and interfaces for the Java Card security framework.
javacardx.crypto	Extension package containing security classes and interfaces for export-controlled functionality.

This document is the specification for the Java Card 2.1 Application Programming Interface.

Java Card 2.1 API Notes

Referenced Standards

ISO - International Standards Organization

- Information Technology - Identification cards - integrated circuit cards with contacts: ISO 7816
- Information Technology - Security Techniques - Digital Signature Scheme Giving Message Recovery: ISO 9796
- Information Technology - Data integrity mechanism using a cryptographic check function employing a block cipher algorithm: ISO 9797
- Information technology - Security techniques - Digital signatures with appendix : ISO 14888

RSA Data Security, Inc.

- RSA Encryption Standard: PKCS #1 Version 2.0
- Password-Based Encryption Standard: PKCS #5 Version 1.5

EMV

- The EMV ’96 ICC Specifications for Payments systems Version 3.0

IPSec

- The Internet Key Exchange (IKE) document RFC 2409 (STD 1)

Standard Names for Security and Crypto

- SHA (also SHA-1): Secure Hash Algorithm, as defined in Secure Hash Standard, NIST FIPS 180-1.
- MD5: The Message Digest algorithm RSA-MD5, as defined by RSA DSI in RFC 1321.
- RIPEMD-160 : as defined in ISO/IEC 10118-3:1998 Information technology -- Security techniques -- Hash-functions -- Part 3: Dedicated hash-functions
- DSA: Digital Signature Algorithm, as defined in Digital Signature Standard, NIST FIPS 186.
- DES: The Data Encryption Standard, as defined by NIST in FIPS 46-1 and 46-2.
- RSA: The Rivest, Shamir and Adleman Asymmetric Cipher algorithm.

Parameter Checking

Policy

All Java Card API implementations must conform to the Java model of parameter checking. That is, the API code should not check for those parameter errors which the VM is expected to detect. These include all parameter errors, such as null pointers, index out of bounds, and so forth, that result in standard runtime exceptions. The runtime exceptions that are thrown by the Java Card VM are:

- ArithmeticException
- ArrayStoreException
- ClassCastException
- IllegalArgumentException
- IllegalStateException
- IndexOutOfBoundsException
- ArrayIndexOutOfBoundsException
- NegativeArraySizeException
- NullPointerException
- SecurityException

Exceptions to the Policy

In some cases, it may be necessary to explicitly check parameters. These exceptions to the policy are documented in the Java Card API specification. A Java Card API implementation must not perform parameter checking with the intent to avoid runtime exceptions, unless this is clearly specified by the Java Card API specification.

Note: If multiple erroneous input parameters exist, any one of several runtime exceptions will be thrown by the VM. Java programmers rely on this behavior, but they do not rely on getting a specific exception. It is not necessary (nor is it reasonable or practical) to document the precise error handling for all possible combinations of equivalence classes of erroneous inputs. The value of this behavior is that the logic error in the calling program is detected and exposed via the runtime exception mechanism, rather than being masked by a normal return.

Hierarchy For All Packages

Package Hierarchies:

java.lang, javacard.framework, javacard.security, javacardx.crypto

Class Hierarchy

- class java.lang.**Object**
  - class javacard.framework.**AID**
  - class javacard.framework.**APDU**
  - class javacard.framework.**Applet**
  - class javacardx.crypto.**Cipher**
  - class javacard.framework.**JCSystem**
  - class javacard.security.**KeyBuilder**
  - class javacard.security.**MessageDigest**
  - class javacard.framework.**OwnerPIN** (implements javacard.framework.PIN)
  - class javacard.security.**RandomData**
  - class javacard.security.**Signature**
  - class java.lang.**Throwable**
    - class java.lang.**Exception**
      - class javacard.framework.**CardException**
        - class javacard.framework.**UserException**
      - class java.lang.**RuntimeException**
        - class java.lang.**ArithmeticException**
        - class java.lang.**ArrayStoreException**
        - class javacard.framework.**CardRuntimeException**
          - class javacard.framework.**APDUException**
          - class javacard.security.**CryptoException**
          - class javacard.framework.**ISOException**
          - class javacard.framework.**PINException**
          - class javacard.framework.**SystemException**
          - class javacard.framework.**TransactionException**
        - class java.lang.**ClassCastException**
        - class java.lang.**IndexOutOfBoundsException**
          - class java.lang.**ArrayIndexOutOfBoundsException**
        - class java.lang.**NegativeArraySizeException**
        - class java.lang.**NullPointerException**
        - class java.lang.**SecurityException**
  - class javacard.framework.**Util**

Interface Hierarchy

- interface javacard.security.**DSAKey**
  - interface javacard.security.**DSAPrivateKey**(also extends javacard.security.PrivateKey)
  - interface javacard.security.**DSAPublicKey**(also extends javacard.security.PublicKey)
- interface javacard.framework.**ISO7816**
- interface javacard.security.**Key**
  - interface javacard.security.**PrivateKey**
    - interface javacard.security.**DSAPrivateKey**(also extends javacard.security.DSAKey)
    - interface javacard.security.**RSAPrivateCrtKey**
    - interface javacard.security.**RSAPrivateKey**
  - interface javacard.security.**PublicKey**
    - interface javacard.security.**DSAPublicKey**(also extends javacard.security.DSAKey)
    - interface javacard.security.**RSAPublicKey**
  - interface javacard.security.**SecretKey**
    - interface javacard.security.**DESKey**
- interface javacardx.crypto.**KeyEncryption**
- interface javacard.framework.**PIN**
- interface javacard.framework.**Shareable**

Package java.lang

Provides classes that are fundamental to the design of the Java Card technology subset of the Java programming language.

See:

Description

Class Summary	
Object	Class `Object` is the root of the Java Card class hierarchy.
Throwable	The Throwable class is the superclass of all errors and exceptions in the Java Card subset of the Java language.

Exception Summary	
ArithmeticException	A JCRE owned instance of `ArithmethicException` is thrown when an exceptional arithmetic condition has occurred.
ArrayIndexOutOfBoundsException	A JCRE owned instance of `IndexOutOfBoundsException` is thrown to indicate that an array has been accessed with an illegal index.
ArrayStoreException	A JCRE owned instance of `ArrayStoreException` is thrown to indicate that an attempt has been made to store the wrong type of object into an array of objects.
ClassCastException	A JCRE owned instance of `ClassCastException` is thrown to indicate that the code has attempted to cast an object to a subclass of which it is not an instance.
Exception	The class `Exception` and its subclasses are a form of `Throwable` that indicates conditions that a reasonable applet might want to catch.
IndexOutOfBoundsException	A JCRE owned instance of `IndexOutOfBoundsException` is thrown to indicate that an index of some sort (such as to an array) is out of range.
NegativeArraySizeException	A JCRE owned instance of `NegativeArraySizeException` is thrown if an applet tries to create an array with negative size.
NullPointerException	A JCRE owned instance of `NullPointerException`is thrown when an applet attempts to use `null` in a case where an object is required.
RuntimeException	`RuntimeException` is the superclass of those exceptions that can be thrown during the normal operation of the Java Card Virtual Machine. A method is not required to declare in its throws clause any subclasses of `RuntimeException` that might be thrown during the execution of the method but not caught.
SecurityException	A JCRE owned instance of `SecurityException` is thrown by the Java Card Virtual Machine to indicate a security violation. This exception is thrown when an attempt is made to illegally access an object belonging to a another applet.

Package java.lang Description

Provides classes that are fundamental to the design of the Java Card technology subset of the Java programming language.

java.lang
Class ArithmeticException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.ArithmeticException
```

public class **ArithmeticException**
extends RuntimeException

A JCRE owned instance of `ArithmethicException` is thrown when an exceptional arithmetic condition has occurred. For example, a "divide by zero" is an exceptional arithmentic condition.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`ArithmeticException``()` Constructs an `ArithmeticException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

ArithmeticException

`public` **`ArithmeticException`**`()`

Constructs an `ArithmeticException`.

---

java.lang
Class ArrayIndexOutOfBoundsException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.IndexOutOfBoundsException
                          |
                          +--java.lang.ArrayIndexOutOfBoundsException
```

public class **ArrayIndexOutOfBoundsException**
extends IndexOutOfBoundsException

A JCRE owned instance of `IndexOutOfBoundsException` is thrown to indicate that an array has been accessed with an illegal index. The index is either negative or greater than or equal to the size of the array.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`ArrayIndexOutOfBoundsException``()` Constructs an `ArrayIndexOutOfBoundsException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

ArrayIndexOutOfBoundsException

`public` **`ArrayIndexOutOfBoundsException`**`()`

Constructs an `ArrayIndexOutOfBoundsException`.

---

java.lang
Class ArrayStoreException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.ArrayStoreException
```

public class **ArrayStoreException**
extends RuntimeException

A JCRE owned instance of `ArrayStoreException` is thrown to indicate that an attempt has been made to store the wrong type of object into an array of objects. For example, the following code generates an `ArrayStoreException`:

```
Object x[] = new AID[3];
x[0] = new OwnerPIN( (byte) 3, (byte) 8);
```

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`ArrayStoreException``()` Constructs an `ArrayStoreException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

ArrayStoreException

`public` **`ArrayStoreException`**`()`

Constructs an `ArrayStoreException`.

---

java.lang
Class ClassCastException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.ClassCastException
```

public class **ClassCastException**
extends RuntimeException

A JCRE owned instance of `ClassCastException` is thrown to indicate that the code has attempted to cast an object to a subclass of which it is not an instance. For example, the following code generates a `ClassCastException`:

```
Object x = new OwnerPIN( (byte)3, (byte)8);
JCSystem.getAppletShareableInterfaceObject( (AID)x, (byte)5 );
```

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`ClassCastException``()` Constructs a `ClassCastException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

ClassCastException

`public` **`ClassCastException`**`()`

Constructs a `ClassCastException`.

---

java.lang
Class Exception

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
```

Direct Known Subclasses:
CardException, RuntimeException

---

public class **Exception**
extends Throwable

The class `Exception` and its subclasses are a form of `Throwable` that indicates conditions that a reasonable applet might want to catch.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`Exception``()` Constructs an `Exception` instance.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

Exception

```
public Exception()
```

Constructs an `Exception` instance.

java.lang
Class IndexOutOfBoundsException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.IndexOutOfBoundsException
```

Direct Known Subclasses:
ArrayIndexOutOfBoundsException

---

public class **IndexOutOfBoundsException**
extends RuntimeException

A JCRE owned instance of `IndexOutOfBoundsException` is thrown to indicate that an index of some sort (such as to an array) is out of range.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`IndexOutOfBoundsException``()` Constructs an `IndexOutOfBoundsException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

IndexOutOfBoundsException

`public` **`IndexOutOfBoundsException`**`()`

Constructs an `IndexOutOfBoundsException`.

---

java.lang
Class NegativeArraySizeException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.NegativeArraySizeException
```

public class **NegativeArraySizeException**
extends RuntimeException

A JCRE owned instance of `NegativeArraySizeException` is thrown if an applet tries to create an array with negative size.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`NegativeArraySizeException``()` Constructs a `NegativeArraySizeException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

NegativeArraySizeException

`public` **`NegativeArraySizeException`**`()`

Constructs a `NegativeArraySizeException`.

---

java.lang
Class NullPointerException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.NullPointerException
```

public class **NullPointerException**
extends RuntimeException

A JCRE owned instance of `NullPointerException`is thrown when an applet attempts to use `null` in a case where an object is required. These include:

- Calling the instance method of a `null` object.
- Accessing or modifying the field of a `null` object.
- Taking the length of `null` as if it were an array.
- Accessing or modifying the slots of `null` as if it were an array.
- Throwing `null` as if it were a `Throwable` value.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`NullPointerException``()` Constructs a `NullPointerException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

NullPointerException

```
public NullPointerException()
```

Constructs a `NullPointerException`.

---

java.lang
Class Object

java.lang.Object

public class **Object**

Class `Object` is the root of the Java Card class hierarchy. Every class has `Object` as a superclass. All objects, including arrays, implement the methods of this class.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

Object()	

Method Summary

boolean	**equals**(Object obj) Compares two Objects for equality.

Constructor Detail

Object

public **Object**()

Method Detail

equals

public boolean **equals**(Object obj)

Compares two Objects for equality.

The `equals` method implements an equivalence relation:

- It is *reflexive*: for any reference value `x`, `x.equals(x)` should return `true`.
- It is *symmetric*: for any reference values `x` and `y`, `x.equals(y)` should return `true` if and only if `y.equals(x)` returns `true`.
- It is *transitive*: for any reference values `x`, `y`, and `z`, if `x.equals(y)` returns `true` and `y.equals(z)` returns `true`, then `x.equals(z)` should return `true`.
- It is *consistent*: for any reference values `x` and `y`, multiple invocations of `x.equals(y)` consistently return `true` or consistently return `false`.
- For any reference value `x`, `x.equals(null)` should return `false`.

The equals method for class `Object` implements the most discriminating possible equivalence relation on objects; that is, for any reference values `x` and `y`, this method returns `true` if and only if `x` and `y` refer to the same object (`x==y` has the value `true`).

Parameters:

`obj` - the reference object with which to compare.

Returns:

`true` if this object is the same as the obj argument; `false` otherwise.

---

java.lang
Class RuntimeException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
```

Direct Known Subclasses:

ArithmeticException, ArrayStoreException, CardRuntimeException, ClassCastException, IndexOutOfBoundsException, NegativeArraySizeException, NullPointerException, SecurityException

---

public class **RuntimeException**
extends Exception

`RuntimeException` is the superclass of those exceptions that can be thrown during the normal operation of the Java Card Virtual Machine.

A method is not required to declare in its throws clause any subclasses of `RuntimeException` that might be thrown during the execution of the method but not caught.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`RuntimeException``()` Constructs a `RuntimeException` instance.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

RuntimeException

`public` **`RuntimeException`**`()`

Constructs a `RuntimeException` instance.

---

java.lang
Class SecurityException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--java.lang.SecurityException
```

public class **SecurityException**
extends RuntimeException

A JCRE owned instance of `SecurityException` is thrown by the Java Card Virtual Machine to indicate a security violation.

This exception is thrown when an attempt is made to illegally access an object belonging to a another applet. It may optionally be thrown by a Java Card VM implementation to indicate fundamental language restrictions, such as attempting to invoke a private method in another class.

For security reasons, the JCRE implementation may mute the card instead of throwing this exception.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`SecurityException``()` Constructs a `SecurityException`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

SecurityException

```
public SecurityException()
```

Constructs a `SecurityException`.

---

java.lang
Class Throwable

```
java.lang.Object
  |
  +--java.lang.Throwable
```

Direct Known Subclasses:
Exception

public class **Throwable**
extends Object

The Throwable class is the superclass of all errors and exceptions in the Java Card subset of the Java language. Only objects that are instances of this class (or of one of its subclasses) are thrown by the Java Card Virtual Machine or can be thrown by the Java `throw` statement. Similarly, only this class or one of its subclasses can be the argument type in a `catch` clause.

This Java Card class's functionality is a strict subset of the definition in the *Java Platform Core API Specification*.

Constructor Summary

`Throwable``()` Constructs a new `Throwable`.

Methods inherited from class java.lang.Object
`equals`

Constructor Detail

Throwable

`public` **`Throwable`**`()`

Constructs a new `Throwable`.

Package javacard.framework

Provides framework of classes and interfaces for the core functionality of a Java Card applet.

See:

Description

Interface Summary	
ISO7816	`ISO7816` encapsulates constants related to ISO 7816-3 and ISO 7816-4.
PIN	This interface represents a PIN.
Shareable	The Shareable interface serves to identify all shared objects.

Class Summary	
AID	This class encapsulates the Application Identifier(AID) associated with an applet.
APDU	Application Protocol Data Unit (APDU) is the communication format between the card and the off-card applications.
Applet	This abstract class defines an applet in Java Card.
JCSystem	The `JCSystem` class includes a collection of methods to control applet execution, resource management, atomic transaction management and inter-applet object sharing in Java Card.
OwnerPIN	This class represents an Owner PIN.
Util	The `Util` class contains common utility functions.

Exception Summary	
APDUException	`APDUException` represents an `APDU` related exception.
CardException	The `CardException` class defines a field `reason` and two accessor methods `getReason()` and `setReason()`.
CardRuntimeException	The `CardRuntimeException` class defines a field `reason` and two accessor methods `getReason()` and `setReason()`.
ISOException	`ISOException` class encapsulates an ISO 7816-4 response status word as its `reason` code.
PINException	`PINException` represents a `OwnerPIN` class access-related exception.
SystemException	`SystemException` represents a `JCSystem` class related exception.
TransactionException	`TransactionException` represents an exception in the transaction subsystem.
UserException	`UserException` represents a User exception.

Package javacard.framework Description

Provides framework of classes and interfaces for the core functionality of a Java Card applet.

javacard.framework
Class AID

```
java.lang.Object
  |
  +--javacard.framework.AID
```

public final class **AID**
extends Object

This class encapsulates the Application Identifier(AID) associated with an applet. An AID is defined in ISO 7816-5 to be a sequence of bytes between 5 and 16 bytes in length.

The JCRE creates instances of `AID` class to identify and manage every applet on the card. Applets need not create instances of this class. An applet may request and use the JCRE owned instances to identify itself and other applet instances.

JCRE owned instances of `AID` are permanent JCRE Entry Point Objects and can be accessed from any applet context. References to these permanent objects can be stored and re-used.

An applet instance can obtain a reference to JCRE owned instances of its own `AID` object by using the `JCSystem.getAID()` method and another applet's `AID` object via the `JCSystem.lookupAID()` method.

An applet uses `AID` instances to request to share another applet's object or to control access to its own shared object from another applet. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

See Also:
`JCSystem`, `SystemException`

Constructor Summary

`AID``(byte[] bArray, short offset, byte length)` The JCRE uses this constructor to create a new `AID` instance encapsulating the specified AID bytes.

Method Summary

boolean	**equals**(byte[] bArray, short offset, byte length) Checks if the specified AID bytes in bArray are the same as those encapsulated in this AID object.
boolean	**equals**(Object anObject) Compares the AID bytes in this AID instance to the AID bytes in the specified object.
byte	**getBytes**(byte[] dest, short offset) Called to get the AID bytes encapsulated within AID object.
boolean	**partialEquals**(byte[] bArray, short offset, byte length) Checks if the specified partial AID byte sequence matches the first length bytes of the encapsulated AID bytes within this AID object.
boolean	**RIDEquals**(AID otherAID) Checks if the RID (National Registered Application provider identifier) portion of the encapsulated AID bytes within the otherAID object matches that of this AID object.

Constructor Detail

AID

```
public AID(byte[] bArray,
           short offset,
           byte length)
    throws SystemException
```

The JCRE uses this constructor to create a new AID instance encapsulating the specified AID bytes.

Parameters:

bArray - the byte array containing the AID bytes.
offset - the start of AID bytes in bArray.
length - the length of the AID bytes in bArray.

Throws:

SystemException - with the following reason code:

- SystemException.ILLEGAL_VALUE if the length parameter is less than 5 or greater than 16.

Method Detail

getBytes

```
public byte getBytes(byte[] dest,
                     short offset)
```

Called to get the AID bytes encapsulated within `AID` object.

Parameters:

`dest` - byte array to copy the AID bytes.

`offset` - within dest where the AID bytes begin.

Returns:

the length of the AID bytes.

---

equals

```
public boolean equals(Object anObject)
```

Compares the AID bytes in `this AID` instance to the AID bytes in the specified object. The result is `true` if and only if the argument is not `null` and is an `AID` object that encapsulates the same AID bytes as `this` object.

This method does not throw `NullPointerException`.

Parameters:

`anObject` - the object to compare `this AID` against.

Returns:

`true` if the AID byte values are equal, `false` otherwise.

Overrides:

equals in class Object

---

equals

```
public boolean equals(byte[] bArray,
                      short offset,
                      byte length)
```

Checks if the specified AID bytes in `bArray` are the same as those encapsulated in `this AID` object. The result is `true` if and only if the `bArray` argument is not `null` and the AID bytes encapsulated in `this AID` object are equal to the specified AID bytes in `bArray`.

This method does not throw `NullPointerException`.

Parameters:

`bArray` - containing the AID bytes

`offset` - within bArray to begin

`length` - of AID bytes in bArray

Returns:

`true` if equal, `false` otherwise.

partialEquals

```
public boolean partialEquals(byte[] bArray,
                             short offset,
                             byte length)
```

Checks if the specified partial AID byte sequence matches the first `length` bytes of the encapsulated AID bytes within `this AID` object. The result is `true` if and only if the `bArray` argument is not `null` and the input `length` is less than or equal to the length of the encapsulated AID bytes within `this AID` object and the specified bytes match.

This method does not throw `NullPointerException`.

Parameters:

`bArray` - containing the partial AID byte sequence
`offset` - within bArray to begin
`length` - of partial AID bytes in bArray

Returns:

`true` if equal, `false` otherwise.

RIDEquals

```
public boolean RIDEquals(AID otherAID)
```

Checks if the RID (National Registered Application provider identifier) portion of the encapsulated AID bytes within the `otherAID` object matches that of `this AID` object. The first 5 bytes of an AID byte sequence is the RID. See ISO 7816-5 for details. The result is `true` if and only if the argument is not `null` and is an `AID` object that encapsulates the same RID bytes as `this` object.

This method does not throw `NullPointerException`.

Parameters:

`otherAID` - the `AID` to compare against.

Returns:

`true` if the RID bytes match, `false` otherwise.

javacard.framework
Class APDU

```
java.lang.Object
  |
  +--javacard.framework.APDU
```

public final class **APDU**
extends Object

Application Protocol Data Unit (APDU) is the communication format between the card and the off-card applications. The format of the APDU is defined in ISO specification 7816-4.

This class only supports messages which conform to the structure of command and response defined in ISO 7816-4. The behavior of messages which use proprietary structure of messages (for example with header CLA byte in range 0xD0-0xFE) is undefined. This class does not support extended length fields.

The APDU object is owned by the JCRE. The APDU class maintains a byte array buffer which is used to transfer incoming APDU header and data bytes as well as outgoing data. The buffer length must be at least 37 bytes (5 bytes of header and 32 bytes of data). The JCRE must zero out the APDU buffer before each new message received from the CAD.

The JCRE designates the APDU object as a temporary JCRE Entry Point Object (See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details). A temporary JCRE Entry Point Object can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components.

The JCRE similarly marks the APDU buffer as a global array (See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details). A global array can be accessed from any applet context. References to global arrays cannot be stored in class variables or instance variables or array components.

The applet receives the APDU instance to process from the JCRE in the `Applet.process(APDU)` method, and the first five bytes [CLA, INS, P1, P2, P3] are available in the APDU buffer.

The APDU class API is designed to be transport protocol independent. In other words, applets can use the same APDU methods regardless of whether the underlying protocol in use is T=0 or T=1 (as defined in ISO 7816-3).

The incoming APDU data size may be bigger than the APDU buffer size and may therefore need to be read in portions by the applet. Similarly, the outgoing response APDU data size may be bigger than the APDU buffer size and may need to be written in portions by the applet. The APDU class has methods to facilitate this.

For sending large byte arrays as response data, the APDU class provides a special method `sendBytesLong()` which manages the APDU buffer.

```
 // The purpose of this example is to show most of the methods
 // in use and not to depict any particular APDU processing

public void process(APDU apdu){
  // ...
  byte[] buffer = apdu.getBuffer();
  byte cla = buffer[ISO7816.OFFSET_CLA];
  byte ins = buffer[ISO7816.OFFSET_INS];
  ...
  // assume this command has incoming data
  // Lc tells us the incoming apdu command length
  short bytesLeft = (short) (buffer[ISO7816.OFFSET_LC] & 0x00FF);
  if (bytesLeft < (short)55) ISOException.throwIt( ISO7816.SW_WRONG_LENGTH );

  short readCount = apdu.setIncomingAndReceive();
  while ( bytesLeft > 0){
      // process bytes in buffer[5] to buffer[readCount+4];
      bytesLeft -= readCount;
      readCount = apdu.receiveBytes ( ISO7816.OFFSET_CDATA );
      }
  //
  //...
  //
  // Note that for a short response as in the case illustrated here
  // the three APDU method calls shown : setOutgoing(),setOutgoingLength() & sendBytes()
  // could be replaced by one APDU method call : setOutgoingAndSend().

  // construct the reply APDU
  short le = apdu.setOutgoing();
  if (le < (short)2) ISOException.throwIt( ISO7816.SW_WRONG_LENGTH );
  apdu.setOutgoingLength( (short)3 );

  // build response data in apdu.buffer[ 0.. outCount-1 ];
  buffer[0] = (byte)1; buffer[1] = (byte)2; buffer[3] = (byte)3;
  apdu.sendBytes ( (short)0 , (short)3 );
  // return good complete status 90 00
  }
```

See Also:

APDUException, ISOException

Field Summary

static byte	**PROTOCOL_T0** ISO 7816 transport protocol type T=0
static byte	**PROTOCOL_T1** ISO 7816 transport protocol type T=1

Method Summary

byte[]	**getBuffer**() Returns the APDU buffer byte array.
static short	**getInBlockSize**() Returns the configured incoming block size. In T=1 protocol, this corresponds to IFSC (information field size for ICC), the maximum size of incoming data blocks into the card. In T=0 protocol, this method returns 1.
byte	**getNAD**() In T=1 protocol, this method returns the Node Address byte, NAD. In T=0 protocol, this method returns 0.
static short	**getOutBlockSize**() Returns the configured outgoing block size. In T=1 protocol, this corresponds to IFSD (information field size for interface device), the maximum size of outgoing data blocks to the CAD. In T=0 protocol, this method returns 258 (accounts for 2 status bytes).
static byte	**getProtocol**() Returns the ISO 7816 transport protocol type, T=1 or T=0 in progress.
short	**receiveBytes**(short bOff) Gets as many data bytes as will fit without APDU buffer overflow, at the specified offset `bOff`. Gets all the remaining bytes if they fit.
void	**sendBytes**(short bOff, short len) Sends `len` more bytes from APDU buffer at specified offset `bOff`.
void	**sendBytesLong**(byte[] outData, short bOff, short len) Sends `len` more bytes from `outData` byte array starting at specified offset `bOff`.
short	**setIncomingAndReceive**() This is the primary receive method.
short	**setOutgoing**() This method is used to set the data transfer direction to outbound and to obtain the expected length of response (Le).
void	**setOutgoingAndSend**(short bOff, short len) This is the "convenience" send method.
void	**setOutgoingLength**(short len) Sets the actual length of response data.
short	**setOutgoingNoChaining**() This method is used to set the data transfer direction to outbound without using BLOCK CHAINING(See ISO 7816-3/4) and to obtain the expected length of response (Le).
static void	**waitExtension**() Requests additional processsing time from CAD.

Methods inherited from class java.lang.Object
equals

Field Detail

PROTOCOL_T0

```
public static final byte PROTOCOL_T0
```

ISO 7816 transport protocol type T=0

PROTOCOL_T1

```
public static final byte PROTOCOL_T1
```

ISO 7816 transport protocol type T=1

Method Detail

getBuffer

```
public byte[] getBuffer()
```

Returns the APDU buffer byte array.

Notes:

- *References to the APDU buffer byte array cannot be stored in class variables or instance variables or array components. See* Java Card Runtime Environment (JCRE) 2.1 Specification *for details.*

Returns:

byte array containing the APDU buffer

---

getInBlockSize

```
public static short getInBlockSize()
```

Returns the configured incoming block size. In T=1 protocol, this corresponds to IFSC (information field size for ICC), the maximum size of incoming data blocks into the card. In T=0 protocol, this method returns 1. IFSC is defined in ISO 7816-3.

This information may be used to ensure that there is enough space remaining in the APDU buffer when `receiveBytes()` is invoked.

Notes:

- *On* `receiveBytes()` *the* `bOff` *param should account for this potential blocksize.*

Returns:

incoming block size setting.

See Also:

`receiveBytes(short)`

---

getOutBlockSize

```
public static short getOutBlockSize()
```

Returns the configured outgoing block size. In T=1 protocol, this corresponds to IFSD (information field size for interface device), the maximum size of outgoing data blocks to the CAD. In T=0 protocol, this method returns 258 (accounts for 2 status bytes). IFSD is defined in ISO 7816-3.

This information may be used prior to invoking the `setOutgoingLength()` method, to limit the length of outgoing messages when BLOCK CHAINING is not allowed.

Notes:

- *On* `setOutgoingLength()` *the* `len` *param should account for this potential blocksize.*

Returns:

outgoing block size setting.

See Also:

`setOutgoingLength(short)`

---

getProtocol

```
public static byte getProtocol()
```

Returns the ISO 7816 transport protocol type, T=1 or T=0 in progress.

Returns:

the protocol type in progress. One of `PROTOCOL_T0`, `PROTOCOL_T1` listed above.

---

getNAD

```
public byte getNAD()
```

In T=1 protocol, this method returns the Node Address byte, NAD. In T=0 protocol, this method returns 0. This may be used as additional information to maintain multiple contexts.

Returns:

NAD transport byte as defined in ISO 7816-3.

setOutgoing

```
public short setOutgoing()
                  throws APDUException
```

This method is used to set the data transfer direction to outbound and to obtain the expected length of response (Le).

Notes.

- *Any remaining incoming data will be discarded.*
- *In T=0 (Case 4) protocol, this method will return 256.*

Returns:

Le, the expected length of response.

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if this method or `setOutgoingNoChaining()` method already invoked.
- `APDUException.IO_ERROR` on I/O error.

setOutgoingNoChaining

```
public short setOutgoingNoChaining()
                           throws APDUException
```

This method is used to set the data transfer direction to outbound without using BLOCK CHAINING(See ISO 7816-3/4) and to obtain the expected length of response (Le). This method should be used in place of the `setOutgoing()` method by applets which need to be compatible with legacy CAD/terminals which do not support ISO 7816-3/4 defined block chaining. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Notes.

- *Any remaining incoming data will be discarded.*
- *In T=0 (Case 4) protocol, this method will return 256.*
- *When this method is used, the* `waitExtension()` *method cannot be used.*
- *In T=1 protocol, retransmission on error may be restricted.*
- *In T=0 protocol, the outbound transfer must be performed without using response status chaining.*
- *In T=1 protocol, the outbound transfer must not set the More(M) Bit in the PCB of the I block. See ISO 7816-3.*

Returns:

Le, the expected length of response data.

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if this method or `setOutgoing()` method already invoked.

- `APDUException.IO_ERROR` on I/O error.

---

setOutgoingLength

```
public void setOutgoingLength(short len)
                      throws APDUException
```

Sets the actual length of response data. Default is 0.

Note:

- *In T=0 (Case 2&4) protocol, the length is used by the JCRE to prompt the CAD for GET RESPONSE commands.*

Parameters:

`len` - the length of response data.

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setOutgoing()` not called or this method already invoked.
- `APDUException.BAD_LENGTH` if `len` is greater than 256 or if non BLOCK CHAINED data transfer is requested and `len` is greater than (IFSD-2), where IFSD is the Outgoing Block Size. The -2 accounts for the status bytes in T=1.
- `APDUException.IO_ERROR` on I/O error.

See Also:

`getOutBlockSize()`

---

receiveBytes

```
public short receiveBytes(short bOff)
                  throws APDUException
```

Gets as many data bytes as will fit without APDU buffer overflow, at the specified offset `bOff`. Gets all the remaining bytes if they fit.

Notes:

- *The space in the buffer must allow for incoming block size.*
- *In T=1 protocol, if all the remaining bytes do not fit in the buffer, this method may return less bytes than the maximum incoming block size (IFSC).*
- *In T=0 protocol, if all the remaining bytes do not fit in the buffer, this method may return less than a full buffer of bytes to optimize and reduce protocol overhead.*
- *In T=1 protocol, if this method throws an* `APDUException` *with* `T1_IFD_ABORT` *reason code, the JCRE will restart APDU command processing using the newly received command. No more input data can be received. No output data can be transmitted. No error status response can be returned.*

Parameters:

`bOff` - the offset into APDU buffer.

Returns:
number of bytes read. Returns 0 if no bytes are available.

Throws:
APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setIncomingAndReceive()` not called or if `setOutgoing()` or `setOutgoingNoChaining()` previously invoked.
- `APDUException.BUFFER_BOUNDS` if not enough buffer space for incoming block size.
- `APDUException.IO_ERROR` on I/O error.
- `APDUException.T1_IFD_ABORT` if T=1 protocol is in use and the CAD sends in an ABORT S-Block command to abort the data transfer.

See Also:
`getInBlockSize()`

---

setIncomingAndReceive

```
public short setIncomingAndReceive()
                            throws APDUException
```

This is the primary receive method. Calling this method indicates that this APDU has incoming data. This method gets as many bytes as will fit without buffer overflow in the APDU buffer following the header. It gets all the incoming bytes if they fit.

Notes:

- *In T=0 (Case 3&4) protocol, the P3 param is assumed to be Lc.*
- *Data is read into the buffer at offset 5.*
- *In T=1 protocol, if all the incoming bytes do not fit in the buffer, this method may return less bytes than the maximum incoming block size (IFSC).*
- *In T=0 protocol, if all the incoming bytes do not fit in the buffer, this method may return less than a full buffer of bytes to optimize and reduce protocol overhead.*
- *This method sets the transfer direction to be inbound and calls* `receiveBytes(5)`.
- *This method may only be called once in a* `Applet.process()` *method.*

Returns:
number of bytes read. Returns 0 if no bytes are available.

Throws:
APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setIncomingAndReceive()` already invoked or if `setOutgoing()` or `setOutgoingNoChaining()` previously invoked.
- `APDUException.IO_ERROR` on I/O error.
- `APDUException.T1_IFD_ABORT` if T=1 protocol is in use and the CAD sends in an ABORT S-Block command to abort the data transfer.

---

sendBytes

```
public void sendBytes(short bOff,
                      short len)
               throws APDUException
```

Sends `len` more bytes from APDU buffer at specified offset `bOff`.

If the last part of the response is being sent by the invocation of this method, the APDU buffer must not be altered. If the data is altered, incorrect output may be sent to the CAD. Requiring that the buffer not be altered allows the implementation to reduce protocol overhead by transmitting the last part of the response along with the status bytes.

Notes:

- *If* `setOutgoingNoChaining()` *was invoked, output block chaining must not be used.*
- *In T=0 protocol, if* `setOutgoingNoChaining()` *was invoked, Le bytes must be transmitted before response status is returned.*
- *In T=0 protocol, if this method throws an* `APDUException` *with* `NO_T0_GETRESPONSE` *reason code, the JCRE will restart APDU command processing using the newly received command. No more output data can be transmitted. No error status response can be returned.*
- *In T=1 protocol, if this method throws an* `APDUException` *with* `T1_IFD_ABORT` *reason code, the JCRE will restart APDU command processing using the newly received command. No more output data can be transmitted. No error status response can be returned.*

Parameters:

`bOff` - the offset into APDU buffer.
`len` - the length of the data in bytes to send.

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setOutgoingLen()` not called or `setOutgoingAndSend()` previously invoked or response byte count exeeded or if `APDUException.NO_T0_GETRESPONSE` previously thrown.
- `APDUException.BUFFER_BOUNDS` if the sum of `bOff` and `len` exceeds the buffer size.
- `APDUException.IO_ERROR` on I/O error.
- `APDUException.NO_T0_GETRESPONSE` if T=0 protocol is in use and the CAD does not respond to response status with GET RESPONSE command.
- `APDUException.T1_IFD_ABORT` if T=1 protocol is in use and the CAD sends in an ABORT S-Block command to abort the data transfer.

See Also:

`setOutgoing()`, `setOutgoingNoChaining()`

---

sendBytesLong

```
public void sendBytesLong(byte[] outData,
                          short bOff,
                          short len)
                   throws APDUException
```

Sends `len` more bytes from `outData` byte array starting at specified offset `bOff`.

If the last of the response is being sent by the invocation of this method, the APDU buffer must not be altered. If the data is altered, incorrect output may be sent to the CAD. Requiring that the buffer not be altered allows the implementation to reduce protocol overhead by transmitting the last part of the response along with the status bytes.

The JCRE may use the APDU buffer to send data to the CAD.

Notes:

- *If* `setOutgoingNoChaining()` *was invoked, output block chaining must not be used.*
- *In T=0 protocol, if* `setOutgoingNoChaining()` *was invoked, Le bytes must be transmitted before response status is returned.*
- *In T=0 protocol, if this method throws an* `APDUException` *with NO_T0_GETRESPONSE reason code, the JCRE will restart APDU command processing using the newly received command. No more output data can be transmitted. No error status response can be returned.*
- *In T=1 protocol, if this method throws an* `APDUException` *with* `T1_IFD_ABORT` *reason code, the JCRE will restart APDU command processing using the newly received command. No more output data can be transmitted. No error status response can be returned.*

Parameters:

`outData` - the source data byte array.
`bOff` - the offset into OutData array.
`len` - the bytelength of the data to send.

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setOutgoingLen()` not called or `setOutgoingAndSend()` previously invoked or response byte count exeeded or if `APDUException.NO_T0_GETRESPONSE` previously thrown.
- `APDUException.IO_ERROR` on I/O error.
- `APDUException.NO_T0_GETRESPONSE` if T=0 protocol is in use and CAD does not respond to response status with GET RESPONSE command.
- `APDUException.T1_IFD_ABORT` if T=1 protocol is in use and the CAD sends in an ABORT S-Block command to abort the data transfer.

See Also:

`setOutgoing()`, `setOutgoingNoChaining()`

---

setOutgoingAndSend

```
public void setOutgoingAndSend(short bOff,
                               short len)
                        throws APDUException
```

This is the "convenience" send method. It provides for the most efficient way to send a short response which fits in the buffer and needs the least protocol overhead. This method is a combination of `setOutgoing(), setOutgoingLength( len )` followed by `sendBytes ( bOff, len )`. In addition, once this method is invoked, `sendBytes()` and `sendBytesLong()` methods cannot be invoked and the APDU buffer must not be altered.

Sends `len` byte response from the APDU buffer at starting specified offset `bOff`.

Notes:

- *No other* `APDU` *send methods can be invoked.*
- *The APDU buffer must not be altered. If the data is altered, incorrect output may be sent to the CAD.*
- *The actual data transmission may only take place on return from* `Applet.process()`

Parameters:

`bOff` - the offset into APDU buffer.
`len` - the bytelength of the data to send.

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setOutgoing()` or `setOutgoingAndSend()` previously invoked or response byte count exeeded.
- `APDUException.IO_ERROR` on I/O error.

---

waitExtension

```
public static void waitExtension()
                          throws APDUException
```

Requests additional processsing time from CAD. The implementation should ensure that this method needs to be invoked only under unusual conditions requiring excessive processing times.

Notes:

- *In T=0 protocol, a NULL procedure byte is sent to reset the work waiting time (see ISO 7816-3).*
- *In T=1 protocol, the implementation needs to request the same T=0 protocol work waiting time quantum by sending a T=1 protocol request for wait time extension(see ISO 7816-3).*
- *If the implementation uses an automatic timer mechanism instead, this method may do nothing.*

Throws:

APDUException - with the following reason codes:

- `APDUException.ILLEGAL_USE` if `setOutgoingNoChaining()` previously

invoked.

- `APDUException.IO_ERROR` on I/O error.

---

javacard.framework
Class APDUException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
                          |
                          +--javacard.framework.APDUException
```

public class **APDUException**
extends CardRuntimeException

`APDUException` represents an `APDU` related exception.

The `APDU` class throws JCRE owned instances of `APDUException`.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

See Also:
`APDU`

Field Summary

static short	**BAD_LENGTH** This reason code is used by the APDU.setOutgoingLength() method to indicate that the length parameter is greater that 256 or if non BLOCK CHAINED data transfer is requested and len is greater than (IFSD-2), where IFSD is the Outgoing Block Size.
static short	**BUFFER_BOUNDS** This reason code is used by the APDU.sendBytes() method to indicate that the sum of buffer offset parameter and the byte length parameter exceeds the APDU buffer size.
static short	**ILLEGAL_USE** This APDUException reason code indicates that the method should not be invoked based on the current state of the APDU.
static short	**IO_ERROR** This reason code indicates that an unrecoverable error occurred in the I/O transmission layer.
static short	**NO_T0_GETRESPONSE** This reason code indicates that during T=0 protocol, the CAD did not return a GET RESPONSE command in response to a <61xx> response status to send additional data.
static short	**T1_IFD_ABORT** This reason code indicates that during T=1 protocol, the CAD returned an ABORT S-Block command and aborted the data transfer.

Constructor Summary

APDUException(short reason) Constructs an APDUException.

Method Summary

static void	**throwIt**(short reason) Throws the JCRE owned instance of APDUException with the specified reason.

Methods inherited from class javacard.framework.CardRuntimeException
`getReason, setReason`

Methods inherited from class java.lang.Object
`equals`

Field Detail

ILLEGAL_USE

```
public static final short ILLEGAL_USE
```

This APDUException reason code indicates that the method should not be invoked based on the current state of the APDU.

BUFFER_BOUNDS

```
public static final short BUFFER_BOUNDS
```

This reason code is used by the `APDU.sendBytes()` method to indicate that the sum of buffer offset parameter and the byte length parameter exceeds the APDU buffer size.

BAD_LENGTH

```
public static final short BAD_LENGTH
```

This reason code is used by the `APDU.setOutgoingLength()` method to indicate that the length parameter is greater that 256 or if non BLOCK CHAINED data transfer is requested and `len` is greater than (IFSD-2), where IFSD is the Outgoing Block Size.

IO_ERROR

```
public static final short IO_ERROR
```

This reason code indicates that an unrecoverable error occurred in the I/O transmission layer.

NO_T0_GETRESPONSE

```
public static final short NO_T0_GETRESPONSE
```

This reason code indicates that during T=0 protocol, the CAD did not return a GET RESPONSE command in response to a <61xx> response status to send additional data. The outgoing transfer has been aborted. No more data or status can be sent to the CAD in this `APDU.process()` method.

T1_IFD_ABORT

```
public static final short T1_IFD_ABORT
```

This reason code indicates that during T=1 protocol, the CAD returned an ABORT S-Block command and aborted the data transfer. The incoming or outgoing transfer has been aborted. No more data can be received from the CAD. No more data or status can be sent to the CAD in this `APDU.process()` method.

Constructor Detail

APDUException

```
public APDUException(short reason)
```

Constructs an APDUException. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception.

Method Detail

throwIt

```
public static void throwIt(short reason)
```

Throws the JCRE owned instance of `APDUException` with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`reason` - the reason for the exception.

Throws:

APDUException - always.

javacard.framework
Class Applet

```
java.lang.Object
  |
  +--javacard.framework.Applet
```

public abstract class **Applet**
extends Object

This abstract class defines an applet in Java Card.

The `Applet` class should be extended by any applet that is intended to be loaded onto, installed into and executed on a Java Card compliant smart card.

Example usage of `Applet`

```
public class MyApplet extends javacard.framework.Applet{
static byte someByteArray[];

public static void install( byte[] bArray, short bOffset, byte bLength  ) throws ISOException {
  // make all my allocations here, so I do not run
  // out of memory later
  MyApplet theApplet = new MyApplet();

  // check incoming parameter
  byte bLen = bArray[bOffset];
  if ( bLen!=0 ) { someByteArray = new byte[bLen]; theApplet.register(); return; }
  else ISOException.throwIt(ISO7816.SW_FUNC_NOT_SUPPORTED);
  }

public boolean select(){
  // selection initialization
  someByteArray[17] = 42; // set selection state
  return true;
  }

public void process(APDU apdu) throws ISOException{
 byte[] buffer = apdu.getBuffer();
 // .. process the incoming data and reply
 if ( buffer[ISO7816.OFFSET_CLA] == (byte)0 ) {
    switch ( buffer[ISO7816.OFFSET_INS] ) {
        case ISO.INS_SELECT:
            ...
            // send response data to select command
            short Le =  apdu.setOutgoing();
            // assume data containing response bytes in replyData[] array.
            if ( Le < ..) ISOException.throwIt( ISO7816.SW_WRONG_LENGTH);
            apdu.setOutgoingLength( (short)replyData.length );
            apdu.sendBytesLong(replyData, (short) 0, (short)replyData.length);
            break;
        case ...
        }
     }
```

```
  }
}
```

See Also:
`SystemException`, `JCSystem`

Constructor Summary

protected	**`Applet`**`()` Only this class's `install()` method should create the applet object.

Method Summary

void	**`deselect`**`()` Called by the JCRE to inform this currently selected applet that another (or the same) applet will be selected.
Shareable	**`getShareableInterfaceObject`**`(AID clientAID, byte parameter)` Called by the JCRE to obtain a shareable interface object from this server applet, on behalf of a request from a client applet.
static void	**`install`**`(byte[] bArray, short bOffset, byte bLength)` To create an instance of the `Applet` subclass, the JCRE will call this static method first.
abstract void	**`process`**`(APDU apdu)` Called by the JCRE to process an incoming APDU command.
protected void	**`register`**`()` This method is used by the applet to register `this` applet instance with the JCRE and to assign the `Applet` subclass AID bytes as its instance AID bytes.
protected void	**`register`**`(byte[] bArray, short bOffset, byte bLength)` This method is used by the applet to register `this` applet instance with the JCRE and assign the specified AID bytes as its instance AID bytes.
boolean	**`select`**`()` Called by the JCRE to inform this applet that it has been selected.
protected boolean	**`selectingApplet`**`()` This method is used by the applet `process()` method to distinguish the SELECT APDU command which selected `this` applet, from all other other SELECT APDU commands which may relate to file or internal applet state selection.

Methods inherited from class java.lang.Object
equals

Constructor Detail

Applet

```
protected Applet()
```

Only this class's `install()` method should create the applet object.

Method Detail

install

```
public static void install(byte[] bArray,
                           short bOffset,
                           byte bLength)
                    throws ISOException
```

To create an instance of the `Applet` subclass, the JCRE will call this static method first.

The applet should perform any necessary initializations and must call one of the `register()` methods. The installation is considered successful when the call to `register()` completes without an exception. The installation is deemed unsuccessful if the `install` method does not call a `register()` method, or if an exception is thrown from within the `install` method prior to the call to a `register()` method, or if the `register()` method throws an exception. If the installation is unsuccessful, the JCRE must perform all the necessary clean up when it receives control. Successful installation makes the applet instance capable of being selected via a SELECT APDU command.

Installation parameters are supplied in the byte array parameter and must be in a format defined by the applet. The `bArray` object is a global array. If the applet desires to preserve any of this data, it should copy the data into its own object.

`bArray` is zeroed by the JCRE after the return from the `install()` method.

References to the `bArray` object cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

The implementation of this method provided by `Applet` class throws an `ISOException` with reason code = `ISO7816.SW_FUNC_NOT_SUPPORTED`.

Note:

- *Exceptions thrown by this method after successful installation are caught by the JCRE and processed by the Installer.*

Parameters:

`bArray` - the array containing installation parameters.
`bOffset` - the starting offset in bArray.
`bLength` - the length in bytes of the parameter data in bArray. The maximum value of bLength is 32.

---

process

```
public abstract void process(APDU apdu)
                     throws ISOException
```

Called by the JCRE to process an incoming APDU command. An applet is expected to perform the action requested and return response data if any to the terminal.

Upon normal return from this method the JCRE sends the ISO 7816-4 defined success status (90 00) in APDU response. If this method throws an `ISOException` the JCRE sends the associated reason code as the response status instead.

The JCRE zeroes out the APDU buffer before receiving a new APDU command from the CAD. The five header bytes of the APDU command are available in APDU buffer[0..4] at the time this method is called.

The `APDU` object parameter is a temporary JCRE Entry Point Object. A temporary JCRE Entry Point Object can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components.

Notes:

- *APDU buffer[5..] is undefined and should not be read or written prior to invoking the* `APDU.setIncomingAndReceive()` *method if incoming data is expected. Altering the APDU buffer[5..] could corrupt incoming data.*

Parameters:

`apdu` - the incoming `APDU` object

Throws:

ISOException - with the response bytes per ISO 7816-4

See Also:

`APDU`

---

select

```
public boolean select()
```

Called by the JCRE to inform this applet that it has been selected.

It is called when a SELECT APDU command is received and before the applet is selected. SELECT APDU commands use instance AID bytes for applet selection. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

A subclass of `Applet` should override this method if it should perform any initialization that may be required to process APDU commands that may follow. This method returns a boolean to indicate that it is ready to accept incoming APDU commands via its `process()` method. If this method returns false, it indicates to the JCRE that this Applet declines to be selected.

The implementation of this method provided by `Applet` class returns `true`.

Returns:

`true` to indicate success, `false` otherwise.

---

deselect

```
public void deselect()
```

Called by the JCRE to inform this currently selected applet that another (or the same) applet will be selected. It is called when a SELECT APDU command is received by the JCRE. This method is invoked prior to another applets or this very applets `select()` method being invoked.

A subclass of `Applet` should override this method if it has any cleanup or bookkeeping work to be performed before another applet is selected.

The default implementation of this method provided by `Applet` class does nothing.

Notes:

- *Unchecked exceptions thrown by this method are caught by the JCRE but the applet is deselected.*
- *Transient objects of* `JCSystem.CLEAR_ON_DESELECT` *clear event type are cleared to their default value by the JCRE after this method.*
- *This method is NOT called on reset or power loss.*

---

getShareableInterfaceObject

```
public Shareable getShareableInterfaceObject(AID clientAID,
                                             byte parameter)
```

Called by the JCRE to obtain a shareable interface object from this server applet, on behalf of a request from a client applet. This method executes in the applet context of `this` applet instance. The client applet initiated this request by calling the `JCSystem.getAppletShareableInterfaceObject()` method. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`clientAID` - the `AID` object of the client applet.

`parameter` - optional parameter byte. The parameter byte may be used by the client to specify which shareable interface object is being requested.

Returns:

the shareable interface object or `null`. Note:

- *The* `clientAID` *parameter is a JCRE owned* `AID` *instance. JCRE owned instances of* `AID` *are permanent JCRE Entry Point Objects and can be accessed from any applet context. References to these permanent objects can be stored and re-used.*

See Also:

`JCSystem.getAppletShareableInterfaceObject(AID, byte)`

---

register

```
protected final void register()
                       throws SystemException
```

This method is used by the applet to register `this` applet instance with the JCRE and to assign the `Applet` subclass AID bytes as its instance AID bytes. One of the `register()` methods must be called from within `install()` to be registered with the JCRE. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Throws:

SystemException - with the following reason codes:

- `SystemException.ILLEGAL_AID` if the `Applet` subclass AID bytes are in use or if the applet instance has previously called one of the `register()` methods.

---

register

```
protected final void register(byte[] bArray,
                              short bOffset,
                              byte bLength)
                       throws SystemException
```

This method is used by the applet to register `this` applet instance with the JCRE and assign the specified AID bytes as its instance AID bytes. One of the `register()` methods must be called from within `install()` to be registered with the JCRE. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`bArray` - the byte array containing the AID bytes.

`bOffset` - the start of AID bytes in bArray.

`bLength` - the length of the AID bytes in bArray.

Throws:

APDUException - with the following reason codes:

SystemException - with the following reason code:

- `SystemException.ILLEGAL_VALUE` if the `bLength` parameter is less than `5` or greater than `16`.
- `SystemException.ILLEGAL_AID` if the specified instance AID bytes are in use or if the RID portion of the AID bytes in the `bArray` parameter does not match the RID portion of the `Applet` subclass AID bytes or if the applet instance has previously called one of the `register()` methods.

---

selectingApplet

```
protected final boolean selectingApplet()
```

This method is used by the applet `process()` method to distinguish the SELECT APDU command which selected `this` applet, from all other other SELECT APDU commands which may relate to file or internal applet state selection.

Returns:

`true` if `this` applet is being selected.

---

javacard.framework
Class CardException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--javacard.framework.CardException
```

Direct Known Subclasses:
UserException

public class **CardException**
extends Exception

The `CardException` class defines a field `reason` and two accessor methods `getReason()` and `setReason()`. The `reason` field encapsulates exception cause identifier in Java Card. All Java Card checked Exception classes should extend `CardException`. This class also provides a resource-saving mechanism (`throwIt()` method) for using a JCRE owned instance of this class.

Constructor Summary

`CardException``(short reason)` Construct a CardException instance with the specified reason.	

Method Summary

`short`	**`getReason`**`()` Get reason code
`void`	**`setReason`**`(short reason)` Set reason code
`static void`	**`throwIt`**`(short reason)` Throw the JCRE owned instance of `CardException` class with the specified reason.

Methods inherited from class java.lang.Object
equals

Constructor Detail

CardException

```
public CardException(short reason)
```

Construct a CardException instance with the specified reason. To conserve on resources, use the `throwIt()` method to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception

Method Detail

getReason

```
public short getReason()
```

Get reason code

Returns:

the reason for the exception

---

setReason

```
public void setReason(short reason)
```

Set reason code

Parameters:

`reason` - the reason for the exception

---

throwIt

```
public static void throwIt(short reason)
                    throws CardException
```

Throw the JCRE owned instance of `CardException` class with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1*

Specification for details.

Parameters:

`reason` - the reason for the exception

Throws:

CardException - always.

---

javacard.framework
Class CardRuntimeException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
```

Direct Known Subclasses:
APDUException, CryptoException, ISOException, PINException, SystemException, TransactionException

---

public class **CardRuntimeException**
extends RuntimeException

The `CardRuntimeException` class defines a field `reason` and two accessor methods `getReason()` and `setReason()`. The `reason` field encapulates exception cause identifier in Java Card. All Java Card unchecked Exception classes should extend `CardRuntimeException`. This class also provides a resource-saving mechanism (`throwIt()` method) for using a JCRE owned instance of this class.

Constructor Summary

`CardRuntimeException``(short reason)` Construct a CardRuntimeException instance with the specified reason.	

Method Summary

`short`	**`getReason`**`()` Get reason code
`void`	**`setReason`**`(short reason)` Set reason code
`static void`	**`throwIt`**`(short reason)` Throw the JCRE owned instance of the `CardRuntimeException` class with the specified reason.

Methods inherited from class java.lang.Object
equals

Constructor Detail

CardRuntimeException

```
public CardRuntimeException(short reason)
```

Construct a CardRuntimeException instance with the specified reason. To conserve on resources, use `throwIt()` method to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception

Method Detail

getReason

```
public short getReason()
```

Get reason code

Returns:

the reason for the exception

---

setReason

```
public void setReason(short reason)
```

Set reason code

Parameters:

`reason` - the reason for the exception

---

throwIt

```
public static void throwIt(short reason)
                    throws CardRuntimeException
```

Throw the JCRE owned instance of the `CardRuntimeException` class with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`reason` - the reason for the exception

Throws:

CardRuntimeException - always.

---

javacard.framework
Interface ISO7816

public abstract interface **ISO7816**

`ISO7816` encapsulates constants related to ISO 7816-3 and ISO 7816-4. `ISO7816` interface contains only static fields.

The static fields with `SW_` prefixes define constants for the ISO 7816-4 defined response status word. The fields which use the `_00` suffix require the low order byte to be customized appropriately e.g (ISO7816.SW_CORRECT_LENGTH_00 + (0x0025 & 0xFF)).

The static fields with `OFFSET_` prefixes define constants to be used to index into the APDU buffer byte array to access ISO 7816-4 defined header information.

Field Summary

static byte	**CLA_ISO7816** APDU command CLA : ISO 7816 = 0x00
static byte	**INS_EXTERNAL_AUTHENTICATE** APDU command INS : EXTERNAL AUTHENTICATE = 0x82
static byte	**INS_SELECT** APDU command INS : SELECT = 0xA4
static byte	**OFFSET_CDATA** APDU command data offset : CDATA = 5
static byte	**OFFSET_CLA** APDU header offset : CLA = 0
static byte	**OFFSET_INS** APDU header offset : INS = 1
static byte	**OFFSET_LC** APDU header offset : LC = 4
static byte	**OFFSET_P1** APDU header offset : P1 = 2
static byte	**OFFSET_P2** APDU header offset : P2 = 3
static short	**SW_APPLET_SELECT_FAILED** Response status : Applet selection failed = 0x6999;

static short	**SW_BYTES_REMAINING_00** Response status : Response bytes remaining = 0x6100
static short	**SW_CLA_NOT_SUPPORTED** Response status : CLA value not supported = 0x6E00
static short	**SW_COMMAND_NOT_ALLOWED** Response status : Command not allowed (no current EF) = 0x6986
static short	**SW_CONDITIONS_NOT_SATISFIED** Response status : Conditions of use not satisfied = 0x6985
static short	**SW_CORRECT_LENGTH_00** Response status : Correct Expected Length (Le) = 0x6C00
static short	**SW_DATA_INVALID** Response status : Data invalid = 0x6984
static short	**SW_FILE_FULL** Response status : Not enough memory space in the file = 0x6A84
static short	**SW_FILE_INVALID** Response status : File invalid = 0x6983
static short	**SW_FILE_NOT_FOUND** Response status : File not found = 0x6A82
static short	**SW_FUNC_NOT_SUPPORTED** Response status : Function not supported = 0x6A81
static short	**SW_INCORRECT_P1P2** Response status : Incorrect parameters (P1,P2) = 0x6A86
static short	**SW_INS_NOT_SUPPORTED** Response status : INS value not supported = 0x6D00
static short	**SW_NO_ERROR** Response status : No Error = (short)0x9000
static short	**SW_RECORD_NOT_FOUND** Response status : Record not found = 0x6A83
static short	**SW_SECURITY_STATUS_NOT_SATISFIED** Response status : Security condition not satisfied = 0x6982
static short	**SW_UNKNOWN** Response status : No precise diagnosis = 0x6F00
static short	**SW_WRONG_DATA** Response status : Wrong data = 0x6A80
static short	**SW_WRONG_LENGTH** Response status : Wrong length = 0x6700

static short	**SW_WRONG_P1P2** Response status : Incorrect parameters (P1,P2) = 0x6B00

Field Detail

SW_NO_ERROR

```
public static final short SW_NO_ERROR
```

Response status : No Error = (short)0x9000

SW_BYTES_REMAINING_00

```
public static final short SW_BYTES_REMAINING_00
```

Response status : Response bytes remaining = 0x6100

SW_WRONG_LENGTH

```
public static final short SW_WRONG_LENGTH
```

Response status : Wrong length = 0x6700

SW_SECURITY_STATUS_NOT_SATISFIED

```
public static final short SW_SECURITY_STATUS_NOT_SATISFIED
```

Response status : Security condition not satisfied = 0x6982

SW_FILE_INVALID

```
public static final short SW_FILE_INVALID
```

Response status : File invalid = 0x6983

SW_DATA_INVALID

```
public static final short SW_DATA_INVALID
```

Response status : Data invalid = 0x6984

SW_CONDITIONS_NOT_SATISFIED

```
public static final short SW_CONDITIONS_NOT_SATISFIED
```

Response status : Conditions of use not satisfied = 0x6985

SW_COMMAND_NOT_ALLOWED

```
public static final short SW_COMMAND_NOT_ALLOWED
```

Response status : Command not allowed (no current EF) = 0x6986

SW_APPLET_SELECT_FAILED

```
public static final short SW_APPLET_SELECT_FAILED
```

Response status : Applet selection failed = 0x6999;

SW_WRONG_DATA

```
public static final short SW_WRONG_DATA
```

Response status : Wrong data = 0x6A80

SW_FUNC_NOT_SUPPORTED

```
public static final short SW_FUNC_NOT_SUPPORTED
```

Response status : Function not supported = 0x6A81

SW_FILE_NOT_FOUND

```
public static final short SW_FILE_NOT_FOUND
```

Response status : File not found = 0x6A82

SW_RECORD_NOT_FOUND

```
public static final short SW_RECORD_NOT_FOUND
```

Response status : Record not found = 0x6A83

SW_INCORRECT_P1P2

```
public static final short SW_INCORRECT_P1P2
```

Response status : Incorrect parameters (P1,P2) = 0x6A86

SW_WRONG_P1P2

```
public static final short SW_WRONG_P1P2
```

Response status : Incorrect parameters (P1,P2) = 0x6B00

SW_CORRECT_LENGTH_00

```
public static final short SW_CORRECT_LENGTH_00
```

Response status : Correct Expected Length (Le) = 0x6C00

SW_INS_NOT_SUPPORTED

```
public static final short SW_INS_NOT_SUPPORTED
```

Response status : INS value not supported = 0x6D00

SW_CLA_NOT_SUPPORTED

```
public static final short SW_CLA_NOT_SUPPORTED
```

Response status : CLA value not supported = 0x6E00

SW_UNKNOWN

```
public static final short SW_UNKNOWN
```

Response status : No precise diagnosis = 0x6F00

SW_FILE_FULL

```
public static final short SW_FILE_FULL
```

Response status : Not enough memory space in the file = 0x6A84

OFFSET_CLA

```
public static final byte OFFSET_CLA
```

APDU header offset : CLA = 0

OFFSET_INS

```
public static final byte OFFSET_INS
```

APDU header offset : INS = 1

OFFSET_P1

```
public static final byte OFFSET_P1
```

APDU header offset : P1 = 2

OFFSET_P2

```
public static final byte OFFSET_P2
```

APDU header offset : P2 = 3

OFFSET_LC

```
public static final byte OFFSET_LC
```

APDU header offset : LC = 4

OFFSET_CDATA

```
public static final byte OFFSET_CDATA
```

APDU command data offset : CDATA = 5

CLA_ISO7816

```
public static final byte CLA_ISO7816
```

APDU command CLA : ISO 7816 = 0x00

INS_SELECT

```
public static final byte INS_SELECT
```

APDU command INS : SELECT = 0xA4

INS_EXTERNAL_AUTHENTICATE

```
public static final byte INS_EXTERNAL_AUTHENTICATE
```

APDU command INS : EXTERNAL AUTHENTICATE = 0x82

javacard.framework
Class ISOException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
                          |
                          +--javacard.framework.ISOException
```

public class **ISOException**
extends CardRuntimeException

`ISOException` class encapsulates an ISO 7816-4 response status word as its `reason` code.

The `APDU` class throws JCRE owned instances of `ISOException`.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Constructor Summary

`ISOException``(short sw)` Constructs an ISOException instance with the specified status word.	

Method Summary

`static void`	**`throwIt`**`(short sw)` Throws the JCRE owned instance of the ISOException class with the specified status word.

Methods inherited from class javacard.framework.CardRuntimeException
`getReason, setReason`

Methods inherited from class java.lang.Object
equals

Constructor Detail

ISOException

```
public ISOException(short sw)
```

Constructs an ISOException instance with the specified status word. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Parameters:

`sw` - the ISO 7816-4 defined status word

Method Detail

throwIt

```
public static void throwIt(short sw)
```

Throws the JCRE owned instance of the ISOException class with the specified status word.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`sw` - ISO 7816-4 defined status word

Throws:

ISOException - always.

javacard.framework
Class JCSystem

```
java.lang.Object
  |
  +--javacard.framework.JCSystem
```

public final class **JCSystem**
extends Object

The `JCSystem` class includes a collection of methods to control applet execution, resource management, atomic transaction management and inter-applet object sharing in Java Card. All methods in `JCSystem` class are static methods.

The JCSystem class also includes methods to control the persistence and transience of objects. The term *persistent* means that objects and their values persist from one CAD session to the next, indefinitely. Persistent object values are updated atomically using transactions.

The `makeTransient...Array()` methods can be used to create *transient* arrays with primitive data components. Transient array data is lost (in an undefined state, but the real data is unavailable) immediately upon power loss, and is reset to the default value at the occurrence of certain events such as card reset or deselect. Updates to the values of transient arrays are not atomic and are not affected by transactions.

The JCRE maintains an atomic transaction commit buffer which is initialized on card reset (or power on). When a transaction is in progress, the JCRE journals all updates to persistent data space into this buffer so that it can always guarantee, at commit time, that everything in the buffer is written or nothing at all is written. The `JCSystem` includes methods to control an atomic transaction. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

See Also:
`SystemException`, `TransactionException`, `Applet`

Field Summary

`static byte`	**`CLEAR_ON_DESELECT`** This event code indicates that the contents of the transient object are cleared to the default value on applet deselection event or in `CLEAR_ON_RESET` cases.
`static byte`	**`CLEAR_ON_RESET`** This event code indicates that the contents of the transient object are cleared to the default value on card reset (or power on) event.
`static byte`	**`NOT_A_TRANSIENT_OBJECT`** This event code indicates that the object is not transient.

Method Summary

static void	**abortTransaction**() Aborts the atomic transaction.
static void	**beginTransaction**() Begins an atomic transaction.
static void	**commitTransaction**() Commits an atomic transaction.
static AID	**getAID**() Returns the JCRE owned instance of the AID object associated with the current applet context.
static Shareable	**getAppletShareableInterfaceObject**(AID serverAID, byte parameter) This method is called by a client applet to get a server applet's shareable interface object.
static short	**getMaxCommitCapacity**() Returns the total number of bytes in the commit buffer.
static AID	**getPreviousContextAID**() This method is called to obtain the JCRE owned instance of the AID object associated with the previously active applet context.
static byte	**getTransactionDepth**() Returns the current transaction nesting depth level.
static short	**getUnusedCommitCapacity**() Returns the number of bytes left in the commit buffer.
static short	**getVersion**() Returns the current major and minor version of the Java Card API.
static byte	**isTransient**(Object theObj) Used to check if the specified object is transient.
static AID	**lookupAID**(byte[] buffer, short offset, byte length) Returns the JCRE owned instance of the AID object, if any, encapsulating the specified AID bytes in the buffer parameter if there exists a successfully installed applet on the card whose instance AID exactly matches that of the specified AID bytes.
static boolean[]	**makeTransientBooleanArray**(short length, byte event) Create a transient boolean array with the specified array length.
static byte[]	**makeTransientByteArray**(short length, byte event) Create a transient byte array with the specified array length.

`static Object[]`	**`makeTransientObjectArray`**`(short length, byte event)` Create a transient array of `Object` with the specified array length.
`static short[]`	**`makeTransientShortArray`**`(short length, byte event)` Create a transient short array with the specified array length.

Methods inherited from class java.lang.Object
`equals`

Field Detail

NOT_A_TRANSIENT_OBJECT

`public static final byte` **`NOT_A_TRANSIENT_OBJECT`**

This event code indicates that the object is not transient.

CLEAR_ON_RESET

`public static final byte` **`CLEAR_ON_RESET`**

This event code indicates that the contents of the transient object are cleared to the default value on card reset (or power on) event.

CLEAR_ON_DESELECT

`public static final byte` **`CLEAR_ON_DESELECT`**

This event code indicates that the contents of the transient object are cleared to the default value on applet deselection event or in `CLEAR_ON_RESET` cases.

Notes:

- `CLEAR_ON_DESELECT` *transient objects can be accessed only when the applet which created the object is the currently the selected applet.*
- *The JCRE will throw a* `SecurityException` *if a* `CLEAR_ON_DESELECT` *transient object is accessed when the currently selected applet is not the applet which created the object.*

Method Detail

isTransient

```
public static byte isTransient(Object theObj)
```

Used to check if the specified object is transient.

Notes:

This method returns `NOT_A_TRANSIENT_OBJECT` *if the specified object is* `null` *or is not an array type.*

Parameters:

`theObj` - the object being queried.

Returns:

`NOT_A_TRANSIENT_OBJECT`, `CLEAR_ON_RESET`, or `CLEAR_ON_DESELECT`.

See Also:

`makeTransientBooleanArray(short, byte)`,
`makeTransientByteArray(short, byte)`,
`makeTransientShortArray(short, byte)`,
`makeTransientObjectArray(short, byte)`

---

makeTransientBooleanArray

```
public static boolean[] makeTransientBooleanArray(short length,
                                                  byte event)
                                           throws SystemException
```

Create a transient boolean array with the specified array length.

Parameters:

`length` - the length of the boolean array.

`event` - the `CLEAR_ON...` event which causes the array elements to be cleared.

Throws:

SystemException - with the following reason codes:

- `SystemException.ILLEGAL_VALUE` if event is not a valid event code.
- `SystemException.NO_TRANSIENT_SPACE` if sufficient transient space is not available.
- `SystemException.ILLEGAL_TRANSIENT` if the current applet context is not the currently selected applet context and `CLEAR_ON_DESELECT` is specified.

---

makeTransientByteArray

```
public static byte[] makeTransientByteArray(short length,
                                            byte event)
                                     throws SystemException
```

Create a transient byte array with the specified array length.

Parameters:

`length` - the length of the byte array.

`event` - the `CLEAR_ON...` event which causes the array elements to be cleared.

Throws:

SystemException - with the following reason codes:

- `SystemException.ILLEGAL_VALUE` if event is not a valid event code.
- `SystemException.NO_TRANSIENT_SPACE` if sufficient transient space is not available.
- `SystemException.ILLEGAL_TRANSIENT` if the current applet context is not the currently selected applet context and `CLEAR_ON_DESELECT` is specified.

---

makeTransientShortArray

```
public static short[] makeTransientShortArray(short length,
                                              byte event)
                                       throws SystemException
```

Create a transient short array with the specified array length.

Parameters:

`length` - the length of the short array.

`event` - the `CLEAR_ON...` event which causes the array elements to be cleared.

Throws:

SystemException - with the following reason codes:

- `SystemException.ILLEGAL_VALUE` if event is not a valid event code.
- `SystemException.NO_TRANSIENT_SPACE` if sufficient transient space is not available.
- `SystemException.ILLEGAL_TRANSIENT` if the current applet context is not the currently selected applet context and `CLEAR_ON_DESELECT` is specified.

---

makeTransientObjectArray

```
public static Object[] makeTransientObjectArray(short length,
                                                byte event)
                                         throws SystemException
```

Create a transient array of `Object` with the specified array length.

Parameters:

`length` - the length of the `Object` array.

`event` - the `CLEAR_ON...` event which causes the array elements to be cleared.

Throws:

SystemException - with the following reason codes:

- `SystemException.ILLEGAL_VALUE` if event is not a valid event code.
- `SystemException.NO_TRANSIENT_SPACE` if sufficient transient space is not available.
- `SystemException.ILLEGAL_TRANSIENT` if the current applet context is not the currently selected applet context and `CLEAR_ON_DESELECT` is specified.

getVersion

```
public static short getVersion()
```

Returns the current major and minor version of the Java Card API.

Returns:

version number as byte.byte (major.minor)

getAID

```
public static AID getAID()
```

Returns the JCRE owned instance of the `AID` object associated with the current applet context. Returns `null` if the `Applet.register()` method has not yet been invoked.

JCRE owned instances of `AID` are permanent JCRE Entry Point Objects and can be accessed from any applet context. References to these permanent objects can be stored and re-used.

See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Returns:

the `AID` object.

lookupAID

```
public static AID lookupAID(byte[] buffer,
                            short offset,
                            byte length)
```

Returns the JCRE owned instance of the `AID` object, if any, encapsulating the specified AID bytes in the `buffer` parameter if there exists a successfully installed applet on the card whose instance AID exactly matches that of the specified AID bytes.

JCRE owned instances of `AID` are permanent JCRE Entry Point Objects and can be accessed from any applet context. References to these permanent objects can be stored and re-used.

See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`buffer` - byte array containing the AID bytes.
`offset` - offset within buffer where AID bytes begin.
`length` - length of AID bytes in buffer.

Returns:

the `AID` object, if any; `null` otherwise. A VM exception is thrown if `buffer` is `null`, or if `offset` or `length` are out of range.

---

beginTransaction

```
public static void beginTransaction()
                            throws TransactionException
```

Begins an atomic transaction. If a transaction is already in progress (transactionDepth != 0), a TransactionException is thrown.

Throws:

TransactionException - with the following reason codes:

- `TransactionException.IN_PROGRESS` if a transaction is already in progress.

See Also:

`commitTransaction()`, `abortTransaction()`

---

abortTransaction

```
public static void abortTransaction()
                            throws TransactionException
```

Aborts the atomic transaction. The contents of the commit buffer is discarded.

Notes:

- *Do not call this method from within a transaction which creates new objects because the JCRE may not recover the heap space used by the new object instances.*
- *The JCRE ensures that any variable of reference type which references an object instantiated from within this aborted transaction is equivalent to a* `null` *reference.*

Throws:

TransactionException - with the following reason codes:

- `TransactionException.NOT_IN_PROGRESS` if a transaction is not in progress.

See Also:

`beginTransaction()`, `commitTransaction()`

---

commitTransaction

```
public static void commitTransaction()
                             throws TransactionException
```

Commits an atomic transaction. The contents of commit buffer is atomically commited. If a transaction is not in progress (transactionDepth == 0) then a TransactionException is thrown.

Throws:

TransactionException - with the following reason codes:

- `TransactionException.NOT_IN_PROGRESS` if a transaction is not in progress.

See Also:

`beginTransaction()`, `abortTransaction()`

getTransactionDepth

```
public static byte getTransactionDepth()
```

Returns the current transaction nesting depth level. At present, only 1 transaction can be in progress at a time.

Returns:

1 if transaction in progress, 0 if not.

getUnusedCommitCapacity

```
public static short getUnusedCommitCapacity()
```

Returns the number of bytes left in the commit buffer.

Returns:

the number of bytes left in the commit buffer

See Also:

`getMaxCommitCapacity()`

getMaxCommitCapacity

```
public static short getMaxCommitCapacity()
```

Returns the total number of bytes in the commit buffer. This is approximately the maximum number of bytes of persistent data which can be modified during a transaction. However, the transaction subsystem requires additional bytes of overhead data to be included in the commit buffer, and this depends on the number of fields modified and the implementation of the transaction subsystem. The application cannot determine the actual maximum amount of data which can be modified during a transaction without taking these overhead bytes into consideration.

Returns:

the total number of bytes in the commit buffer

See Also:

`getUnusedCommitCapacity()`

getPreviousContextAID

```
public static AID getPreviousContextAID()
```

This method is called to obtain the JCRE owned instance of the `AID` object associated with the previously active applet context. This method is typically used by a server applet, while executing a shareable interface method to determine the identity of its client and thereby control access privileges.

JCRE owned instances of `AID` are permanent JCRE Entry Point Objects and can be accessed from any applet context. References to these permanent objects can be stored and re-used.

See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Returns:

the `AID` object of the previous context, or `null` if JCRE.

---

getAppletShareableInterfaceObject

```
public static Shareable getAppletShareableInterfaceObject(AID serverAID,
                                                          byte parameter)
```

This method is called by a client applet to get a server applet's shareable interface object.

This method returns `null` if the `Applet.register()` has not yet been invoked or if the server does not exist or if the server returns `null`.

Parameters:

`serverAID` - the AID of the server applet.

`parameter` - optional parameter data.

Returns:

the shareable interface object or `null`.

See Also:

`Applet.getShareableInterfaceObject(AID, byte)`

---

javacard.framework
Class OwnerPIN

```
java.lang.Object
  |
  +--javacard.framework.OwnerPIN
```

public class **OwnerPIN**
extends Object
implements PIN

This class represents an Owner PIN. It implements Personal Identification Number functionality as defined in the `PIN` interface. It provides the ability to update the PIN and thus owner functionality.

The implementation of this class must protect against attacks based on program flow prediction.Even if a transaction is in progress, internal state such as the try counter, the validated flag and the blocking state must not be conditionally updated during PIN presentation.

If an implementation of this class creates transient arrays, it must ensure that they are `CLEAR_ON_RESET` transient objects.

The protected methods `getValidatedFlag` and `setValidatedFlag` allow a subclass of this class to optimize the storage for the validated boolean state.

Some methods of instances of this class are only suitable for sharing when there exists a trust relationship among the applets. A typical shared usage would use a proxy PIN interface which implements both the `PIN` interface and the `Shareable` interface.

Any of the methods of the `OwnerPIN` may be called with a transaction in progress. None of the methods of `OwnerPIN` class initiate or alter the state of the transaction if one is in progress.

See Also:
`PINException`, `PIN`, `Shareable`, `JCSystem`

Constructor Summary

`OwnerPIN``(byte tryLimit, byte maxPINSize)` Constructor.

Method Summary

boolean	**check**(byte[] pin, short offset, byte length) Compares pin against the PIN value.
byte	**getTriesRemaining**() Returns the number of times remaining that an incorrect PIN can be presented before the PIN is blocked.
protected boolean	**getValidatedFlag**() This protected method returns the validated flag.
boolean	**isValidated**() Returns true if a valid PIN has been presented since the last card reset or last call to reset().
void	**reset**() If the validated flag is set, this method resets it.
void	**resetAndUnblock**() This method resets the validated flag and resets the PIN try counter to the value of the PIN try limit.
protected void	**setValidatedFlag**(boolean value) This protected method sets the value of the validated flag.
void	**update**(byte[] pin, short offset, byte length) This method sets a new value for the PIN and resets the PIN try counter to the value of the PIN try limit.

Methods inherited from class java.lang.Object
equals

Constructor Detail

OwnerPIN

```
public OwnerPIN(byte tryLimit,
                byte maxPINSize)
         throws PINException
```

Constructor. Allocates a new PIN instance with validated flag set to false.

Parameters:

tryLimit - the maximum number of times an incorrect PIN can be presented.

maxPINSize - the maximum allowed PIN size. maxPINSize must be >=1.

Throws:

PINException - with the following reason codes:

- `PINException.ILLEGAL_VALUE` if `maxPINSize` parameter is less than 1.

Method Detail

getValidatedFlag

```
protected boolean getValidatedFlag()
```

This protected method returns the validated flag. This method is intended for subclass of this `OwnerPIN` to access or override the internal PIN state of the `OwnerPIN`.

Returns:

the boolean state of the PIN validated flag.

---

setValidatedFlag

```
protected void setValidatedFlag(boolean value)
```

This protected method sets the value of the validated flag. This method is intended for subclass of this `OwnerPIN` to control or override the internal PIN state of the `OwnerPIN`.

Parameters:

`value` - the new value for the validated flag.

---

getTriesRemaining

```
public byte getTriesRemaining()
```

Returns the number of times remaining that an incorrect PIN can be presented before the `PIN` is blocked.

Specified by:

getTriesRemaining in interface PIN

Returns:

the number of times remaining

---

check

```
public boolean check(byte[] pin,
                     short offset,
                     byte length)
              throws ArrayIndexOutOfBoundsException,
                     NullPointerException
```

Compares `pin` against the PIN value. If they match and the `PIN` is not blocked, it sets the validated flag and resets the try counter to its maximum. If it does not match, it decrements the try counter and, if the counter has reached zero, blocks the `PIN`. Even if a transaction is in progress, internal state such as the try counter, the validated flag and the blocking state must not be conditionally updated.

Notes:

- *If* `NullPointerException` *or* `ArrayIndexOutOfBoundsException` *is thrown, the validated flag must be set to false, the try counter must be decremented and, the* `PIN` *blocked if the counter reaches zero.*
- *If* `offset` *or* `length` *parameter is negative an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `offset+length` *is greater than* `pin.length`*, the length of the* `pin` *array, an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `pin` *parameter is* `null` *a* `NullPointerException` *exception is thrown.*

Specified by:

check in interface PIN

Parameters:

`pin` - the byte array containing the PIN value being checked
`offset` - the starting offset in the `pin` array
`length` - the length of `pin`.

Returns:

`true` if the PIN value matches; `false` otherwise

Throws:

ArrayIndexOutOfBoundsException - - if the check operation would cause access of data outside array bounds.
NullPointerException - - if `pin` is `null`

---

isValidated

```
public boolean isValidated()
```

Returns `true` if a valid PIN has been presented since the last card reset or last call to `reset()`.

Specified by:

isValidated in interface PIN

Returns:

`true` if validated; `false` otherwise

---

reset

```
public void reset()
```

If the validated flag is set, this method resets it. If the validated flag is not set, this method does nothing.

Specified by:

reset in interface PIN

update

```
public void update(byte[] pin,
                   short offset,
                   byte length)
            throws PINException
```

This method sets a new value for the PIN and resets the PIN try counter to the value of the PIN try limit. It also resets the validated flag.

This method copies the input pin parameter into an internal representation. If a transaction is in progress, the new pin and try counter update must be conditional i.e the copy operation must use the transaction facility.

Parameters:

pin - the byte array containing the new PIN value
offset - the starting offset in the pin array
length - the length of the new PIN.

Throws:

PINException - with the following reason codes:

- PINException.ILLEGAL_VALUE if length is greater than configured maximum PIN size.

See Also:

JCSystem.beginTransaction()

---

resetAndUnblock

```
public void resetAndUnblock()
```

This method resets the validated flag and resets the PIN try counter to the value of the PIN try limit. This method is used by the owner to re-enable the blocked PIN.

---

javacard.framework
Interface PIN

All Known Implementing Classes:
OwnerPIN

---

public abstract interface **PIN**

This interface represents a PIN. An implementation must maintain these internal values:

- PIN value
- try limit, the maximum number of times an incorrect PIN can be presented before the PIN is blocked. When the PIN is blocked, it cannot be validated even on valid PIN presentation.
- max PIN size, the maximum length of PIN allowed
- try counter, the remaining number of times an incorrect PIN presentation is permitted before the `PIN` becomes blocked.
- validated flag, true if a valid PIN has been presented. This flag is reset on every card reset.

This interface does not make any assumptions about where the data for the PIN value comparison is stored.

An owner implementation of this interface must provide a way to initialize/update the PIN value.The owner implemention of the interface must protect against attacks based on program flow prediction. Even if a transaction is in progress, internal state such as the try counter, the validated flag and the blocking state must not be conditionally updated during PIN presentation.

A typical card global PIN usage will combine an instance of `OwnerPIN` class and a a Proxy PIN interface which implements both the `PIN` and the `Shareable` interfaces. The `OwnerPIN` instance would be manipulated only by the owner who has update privilege. All others would access the global PIN functionality via the proxy PIN interface.

See Also:
`OwnerPIN`, `Shareable`

Method Summary

boolean	**check**(byte[] pin, short offset, byte length) Compares pin against the PIN value.
byte	**getTriesRemaining**() Returns the number of times remaining that an incorrect PIN can be presented before the PIN is blocked.
boolean	**isValidated**() Returns true if a valid PIN value has been presented since the last card reset or last call to reset().
void	**reset**() If the validated flag is set, this method resets it.

Method Detail

getTriesRemaining

```
public byte getTriesRemaining()
```

Returns the number of times remaining that an incorrect PIN can be presented before the PIN is blocked.

Returns:

the number of times remaining

---

check

```
public boolean check(byte[] pin,
                     short offset,
                     byte length)
              throws ArrayIndexOutOfBoundsException,
                     NullPointerException
```

Compares pin against the PIN value. If they match and the PIN is not blocked, it sets the validated flag and resets the try counter to its maximum. If it does not match, it decrements the try counter and, if the counter has reached zero, blocks the PIN. Even if a transaction is in progress, internal state such as the try counter, the validated flag and the blocking state must not be conditionally updated.

Notes:

- *If* NullPointerException *or* ArrayIndexOutOfBoundsException *is thrown, the validated flag must be set to false, the try counter must be decremented and, the* PIN *blocked if the counter reaches zero.*
- *If* offset *or* length *parameter is negative an* ArrayIndexOutOfBoundsException

exception is thrown.

- *If* `offset+length` *is greater than* `pin.length`*, the length of the* `pin` *array, an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `pin` *parameter is* `null` *a* `NullPointerException` *exception is thrown.*

Parameters:

`pin` - the byte array containing the PIN value being checked
`offset` - the starting offset in the `pin` array
`length` - the length of `pin`.

Returns:

`true` if the PIN value matches; `false` otherwise

Throws:

ArrayIndexOutOfBoundsException - - if the check operation would cause access of data outside array bounds.
NullPointerException - - if `pin` is `null`

---

isValidated

```
public boolean isValidated()
```

Returns `true` if a valid PIN value has been presented since the last card reset or last call to `reset()`.

Returns:

`true` if validated; `false` otherwise

---

reset

```
public void reset()
```

If the validated flag is set, this method resets it. If the validated flag is not set, this method does nothing.

---

javacard.framework
Class PINException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
                          |
                          +--javacard.framework.PINException
```

public class **PINException**
extends CardRuntimeException

`PINException` represents a `OwnerPIN` class access-related exception.

The `OwnerPIN` class throws JCRE owned instances of `PINException`.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

See Also:
`OwnerPIN`

Field Summary

`static short`	**`ILLEGAL_VALUE`** This reason code is used to indicate that one or more input parameters is out of allowed bounds.

Constructor Summary

`PINException``(short reason)` Constructs a PINException.	

Method Summary

static void	**throwIt**(short reason) Throws the JCRE owned instance of PINException with the specified reason.

Methods inherited from class javacard.framework.CardRuntimeException

getReason, setReason

Methods inherited from class java.lang.Object

equals

Field Detail

ILLEGAL_VALUE

```
public static final short ILLEGAL_VALUE
```

This reason code is used to indicate that one or more input parameters is out of allowed bounds.

Constructor Detail

PINException

```
public PINException(short reason)
```

Constructs a PINException. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception.

Method Detail

throwIt

```
public static void throwIt(short reason)
```

Throws the JCRE owned instance of `PINException` with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`reason` - the reason for the exception.

Throws:

PINException - always.

---

javacard.framework
Interface Shareable

public abstract interface **Shareable**

The Shareable interface serves to identify all shared objects. Any object that needs to be shared through the applet firewall must directly or indirectly implement this interface. Only those methods specified in a shareable interface are available through the firewall. Implementation classes can implement any number of shareable interfaces and can extend other shareable implementation classes.

javacard.framework
Class SystemException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
                          |
                          +--javacard.framework.SystemException
```

public class **SystemException**
extends CardRuntimeException

`SystemException` represents a `JCSystem` class related exception. It is also thrown by the `javacard.framework.Applet.register()` methods and by the `AID` class constructor.

These API classes throw JCRE owned instances of `SystemException`.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

See Also:
`JCSystem`, `Applet`, `AID`

Field Summary

static short	**ILLEGAL_AID** This reason code is used by the `javacard.framework.Applet.register()` method to indicate that the input AID parameter is not a legal AID value.
static short	**ILLEGAL_TRANSIENT** This reason code is used to indicate that the request to create a transient object is not allowed in the current applet context.
static short	**ILLEGAL_VALUE** This reason code is used to indicate that one or more input parameters is out of allowed bounds.
static short	**NO_RESOURCE** This reason code is used to indicate that there is insufficient resource in the Card for the request.
static short	**NO_TRANSIENT_SPACE** This reason code is used by the `makeTransient..()` methods to indicate that no room is available in volatile memory for the requested object.

Constructor Summary

SystemException(short reason) Constructs a SystemException.	

Method Summary

static void	**throwIt**(short reason) Throws the JCRE owned instance of `SystemException` with the specified reason.

Methods inherited from class javacard.framework.CardRuntimeException
`getReason, setReason`

Methods inherited from class java.lang.Object
equals

Field Detail

ILLEGAL_VALUE

```
public static final short ILLEGAL_VALUE
```

This reason code is used to indicate that one or more input parameters is out of allowed bounds.

NO_TRANSIENT_SPACE

```
public static final short NO_TRANSIENT_SPACE
```

This reason code is used by the `makeTransient..()` methods to indicate that no room is available in volatile memory for the requested object.

ILLEGAL_TRANSIENT

```
public static final short ILLEGAL_TRANSIENT
```

This reason code is used to indicate that the request to create a transient object is not allowed in the current applet context. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

ILLEGAL_AID

```
public static final short ILLEGAL_AID
```

This reason code is used by the `javacard.framework.Applet.register()` method to indicate that the input AID parameter is not a legal AID value.

NO_RESOURCE

```
public static final short NO_RESOURCE
```

This reason code is used to indicate that there is insufficient resource in the Card for the request.

For example, the Java Card Virtual Machine may `throw` this exception reason when there is insufficient heap space to create a new instance.

Constructor Detail

SystemException

```
public SystemException(short reason)
```

Constructs a SystemException. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception.

Method Detail

throwIt

```
public static void throwIt(short reason)
```

Throws the JCRE owned instance of `SystemException` with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`reason` - the reason for the exception.

Throws:

SystemException - always.

---

javacard.framework
Class TransactionException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
                          |
                          +--javacard.framework.TransactionException
```

public class **TransactionException**
extends CardRuntimeException

`TransactionException` represents an exception in the transaction subsystem. The methods referred to in this class are in the `JCSystem` class.

The `JCSystem` class and the transaction facility throw JCRE owned instances of `TransactionException`.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

See Also:
`JCSystem`

Field Summary

static short	**BUFFER_FULL** This reason code is used during a transaction to indicate that the commit buffer is full.
static short	**IN_PROGRESS** This reason code is used by the `beginTransaction` method to indicate a transaction is already in progress.
static short	**INTERNAL_FAILURE** This reason code is used during a transaction to indicate an internal JCRE problem (fatal error).
static short	**NOT_IN_PROGRESS** This reason code is used by the `abortTransaction` and `commintTransaction` methods when a transaction is not in progress.

Constructor Summary

TransactionException(short reason) Constructs a TransactionException with the specified reason.	

Method Summary

static void	**throwIt**(short reason) Throws the JCRE owned instance of `TransactionException` with the specified reason.

Methods inherited from class javacard.framework.CardRuntimeException

`getReason, setReason`

Methods inherited from class java.lang.Object

`equals`

Field Detail

IN_PROGRESS

```
public static final short IN_PROGRESS
```

This reason code is used by the `beginTransaction` method to indicate a transaction is already in progress.

NOT_IN_PROGRESS

```
public static final short NOT_IN_PROGRESS
```

This reason code is used by the `abortTransaction` and `commintTransaction` methods when a transaction is not in progress.

BUFFER_FULL

```
public static final short BUFFER_FULL
```

This reason code is used during a transaction to indicate that the commit buffer is full.

INTERNAL_FAILURE

```
public static final short INTERNAL_FAILURE
```

This reason code is used during a transaction to indicate an internal JCRE problem (fatal error).

Constructor Detail

TransactionException

```
public TransactionException(short reason)
```

Constructs a TransactionException with the specified reason. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Method Detail

throwIt

```
public static void throwIt(short reason)
```

Throws the JCRE owned instance of `TransactionException` with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Throws:

TransactionException - always.

---

javacard.framework
Class UserException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--javacard.framework.CardException
                    |
                    +--javacard.framework.UserException
```

public class **UserException**
extends CardException

`UserException` represents a User exception. This class also provides a resource-saving mechanism (the `throwIt()` method) for user exceptions by using a JCRE owned instance.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Constructor Summary

`UserException``()` Constructs a `UserException` with reason = 0.
`UserException``(short reason)` Constructs a `UserException` with the specified reason.

Method Summary

`static void`	**`throwIt`**`(short reason)` Throws the JCRE owned instance of `UserException` with the specified reason.

Methods inherited from class javacard.framework.CardException
`getReason, setReason`

Methods inherited from class java.lang.Object
equals

Constructor Detail

UserException

```
public UserException()
```

Constructs a `UserException` with reason = 0. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

UserException

```
public UserException(short reason)
```

Constructs a `UserException` with the specified reason. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception.

Method Detail

throwIt

```
public static void throwIt(short reason)
                    throws UserException
```

Throws the JCRE owned instance of `UserException` with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`reason` - the reason for the exception.

Throws:

UserException - always.

javacard.framework
Class Util

```
java.lang.Object
  |
  +--javacard.framework.Util
```

public class **Util**
extends Object

The `Util` class contains common utility functions. Some of the methods may be implemented as native functions for performance reasons. All methods in `Util`, class are static methods.

Some methods of `Util` namely `arrayCopy()`, `arrayCopyNonAtomic()`, `arrayFillNonAtomic()` and `setShort()`, refer to the persistence of array objects. The term *persistent* means that arrays and their values persist from one CAD session to the next, indefinitely. The `JCSystem` class is used to control the persistence and transience of objects.

See Also:
`JCSystem`

Method Summary

static byte	**arrayCompare**(byte[] src, short srcOff, byte[] dest, short destOff, short length) Compares an array from the specified source array, beginning at the specified position, with the specified position of the destination array from left to right.
static short	**arrayCopy**(byte[] src, short srcOff, byte[] dest, short destOff, short length) Copies an array from the specified source array, beginning at the specified position, to the specified position of the destination array.
static short	**arrayCopyNonAtomic**(byte[] src, short srcOff, byte[] dest, short destOff, short length) Copies an array from the specified source array, beginning at the specified position, to the specified position of the destination array (non-atomically).
static short	**arrayFillNonAtomic**(byte[] bArray, short bOff, short bLen, byte bValue) Fills the byte array (non-atomically) beginning at the specified position, for the specified length with the specified byte value.
static short	**getShort**(byte[] bArray, short bOff) Concatenates two bytes in a byte array to form a short value.
static short	**makeShort**(byte b1, byte b2) Concatenates the two parameter bytes to form a short value.
static short	**setShort**(byte[] bArray, short bOff, short sValue) Deposits the short value as two successive bytes at the specified offset in the byte array.

Methods inherited from class java.lang.Object
equals

Method Detail

arrayCopy

```
public static final short arrayCopy(byte[] src,
                                    short srcOff,
                                    byte[] dest,
                                    short destOff,
```

```
                short length)
        throws ArrayIndexOutOfBoundsException,
                NullPointerException,
                TransactionException
```

Copies an array from the specified source array, beginning at the specified position, to the specified position of the destination array.

Notes:

- *If* `srcOff` *or* `destOff` *or* `length` *parameter is negative an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `srcOff+length` *is greater than* `src.length`*, the length of the* `src` *array a* `ArrayIndexOutOfBoundsException` *exception is thrown and no copy is performed.*
- *If* `destOff+length` *is greater than* `dest.length`*, the length of the* `dest` *array an* `ArrayIndexOutOfBoundsException` *exception is thrown and no copy is performed.*
- *If* `src` *or* `dest` *parameter is* `null` *a* `NullPointerException` *exception is thrown.*
- *If the* `src` *and* `dest` *arguments refer to the same array object, then the copying is performed as if the components at positions* `srcOff` *through* `srcOff+length-1` *were first copied to a temporary array with length components and then the contents of the temporary array were copied into positions* `destOff` *through* `destOff+length-1` *of the argument array.*
- *If the destination array is persistent, the entire copy is performed atomically.*
- *The copy operation is subject to atomic commit capacity limitations. If the commit capacity is exceeded, no copy is performed and a* `TransactionException` *exception is thrown.*

Parameters:

`src` - source byte array.
`srcOff` - offset within source byte array to start copy from.
`dest` - destination byte array.
`destOff` - offset within destination byte array to start copy into.
`length` - byte length to be copied.

Returns:

`destOff+length`

Throws:

ArrayIndexOutOfBoundsException - - if copying would cause access of data outside array bounds.
NullPointerException - - if either `src` or `dest` is `null`.
TransactionException - - if copying would cause the commit capacity to be exceeded.

See Also:

`JCSystem.getUnusedCommitCapacity()`

---

arrayCopyNonAtomic

```
public static final short arrayCopyNonAtomic(byte[] src,
                                             short srcOff,
                                             byte[] dest,
```

```
                short destOff,
                short length)
        throws ArrayIndexOutOfBoundsException,
                NullPointerException
```

Copies an array from the specified source array, beginning at the specified position, to the specified position of the destination array (non-atomically).

This method does not use the transaction facility during the copy operation even if a transaction is in progress. Thus, this method is suitable for use only when the contents of the destination array can be left in a partially modified state in the event of a power loss in the middle of the copy operation.

Notes:

- *If* `srcOff` *or* `destOff` *or* `length` *parameter is negative an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `srcOff+length` *is greater than* `src.length`*, the length of the* `src` *array a* `ArrayIndexOutOfBoundsException` *exception is thrown and no copy is performed.*
- *If* `destOff+length` *is greater than* `dest.length`*, the length of the* `dest` *array an* `ArrayIndexOutOfBoundsException` *exception is thrown and no copy is performed.*
- *If* `src` *or* `dest` *parameter is* `null` *a* `NullPointerException` *exception is thrown.*
- *If the* `src` *and* `dest` *arguments refer to the same array object, then the copying is performed as if the components at positions* `srcOff` *through* `srcOff+length-1` *were first copied to a temporary array with length components and then the contents of the temporary array were copied into positions* `destOff` *through* `destOff+length-1` *of the argument array.*
- *If power is lost during the copy operation and the destination array is persistent, a partially changed destination array could result.*
- *The copy* `length` *parameter is not constrained by the atomic commit capacity limitations.*

Parameters:

`src` - source byte array.
`srcOff` - offset within source byte array to start copy from.
`dest` - destination byte array.
`destOff` - offset within destination byte array to start copy into.
`length` - byte length to be copied.

Returns:

`destOff+length`

Throws:

ArrayIndexOutOfBoundsException - - if copying would cause access of data outside array bounds.
NullPointerException - - if either `src` or `dest` is `null`.

See Also:

`JCSystem.getUnusedCommitCapacity()`

---

arrayFillNonAtomic

```
public static final short arrayFillNonAtomic(byte[] bArray,
                                              short bOff,
                                              short bLen,
                                              byte bValue)
                                       throws ArrayIndexOutOfBoundsException,
                                              NullPointerException
```

Fills the byte array (non-atomically) beginning at the specified position, for the specified length with the specified byte value.

This method does not use the transaction facility during the fill operation even if a transaction is in progress. Thus, this method is suitable for use only when the contents of the byte array can be left in a partially filled state in the event of a power loss in the middle of the fill operation.

Notes:

- *If* `bOff` *or* `bLen` *parameter is negative an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `bOff+bLen` *is greater than* `bArray.length`*, the length of the* `bArray` *array an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `bArray` *parameter is* `null` *a* `NullPointerException` *exception is thrown.*
- *If power is lost during the copy operation and the byte array is persistent, a partially changed byte array could result.*
- *The* `bLen` *parameter is not constrained by the atomic commit capacity limitations.*

Parameters:

`bArray` - the byte array.
`bOff` - offset within byte array to start filling bValue into.
`bLen` - byte length to be filled.
`bValue` - the value to fill the byte array with.

Returns:

`bOff+bLen`

Throws:

ArrayIndexOutOfBoundsException - - if the fill operation would cause access of data outside array bounds.
NullPointerException - - if bArray is `null`

See Also:

`JCSystem.getUnusedCommitCapacity()`

---

arrayCompare

```
public static final byte arrayCompare(byte[] src,
                                       short srcOff,
                                       byte[] dest,
                                       short destOff,
                                       short length)
                                throws ArrayIndexOutOfBoundsException,
                                       NullPointerException
```

Compares an array from the specified source array, beginning at the specified position, with the specified position of the destination array from left to right. Returns the ternary result of the comparison : less than(-1), equal(0) or greater than(1).

Notes:

- *If* `srcOff` *or* `destOff` *or* `length` *parameter is negative an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `srcOff+length` *is greater than* `src.length`*, the length of the* `src` *array a* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `destOff+length` *is greater than* `dest.length`*, the length of the* `dest` *array an* `ArrayIndexOutOfBoundsException` *exception is thrown.*
- *If* `src` *or* `dest` *parameter is* `null` *a* `NullPointerException` *exception is thrown.*

Parameters:

`src` - source byte array.
`srcOff` - offset within source byte array to start compare.
`dest` - destination byte array.
`destOff` - offset within destination byte array to start compare.
`length` - byte length to be compared.

Returns:

the result of the comparison as follows:

- `0` if identical
- `-1` if the first miscomparing byte in source array is less than that in destination array,
- `1` if the first miscomparing byte in source array is greater that that in destination array.

Throws:

ArrayIndexOutOfBoundsException - - if comparing all bytes would cause access of data outside array bounds.
NullPointerException - - if either `src` or `dest` is `null`.

---

makeShort

```
public static final short makeShort(byte b1,
                                    byte b2)
```

Concatenates the two parameter bytes to form a short value.

Parameters:

`b1` - the first byte (high order byte).
`b2` - the second byte (low order byte).

Returns:

the short value - the concatenated result

---

getShort

```
public static final short getShort(byte[] bArray,
                                   short bOff)
```

Concatenates two bytes in a byte array to form a short value.

Parameters:

`bArray` - byte array.

`bOff` - offset within byte array containing first byte (the high order byte).

Returns:

the short value - the concatenated result

---

setShort

```
public static final short setShort(byte[] bArray,
                                   short bOff,
                                   short sValue)
                            throws TransactionException
```

Deposits the short value as two successive bytes at the specified offset in the byte array.

Parameters:

`bArray` - byte array.

`bOff` - offset within byte array to deposit the first byte (the high order byte).

`sValue` - the short value to set into array.

Returns:

`bOff+2`

Note:

- *If the byte array is persistent, this operation is performed atomically. If the commit capacity is exceeded, no operation is performed and a* `TransactionException` *exception is thrown.*

Throws:

TransactionException - - if the operation would cause the commit capacity to be exceeded.

See Also:

`JCSystem.getUnusedCommitCapacity()`

Package javacard.security

Provides the classes and interfaces for the Java Card security framework.

See:

Description

Interface Summary	
DESKey	`DESKey` contains an 8/16/24 byte key for single/2 key triple DES/3 key triple DES operations.
DSAKey	The DSAKey interface is the base interface for the DSA algorithms private and public key implementaions.
DSAPrivateKey	The `DSAPrivateKey` interface is used to sign data using the DSA algorithm.
DSAPublicKey	The `DSAPublicKey` interface is used to verify signatures on signed data using the DSA algorithm.
Key	The `Key` interface is the base interface for all keys.
PrivateKey	The `PrivateKey` class is the base class for private keys used in asymmetric algorithms.
PublicKey	The `PublicKey` class is the base class for public keys used in asymmetric algorithms.
RSAPrivateCrtKey	The `RSAPrivateCrtKey` interface is used to sign data using the RSA algorithm in its Chinese Remainder Theorem form.
RSAPrivateKey	The `RSAPrivateKey` class is used to sign data using the RSA algorithm in its modulus/exponent form.
RSAPublicKey	The `RSAPublicKey` is used to verify signatures on signed data using the RSA algorithm.
SecretKey	The `SecretKey` class is the base interface for keys used in symmetric alogrightms (e.g. DES).

Class Summary	
KeyBuilder	The `KeyBuilder` class is a key object factory.
MessageDigest	The `MessageDigest` class is the base class for hashing algorthims.
RandomData	The `RandomData` abstract class is the base class for random number generation.
Signature	The `Signature` class is the base class for Signature algorthims.

Exception Summary	
CryptoException	`CryptoException` represents a cryptography-related exception.

Package javacard.security Description

Provides the classes and interfaces for the Java Card security framework.

javacard.security
Class CryptoException

```
java.lang.Object
  |
  +--java.lang.Throwable
        |
        +--java.lang.Exception
              |
              +--java.lang.RuntimeException
                    |
                    +--javacard.framework.CardRuntimeException
                          |
                          +--javacard.security.CryptoException
```

public class **CryptoException**
extends CardRuntimeException

`CryptoException` represents a cryptography-related exception.

The API classes throw JCRE owned instances of `SystemException`.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components.

See Also:
`KeyBuilder`, `MessageDigest`, `Signature`, `RandomData`, `Cipher`

Field Summary

static short	**ILLEGAL_USE** This reason code is used to indicate that the signature or cipher algorithm does not pad the incoming message and the input message is not block aligned.
static short	**ILLEGAL_VALUE** This reason code is used to indicate that one or more input parameters is out of allowed bounds.
static short	**INVALID_INIT** This reason code is used to indicate that the signature or cipher object has not been correctly initialized for the requested operation.
static short	**NO_SUCH_ALGORITHM** This reason code is used to indicate that the requested algorithm or key type is not supported.
static short	**UNINITIALIZED_KEY** This reason code is used to indicate that the key is uninitialized.

Constructor Summary

CryptoException(short reason) Constructs a CryptoException with the specified reason.

Method Summary

static void	**throwIt**(short reason) Throws the JCRE owned instance of CryptoException with the specified reason.

Methods inherited from class javacard.framework.CardRuntimeException

getReason, setReason

Methods inherited from class java.lang.Object

equals

Field Detail

ILLEGAL_VALUE

```
public static final short ILLEGAL_VALUE
```

This reason code is used to indicate that one or more input parameters is out of allowed bounds.

UNINITIALIZED_KEY

```
public static final short UNINITIALIZED_KEY
```

This reason code is used to indicate that the key is uninitialized.

NO_SUCH_ALGORITHM

```
public static final short NO_SUCH_ALGORITHM
```

This reason code is used to indicate that the requested algorithm or key type is not supported.

INVALID_INIT

```
public static final short INVALID_INIT
```

This reason code is used to indicate that the signature or cipher object has not been correctly initialized for the requested operation.

ILLEGAL_USE

```
public static final short ILLEGAL_USE
```

This reason code is used to indicate that the signature or cipher algorithm does not pad the incoming message and the input message is not block aligned.

Constructor Detail

CryptoException

```
public CryptoException(short reason)
```

Constructs a `CryptoException` with the specified reason. To conserve on resources use `throwIt()` to use the JCRE owned instance of this class.

Parameters:

`reason` - the reason for the exception.

Method Detail

throwIt

```
public static void throwIt(short reason)
```

Throws the JCRE owned instance of `CryptoException` with the specified reason.

JCRE owned instances of exception classes are temporary JCRE Entry Point Objects and can be accessed from any applet context. References to these temporary objects cannot be stored in class variables or instance variables or array components. See *Java Card Runtime Environment (JCRE) 2.1 Specification* for details.

Parameters:

`reason` - the reason for the exception.

Throws:

CryptoException - always.

javacard.security
Interface DESKey

public abstract interface **DESKey**
extends SecretKey

`DESKey` contains an 8/16/24 byte key for single/2 key triple DES/3 key triple DES operations.

When the key data is set, the key is initialized and ready for use.

See Also:
`KeyBuilder`, `Signature`, `Cipher`, `KeyEncryption`

Method Summary

byte	**`getKey`**`(byte[] keyData, short kOff)` Returns the `Key` data in plain text.
void	**`setKey`**`(byte[] keyData, short kOff)` Sets the `Key` data.

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

Method Detail

setKey

```
public void setKey(byte[] keyData,
                   short kOff)
            throws CryptoException
```

Sets the `Key` data. The plaintext length of input key data is 8 bytes for DES, 16 bytes for 2 key triple DES and 24 bytes for 3 key triple DES. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input key data is copied into the internal representation.

Parameters:

`keyData` - byte array containing key initialization data
`kOff` - offset within keyData to start

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input key data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`, `keyData` *is decrypted using the* `Cipher` *object.*

---

getKey

```
public byte getKey(byte[] keyData,
                   short kOff)
```

Returns the `Key` data in plain text. The length of output key data is 8 bytes for DES, 16 bytes for 2 key triple DES and 24 bytes for 3 key triple DES. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`keyData` - byte array to return key data

`kOff` - offset within `keyData` to start.

Returns:

the byte length of the key data returned.

---

javacard.security
Interface DSAKey

All Known Subinterfaces:
DSAPrivateKey, DSAPublicKey

public abstract interface **DSAKey**

The DSAKey interface is the base interface for the DSA algorithms private and public key implementaions. A DSA private key implementation must also implement the `DSAPrivateKey` interface methods. A DSA public key implementation must also implement the `DSAPublicKey` interface methods.

When all four components of the key (X or Y,P,Q,G) are set, the key is initialized and ready for use.

See Also:
`DSAPublicKey`, `DSAPrivateKey`, `KeyBuilder`, `Signature`, `KeyEncryption`

Method Summary

`short`	**`getG`**`(byte[] buffer, short offset)` Returns the subprime parameter value of the key in plain text.
`short`	**`getP`**`(byte[] buffer, short offset)` Returns the base parameter value of the key in plain text.
`short`	**`getQ`**`(byte[] buffer, short offset)` Returns the prime parameter value of the key in plain text.
`void`	**`setG`**`(byte[] buffer, short offset, short length)` Sets the subprime parameter value of the key.
`void`	**`setP`**`(byte[] buffer, short offset, short length)` Sets the base parameter value of the key.
`void`	**`setQ`**`(byte[] buffer, short offset, short length)` Sets the prime parameter value of the key.

Method Detail

setP

```
public void setP(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the base parameter value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input base parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer
`offset` - the offset into the input buffer at which the base parameter value begins
`length` - the length of the base parameter value

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the base parameter value is decrypted using the* `Cipher` *object.*

---

setQ

```
public void setQ(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the prime parameter value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input prime parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer
`offset` - the offset into the input buffer at which the prime parameter value begins
`length` - the length of the prime parameter value

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the prime parameter value is decrypted using the* `Cipher` *object.*

setG

```
public void setG(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the subprime parameter value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input subprime parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the subprime parameter value begins

`length` - the length of the subprime parameter value

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the subprime parameter value is decrypted using the* `Cipher` *object.*

---

getP

```
public short getP(byte[] buffer,
                  short offset)
```

Returns the base parameter value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the base parameter value starts

Returns:

the byte length of the base parameter value returned

---

getQ

```
public short getQ(byte[] buffer,
                  short offset)
```

Returns the prime parameter value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the prime parameter value begins

Returns:

the byte length of the prime parameter value returned

---

getG

```
public short getG(byte[] buffer,
                  short offset)
```

Returns the subprime parameter value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the subprime parameter value begins

Returns:

the byte length of the subprime parameter value returned

---

javacard.security
Interface DSAPrivateKey

public abstract interface **DSAPrivateKey**
extends PrivateKey, DSAKey

The `DSAPrivateKey` interface is used to sign data using the DSA algorithm. An implementation of `DSAPrivateKey` interface must also implement the `DSAKey` interface methods.

When all four components of the key (X,P,Q,G) are set, the key is initialized and ready for use.

See Also:
`DSAPublicKey`, `KeyBuilder`, `Signature`, `KeyEncryption`

Method Summary

short	**`getX`**`(byte[] buffer, short offset)` Returns the value of the key in plain text.
void	**`setX`**`(byte[] buffer, short offset, short length)` Sets the value of the key.

Methods inherited from interface javacard.security.DSAKey
`getG, getP, getQ, setG, setP, setQ`

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

Method Detail

setX

```
public void setX(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the value of the key. When the base, prime and subprime parameters are intialized and the key value is set, the key is ready for use. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input key data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the modulus value begins

`length` - the length of the modulus

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input key data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the key value is decrypted using the* `Cipher` *object.*

---

getX

```
public short getX(byte[] buffer,
                  short offset)
```

Returns the value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the key value starts

Returns:

the byte length of the key value returned

---

javacard.security
Interface DSAPublicKey

public abstract interface **DSAPublicKey**
extends PublicKey, DSAKey

The `DSAPublicKey` interface is used to verify signatures on signed data using the DSA algorithm. An implementation of `DSAPublicKey` interface must also implement the `DSAKey` interface methods.

When all four components of the key (Y,P,Q,G) are set, the key is initialized and ready for use.

See Also:
`DSAPrivateKey`, `KeyBuilder`, `Signature`, `KeyEncryption`

Method Summary

short	**getY**(byte[] buffer, short offset) Returns the value of the key in plain text.
void	**setY**(byte[] buffer, short offset, short length) Sets the value of the key.

Methods inherited from interface javacard.security.DSAKey
getG, getP, getQ, setG, setP, setQ

Methods inherited from interface javacard.security.Key
clearKey, getSize, getType, isInitialized

Method Detail

setY

```
public void setY(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the value of the key. When the base, prime and subprime parameters are intialized and the key value is set, the key is ready for use. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input key data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the key value begins

`length` - the length of the key value

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input key data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the key value is decrypted using the* `Cipher` *object.*

---

getY

```
public short getY(byte[] buffer,
                  short offset)
```

Returns the value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the input buffer at which the key value starts

Returns:

the byte length of the key value returned

---

javacard.security
Interface Key

All Known Subinterfaces:
DESKey, DSAPrivateKey, DSAPublicKey, PrivateKey, PublicKey, RSAPrivateCrtKey, RSAPrivateKey, RSAPublicKey, SecretKey

public abstract interface **Key**

The `Key` interface is the base interface for all keys.

See Also:
`KeyBuilder`

Method Summary

`void`	**`clearKey`**`()` Clears the key and sets its initialized state to false.
`short`	**`getSize`**`()` Returns the key size in number of bits.
`byte`	**`getType`**`()` Returns the key interface type.
`boolean`	**`isInitialized`**`()` Reports the initialized state of the key.

Method Detail

isInitialized

`public boolean` **`isInitialized`**`()`

Reports the initialized state of the key. Keys must be initialized before being used.

A `Key` object sets its initialized state to true only when all the associated set methods have been invoked at least once since the time the initialized state was set to false.

A newly created `Key` object sets its initialized state to false. Invocation of the `clearKey()` method sets the initialized state to false. A key with transient key data sets its initialized state to false on the associated clear events.

Returns:
`true` if the key has been initialized.

---

clearKey

```
public void clearKey()
```

Clears the key and sets its initialized state to false.

---

getType

```
public byte getType()
```

Returns the key interface type.
Returns:
the key interface type.

See Also:
`KeyBuilder`

---

getSize

```
public short getSize()
```

Returns the key size in number of bits.
Returns:
the key size in number of bits.

---

javacard.security
Class KeyBuilder

```
java.lang.Object
  |
  +--javacard.security.KeyBuilder
```

public class **KeyBuilder**
extends Object

The `KeyBuilder` class is a key object factory.

Field Summary

`static short`	**`LENGTH_DES`** DES Key Length `LENGTH_DES` = 64.
`static short`	**`LENGTH_DES3_2KEY`** DES Key Length `LENGTH_DES3_2KEY` = 128.
`static short`	**`LENGTH_DES3_3KEY`** DES Key Length `LENGTH_DES3_3KEY` = 192.
`static short`	**`LENGTH_DSA_1024`** DSA Key Length `LENGTH_DSA_1024` = 1024.
`static short`	**`LENGTH_DSA_512`** DSA Key Length `LENGTH_DSA_512` = 512.
`static short`	**`LENGTH_DSA_768`** DSA Key Length `LENGTH_DSA_768` = 768.
`static short`	**`LENGTH_RSA_1024`** RSA Key Length `LENGTH_RSA_1024` = 1024.
`static short`	**`LENGTH_RSA_2048`** RSA Key Length `LENGTH_RSA_2048` = 2048.
`static short`	**`LENGTH_RSA_512`** RSA Key Length `LENGTH_RSA_512` = 512.
`static short`	**`LENGTH_RSA_768`** RSA Key Length `LENGTH_RSA_768` = 768.
`static byte`	**`TYPE_DES`** `Key` object which implements interface type `DESKey` with persistent key data.

static byte	**TYPE_DES_TRANSIENT_DESELECT** Key object which implements interface type DESKey with CLEAR_ON_DESELECT transient key data.
static byte	**TYPE_DES_TRANSIENT_RESET** Key object which implements interface type DESKey with CLEAR_ON_RESET transient key data.
static byte	**TYPE_DSA_PRIVATE** Key object which implements the interface type DSAPrivateKey for the DSA algorithm.
static byte	**TYPE_DSA_PUBLIC** Key object which implements the interface type DSAPublicKey for the DSA algorithm.
static byte	**TYPE_RSA_CRT_PRIVATE** Key object which implements interface type RSAPrivateCrtKey which uses Chinese Remainder Theorem.
static byte	**TYPE_RSA_PRIVATE** Key object which implements interface type RSAPrivateKey which uses modulus/exponent form.
static byte	**TYPE_RSA_PUBLIC** Key object which implements interface type RSAPublicKey.

Method Summary

static Key	**buildKey**(byte keyType, short keyLength, boolean keyEncryption) Creates cryptographic keys for signature and cipher algorithms.

Methods inherited from class java.lang.Object
equals

Field Detail

TYPE_DES_TRANSIENT_RESET

`public static final byte` **`TYPE_DES_TRANSIENT_RESET`**

`Key` object which implements interface type `DESKey` with CLEAR_ON_RESET transient key data.

This `Key` object implicitly performs a `clearKey()` on power on or card reset.

TYPE_DES_TRANSIENT_DESELECT

`public static final byte` **`TYPE_DES_TRANSIENT_DESELECT`**

`Key` object which implements interface type `DESKey` with CLEAR_ON_DESELECT transient key data.

This `Key` object implicitly performs a `clearKey()` on power on, card reset and applet deselection.

TYPE_DES

`public static final byte` **`TYPE_DES`**

`Key` object which implements interface type `DESKey` with persistent key data.

TYPE_RSA_PUBLIC

`public static final byte` **`TYPE_RSA_PUBLIC`**

`Key` object which implements interface type `RSAPublicKey`.

TYPE_RSA_PRIVATE

`public static final byte` **`TYPE_RSA_PRIVATE`**

`Key` object which implements interface type `RSAPrivateKey` which uses modulus/exponent form.

TYPE_RSA_CRT_PRIVATE

`public static final byte` **`TYPE_RSA_CRT_PRIVATE`**

`Key` object which implements interface type `RSAPrivateCrtKey` which uses Chinese Remainder Theorem.

TYPE_DSA_PUBLIC

`public static final byte` **`TYPE_DSA_PUBLIC`**

`Key` object which implements the interface type `DSAPublicKey` for the DSA algorithm.

TYPE_DSA_PRIVATE

`public static final byte` **`TYPE_DSA_PRIVATE`**

`Key` object which implements the interface type `DSAPrivateKey` for the DSA algorithm.

LENGTH_DES

`public static final short` **`LENGTH_DES`**

DES Key Length `LENGTH_DES` = 64.

LENGTH_DES3_2KEY

`public static final short` **`LENGTH_DES3_2KEY`**

DES Key Length `LENGTH_DES3_2KEY` = 128.

LENGTH_DES3_3KEY

`public static final short` **`LENGTH_DES3_3KEY`**

DES Key Length `LENGTH_DES3_3KEY` = 192.

LENGTH_RSA_512

`public static final short` **`LENGTH_RSA_512`**

RSA Key Length `LENGTH_RSA_512` = 512.

LENGTH_RSA_768

`public static final short` **`LENGTH_RSA_768`**

RSA Key Length `LENGTH_RSA_768` = 768.

LENGTH_RSA_1024

```
public static final short LENGTH_RSA_1024
```

RSA Key Length `LENGTH_RSA_1024` = 1024.

LENGTH_RSA_2048

```
public static final short LENGTH_RSA_2048
```

RSA Key Length `LENGTH_RSA_2048` = 2048.

LENGTH_DSA_512

```
public static final short LENGTH_DSA_512
```

DSA Key Length `LENGTH_DSA_512` = 512.

LENGTH_DSA_768

```
public static final short LENGTH_DSA_768
```

DSA Key Length `LENGTH_DSA_768` = 768.

LENGTH_DSA_1024

```
public static final short LENGTH_DSA_1024
```

DSA Key Length `LENGTH_DSA_1024` = 1024.

Method Detail

buildKey

```
public static Key buildKey(byte keyType,
                           short keyLength,
                           boolean keyEncryption)
                    throws CryptoException
```

Creates cryptographic keys for signature and cipher algorithms. Instances created by this method may be the only key objects used to initialize instances of `Signature` and `Cipher`. Note that the object returned must be cast to their appropriate key type interface.

Parameters:

`keyType` - the type of key to be generated. Valid codes listed in TYPE.. constants.
`keyLength` - the key size in bits. The valid key bit lengths are key type dependent. See above.
`keyEncryption` - if `true` this boolean requests a key implementation which implements the `javacardx.cipher.KeyEncryption` interface.

Returns:

the key object instance of the requested key type, length and encrypted access.

Throws:

CryptoException - with the following reason codes:

- `CryptoException.NO_SUCH_ALGORITHM` if the requested algorithm associated with the specified type, size of key and key encryption interface is not supported.

---

javacard.security
Class MessageDigest

```
java.lang.Object
  |
  +--javacard.security.MessageDigest
```

public abstract class **MessageDigest**
extends Object

The `MessageDigest` class is the base class for hashing algorthims. Implementations of MessageDigest algorithms must extend this class and implement all the abstract methods.

Field Summary

static byte	**ALG_MD5** Message Digest algorithm MD5.
static byte	**ALG_RIPEMD160** Message Digest algorithm RIPE MD-160.
static byte	**ALG_SHA** Message Digest algorithm SHA.

Constructor Summary

protected	**MessageDigest**() Protected Constructor

Method Summary

abstract short	**doFinal**(byte[] inBuff, short inOffset, short inLength, byte[] outBuff, short outOffset) Generates a hash of all/last input data.
abstract byte	**getAlgorithm**() Gets the Message digest algorithm.
static MessageDigest	**getInstance**(byte algorithm, boolean externalAccess) Creates a MessageDigest object instance of the selected algorithm.
abstract byte	**getLength**() Returns the byte length of the hash.
abstract void	**reset**() Resets the MessageDigest object to the initial state for further use.
abstract void	**update**(byte[] inBuff, short inOffset, short inLength) Accumulates a hash of the input data.

Methods inherited from class java.lang.Object
equals

Field Detail

ALG_SHA

```
public static final byte ALG_SHA
```

Message Digest algorithm SHA.

ALG_MD5

```
public static final byte ALG_MD5
```

Message Digest algorithm MD5.

ALG_RIPEMD160

```
public static final byte ALG_RIPEMD160
```

Message Digest algorithm RIPE MD-160.

Constructor Detail

MessageDigest

```
protected MessageDigest()
```

Protected Constructor

Method Detail

getInstance

```
public static final MessageDigest getInstance(byte algorithm,
                                               boolean externalAccess)
                                        throws CryptoException
```

Creates a `MessageDigest` object instance of the selected algorithm.

Parameters:

`algorithm` - the desired message digest algorithm. Valid codes listed in ALG_.. constants. See above.

`externalAccess` - if `true` indicates that the instance will be shared among multiple applet instances and that the `MessageDigest` instance will also be accessed (via a `Shareable` interface) when the owner of the `MessageDigest` instance is not the currently selected applet.

Returns:

the `MessageDigest` object instance of the requested algorithm.

Throws:

CryptoException - with the following reason codes:

- `CryptoException.NO_SUCH_ALGORITHM` if the requested algorithm is not supported.

---

getAlgorithm

```
public abstract byte getAlgorithm()
```

Gets the Message digest algorithm.

Returns:

the algorithm code defined above.

---

getLength

```
public abstract byte getLength()
```

Returns the byte length of the hash.

Returns:

hash length

---

doFinal

```
public abstract short doFinal(byte[] inBuff,
                              short inOffset,
                              short inLength,
                              byte[] outBuff,
                              short outOffset)
```

Generates a hash of all/last input data. Completes and returns the hash computation after performing final operations such as padding. The `MessageDigest` object is reset after this call is made.

The input and output buffer data may overlap.

Parameters:

`inBuff` - the input buffer of data to be hashed
`inOffset` - the offset into the input buffer at which to begin hash generation
`inLength` - the byte length to hash
`outBuff` - the output buffer, may be the same as the input buffer
`outOffset` - the offset into the output buffer where the resulting hash value begins

Returns:

number of bytes of hash output in `outBuff`

---

update

```
public abstract void update(byte[] inBuff,
                            short inOffset,
                            short inLength)
```

Accumulates a hash of the input data. When this method is used temporary storage of intermediate results is required. This method should only be used if all the input data required for the hash is not available in one byte array. The doFinal() method is recommended whenever possible.

Parameters:

`inBuff` - the input buffer of data to be hashed
`inOffset` - the offset into the input buffer at which to begin hash generation
`inLength` - the byte length to hash

See Also:

`doFinal(byte[], short, short, byte[], short)`

reset

```
public abstract void reset()
```

Resets the `MessageDigest` object to the initial state for further use.

javacard.security
Interface PrivateKey

All Known Subinterfaces:
DSAPrivateKey, RSAPrivateCrtKey, RSAPrivateKey

public abstract interface **PrivateKey**
extends Key

The `PrivateKey` class is the base class for private keys used in asymmetric algorithms.

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

javacard.security
Interface PublicKey

All Known Subinterfaces:
DSAPublicKey, RSAPublicKey

public abstract interface **PublicKey**
extends Key

The `PublicKey` class is the base class for public keys used in asymmetric algorithms.

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

javacard.security
Interface RSAPrivateCrtKey

public abstract interface **RSAPrivateCrtKey**
extends PrivateKey

The `RSAPrivateCrtKey` interface is used to sign data using the RSA algorithm in its Chinese Remainder Theorem form. It may also be used by the `javacardx.crypto.Cipher` class to encrypt/decrypt messages.

Let $S = m^d \bmod n$, where m is the data to be signed, d is the private key exponent, and n is private key modulus composed of two prime numbers p and q. The following names are used in the initializer methods in this interface:

P, the prime factor p
Q, the prime factor q.
PQ = $q^{-1} \bmod p$
DP1 = $d \bmod (p - 1)$
DQ1 = $d \bmod (q - 1)$

When all five components (P,Q,PQ,DP1,DQ1) of the key are set, the key is initialized and ready for use.

See Also:
`RSAPrivateKey`, `RSAPublicKey`, `KeyBuilder`, `Signature`, `Cipher`, `KeyEncryption`

Method Summary

short	**getDP1**(byte[] buffer, short offset) Returns the value of the DP1 parameter in plain text.
short	**getDQ1**(byte[] buffer, short offset) Returns the value of the DQ1 parameter in plain text.
short	**getP**(byte[] buffer, short offset) Returns the value of the P parameter in plain text.
short	**getPQ**(byte[] buffer, short offset) Returns the value of the PQ parameter in plain text.
short	**getQ**(byte[] buffer, short offset) Returns the value of the Q parameter in plain text.
void	**setDP1**(byte[] buffer, short offset, short length) Sets the value of the DP1 parameter.
void	**setDQ1**(byte[] buffer, short offset, short length) Sets the value of the DQ1 parameter.
void	**setP**(byte[] buffer, short offset, short length) Sets the value of the P parameter.
void	**setPQ**(byte[] buffer, short offset, short length) Sets the value of the PQ parameter.
void	**setQ**(byte[] buffer, short offset, short length) Sets the value of the Q parameter.

Methods inherited from interface javacard.security.Key
clearKey, getSize, getType, isInitialized

Method Detail

setP

```
public void setP(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the value of the P parameter. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input P parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer
`offset` - the offset into the input buffer at which the parameter value begins
`length` - the length of the parameter

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the P parameter value is decrypted using the* `Cipher` *object.*

---

setQ

```
public void setQ(byte[] buffer,
                 short offset,
                 short length)
          throws CryptoException
```

Sets the value of the Q parameter. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input Q parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer
`offset` - the offset into the input buffer at which the parameter value begins
`length` - the length of the parameter

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the Q parameter value is decrypted using the* `Cipher` *object.*

---

setDP1

```
public void setDP1(byte[] buffer,
                   short offset,
                   short length)
            throws CryptoException
```

Sets the value of the DP1 parameter. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input DP1 parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the parameter value begins

`length` - the length of the parameter

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the DP1 parameter value is decrypted using the* `Cipher` *object.*

---

setDQ1

```
public void setDQ1(byte[] buffer,
                   short offset,
                   short length)
            throws CryptoException
```

Sets the value of the DQ1 parameter. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input DQ1 parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the parameter value begins

`length` - the length of the parameter

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the DQ1 parameter value is decrypted using the* `Cipher` *object.*

setPQ

```
public void setPQ(byte[] buffer,
                  short offset,
                  short length)
           throws CryptoException
```

Sets the value of the PQ parameter. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input PQ parameter data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the parameter value begins

`length` - the length of the parameter

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input parameter data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the PQ parameter value is decrypted using the* `Cipher` *object.*

---

getP

```
public short getP(byte[] buffer,
                  short offset)
```

Returns the value of the P parameter in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the parameter value begins

Returns:

the byte length of the P parameter value returned

---

getQ

```
public short getQ(byte[] buffer,
                  short offset)
```

Returns the value of the Q parameter in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the parameter value begins

Returns:

the byte length of the Q parameter value returned

---

getDP1

```
public short getDP1(byte[] buffer,
                    short offset)
```

Returns the value of the DP1 parameter in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the parameter value begins

Returns:

the byte length of the DP1 parameter value returned

---

getDQ1

```
public short getDQ1(byte[] buffer,
                    short offset)
```

Returns the value of the DQ1 parameter in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the parameter value begins

Returns:

the byte length of the DQ1 parameter value returned

---

getPQ

```
public short getPQ(byte[] buffer,
                   short offset)
```

Returns the value of the PQ parameter in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the parameter value begins

Returns:

the byte length of the PQ parameter value returned

javacard.security
Interface RSAPrivateKey

public abstract interface **RSAPrivateKey**
extends PrivateKey

The `RSAPrivateKey` class is used to sign data using the RSA algorithm in its modulus/exponent form. It may also be used by the `javacardx.crypto.Cipher` class to encrypt/decrypt messages.

When both the modulus and exponent of the key are set, the key is initialized and ready for use.

See Also:
`RSAPublicKey`, `RSAPrivateCrtKey`, `KeyBuilder`, `Signature`, `Cipher`, `KeyEncryption`

Method Summary

`short`	**`getExponent`**`(byte[] buffer, short offset)` Returns the private exponent value of the key in plain text.
`short`	**`getModulus`**`(byte[] buffer, short offset)` Returns the modulus value of the key in plain text.
`void`	**`setExponent`**`(byte[] buffer, short offset, short length)` Sets the private exponent value of the key.
`void`	**`setModulus`**`(byte[] buffer, short offset, short length)` Sets the modulus value of the key.

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

Method Detail

setModulus

```
public void setModulus(byte[] buffer,
                       short offset,
                       short length)
                throws CryptoException
```

Sets the modulus value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input modulus data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the modulus value begins

`length` - the length of the modulus

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input modulus data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the modulus value is decrypted using the* `Cipher` *object.*

---

setExponent

```
public void setExponent(byte[] buffer,
                        short offset,
                        short length)
                 throws CryptoException
```

Sets the private exponent value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input exponent data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the exponent value begins

`length` - the length of the exponent

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input exponent data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the exponent value is decrypted using the* `Cipher` *object.*

getModulus

```
public short getModulus(byte[] buffer,
                        short offset)
```

Returns the modulus value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the modulus value starts

Returns:

the byte length of the modulus value returned

getExponent

```
public short getExponent(byte[] buffer,
                         short offset)
```

Returns the private exponent value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the exponent value begins

Returns:

the byte length of the private exponent value returned

javacard.security
Interface RSAPublicKey

public abstract interface **RSAPublicKey**
extends PublicKey

The `RSAPublicKey` is used to verify signatures on signed data using the RSA algorithm. It may also used by the `javacardx.crypto.Cipher` class to encrypt/decrypt messages.

When both the modulus and exponent of the key are set, the key is initialized and ready for use.

See Also:
`RSAPrivateKey`, `RSAPrivateCrtKey`, `KeyBuilder`, `Signature`, `Cipher`, `KeyEncryption`

Method Summary

`short`	**`getExponent`**`(byte[] buffer, short offset)` Returns the private exponent value of the key in plain text.
`short`	**`getModulus`**`(byte[] buffer, short offset)` Returns the modulus value of the key in plain text.
`void`	**`setExponent`**`(byte[] buffer, short offset, short length)` Sets the public exponent value of the key.
`void`	**`setModulus`**`(byte[] buffer, short offset, short length)` Sets the modulus value of the key.

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

Method Detail

setModulus

```
public void setModulus(byte[] buffer,
                       short offset,
                       short length)
                throws CryptoException
```

Sets the modulus value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input modulus data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the modulus value begins

`length` - the byte length of the modulus

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input modulus data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the modulus value is decrypted using the* `Cipher` *object.*

---

setExponent

```
public void setExponent(byte[] buffer,
                        short offset,
                        short length)
                 throws CryptoException
```

Sets the public exponent value of the key. The plaintext data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte). Input exponent data is copied into the internal representation.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer at which the exponent value begins

`length` - the byte length of the exponent

Throws:

CryptoException - with the following reason code:

- `CryptoException.ILLEGAL_VALUE` if the input exponent data length is inconsistent with the implementation or if input data decryption is required and fails.

Note:

- *If the key object implements the* `javacardx.crypto.KeyEncryption` *interface and the* `Cipher` *object specified via* `setKeyCipher()` *is not* `null`*, the exponent value is decrypted using the* `Cipher` *object.*

getModulus

```
public short getModulus(byte[] buffer,
                        short offset)
```

Returns the modulus value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the input buffer at which the modulus value starts

Returns:

the byte length of the modulus value returned

getExponent

```
public short getExponent(byte[] buffer,
                         short offset)
```

Returns the private exponent value of the key in plain text. The data format is big-endian and right-aligned (the least significant bit is the least significant bit of last byte).

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer at which the exponent value begins

Returns:

the byte length of the public exponent returned

javacard.security
Class RandomData

```
java.lang.Object
  |
  +--javacard.security.RandomData
```

public abstract class **RandomData**
extends Object

The `RandomData` abstract class is the base class for random number generation. Implementations of RandomData algorithms must extend this class and implement all the abstract methods.

Field Summary

static byte	**ALG_PSEUDO_RANDOM** Utility pseudo random number generation algorithms.
static byte	**ALG_SECURE_RANDOM** Cryptographically secure random number generation algorithms.

Constructor Summary

protected	**RandomData**() Protected constructor for subclassing.

Method Summary

abstract void	**generateData**(byte[] buffer, short offset, short length) Generates random data.
static RandomData	**getInstance**(byte algorithm) Creates a `RandomData` instance of the selected algorithm.
abstract void	**setSeed**(byte[] buffer, short offset, short length) Seeds the random data generator.

Methods inherited from class java.lang.Object
equals

Field Detail

ALG_PSEUDO_RANDOM

```
public static final byte ALG_PSEUDO_RANDOM
```

Utility pseudo random number generation algorithms.

ALG_SECURE_RANDOM

```
public static final byte ALG_SECURE_RANDOM
```

Cryptographically secure random number generation algorithms.

Constructor Detail

RandomData

```
protected RandomData()
```

Protected constructor for subclassing.

Method Detail

getInstance

```
public static final RandomData getInstance(byte algorithm)
                                    throws CryptoException
```

Creates a `RandomData` instance of the selected algorithm. The pseudo random `RandomData` instance's seed is initialized to a internal default value.

Parameters:

`algorithm` - the desired random number algorithm. Valid codes listed in ALG_.. constants. See above.

Returns:

the `RandomData` object instance of the requested algorithm.

Throws:

CryptoException - with the following reason codes:

- `CryptoException.NO_SUCH_ALGORITHM` if the requested algorithm is not supported.

---

generateData

```
public abstract void generateData(byte[] buffer,
                                  short offset,
                                  short length)
```

Generates random data.

Parameters:

`buffer` - the output buffer

`offset` - the offset into the output buffer

`length` - the length of random data to generate

---

setSeed

```
public abstract void setSeed(byte[] buffer,
                             short offset,
                             short length)
```

Seeds the random data generator.

Parameters:

`buffer` - the input buffer

`offset` - the offset into the input buffer

`length` - the length of the seed data

---

javacard.security
Interface SecretKey

All Known Subinterfaces:
DESKey

public abstract interface **SecretKey**
extends Key

The `SecretKey` class is the base interface for keys used in symmetric alogrightms (e.g. DES).

Methods inherited from interface javacard.security.Key
`clearKey, getSize, getType, isInitialized`

javacard.security
Class Signature

```
java.lang.Object
  |
  +--javacard.security.Signature
```

public abstract class **Signature**
extends Object

The `Signature` class is the base class for Signature algorthims. Implementations of Signature algorithms must extend this class and implement all the abstract methods.

The term "pad" is used in the public key signature algorithms below to refer to all the operations specified in the referenced scheme to transform the message digest into the encryption block size.

Field Summary

`static byte`	**`ALG_DES_MAC4_ISO9797_M1`** Signature algorithm `ALG_DES_MAC4_ISO9797_M1` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.
`static byte`	**`ALG_DES_MAC4_ISO9797_M2`** Signature algorithm `ALG_DES_MAC4_ISO9797_M2` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.
`static byte`	**`ALG_DES_MAC4_NOPAD`** Signature algorithm `ALG_DES_MAC4_NOPAD` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data.
`static byte`	**`ALG_DES_MAC4_PKCS5`** Signature algorithm `ALG_DES_MAC4_PKCS5` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.
`static byte`	**`ALG_DES_MAC8_ISO9797_M1`** Signature algorithm `ALG_DES_MAC8_ISO9797_M1` generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.

static byte	**ALG_DES_MAC8_ISO9797_M2** Signature algorithm ALG_DES_MAC8_ISO9797_M2 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.
static byte	**ALG_DES_MAC8_NOPAD** Signature algorithm ALG_DES_MAC_8_NOPAD generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data.
static byte	**ALG_DES_MAC8_PKCS5** Signature algorithm ALG_DES_MAC8_PKCS5 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.
static byte	**ALG_DSA_SHA** Signature algorithm ALG_DSA_SHA signs/verifies the 20 byte SHA digest using DSA.
static byte	**ALG_RSA_MD5_PKCS1** Signature algorithm ALG_RSA_MD5_PKCS1 encrypts the 16 byte MD5 digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.
static byte	**ALG_RSA_MD5_RFC2409** Signature algorithm ALG_RSA_MD5_RFC2409 encrypts the 16 byte MD5 digest using RSA. The digest is padded according to the RFC2409 scheme.
static byte	**ALG_RSA_RIPEMD160_ISO9796** Signature algorithm ALG_RSA_RIPEMD160_ISO9796 encrypts the 20 byte RIPE MD-160 digest using RSA. The digest is padded according to the ISO 9796 scheme.
static byte	**ALG_RSA_RIPEMD160_PKCS1** Signature algorithm ALG_RSA_RIPEMD160_PKCS1 encrypts the 20 byte RIPE MD-160 digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.
static byte	**ALG_RSA_SHA_ISO9796** Signature algorithm ALG_RSA_SHA_ISO9796 encrypts the 20 byte SHA digest using RSA. The digest is padded according to the ISO 9796 (EMV'96) scheme.
static byte	**ALG_RSA_SHA_PKCS1** Signature algorithm ALG_RSA_SHA_PKCS1 encrypts the 20 byte SHA digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.
static byte	**ALG_RSA_SHA_RFC2409** Signature algorithm ALG_RSA_SHA_RFC2409 encrypts the 20 byte SHA digest using RSA. The digest is padded according to the RFC2409 scheme.
static byte	**MODE_SIGN** Used in init() methods to indicate signature sign mode.

static byte	**MODE_VERIFY** Used in `init()` methods to indicate signature verify mode.

Constructor Summary

protected	**Signature**() Protected Constructor

Method Summary

abstract byte	**getAlgorithm**() Gets the Signature algorithm.
static Signature	**getInstance**(byte algorithm, boolean externalAccess) Creates a `Signature` object instance of the selected algorithm.
abstract short	**getLength**() Returns the byte length of the signature data.
abstract void	**init**(Key theKey, byte theMode) Initializes the `Signature` object with the appropriate `Key`.
abstract void	**init**(Key theKey, byte theMode, byte[] bArray, short bOff, short bLen) Initializes the `Signature` object with the appropriate `Key` and algorithm specific parameters.
abstract short	**sign**(byte[] inBuff, short inOffset, short inLength, byte[] sigBuff, short sigOffset) Generates the signature of all/last input data.
abstract void	**update**(byte[] inBuff, short inOffset, short inLength) Accumulates a signature of the input data.
abstract boolean	**verify**(byte[] inBuff, short inOffset, short inLength, byte[] sigBuff, short sigOffset, short sigLength) Verifies the signature of all/last input data against the passed in signature.

Methods inherited from class java.lang.Object
equals

Field Detail

ALG_DES_MAC4_NOPAD

```
public static final byte ALG_DES_MAC4_NOPAD
```

Signature algorithm `ALG_DES_MAC4_NOPAD` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data. If the input data is not (8 byte) block aligned it throws `CryptoExeption` with the reason code `ILLEGAL_USE`.

ALG_DES_MAC8_NOPAD

```
public static final byte ALG_DES_MAC8_NOPAD
```

Signature algorithm `ALG_DES_MAC_8_NOPAD` generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data. If the input data is not (8 byte) block aligned it throws `CryptoExeption` with the reason code `ILLEGAL_USE`.

Note:

- *This algorithm must not be implemented if export restrictions apply.*

ALG_DES_MAC4_ISO9797_M1

```
public static final byte ALG_DES_MAC4_ISO9797_M1
```

Signature algorithm `ALG_DES_MAC4_ISO9797_M1` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.

ALG_DES_MAC8_ISO9797_M1

```
public static final byte ALG_DES_MAC8_ISO9797_M1
```

Signature algorithm `ALG_DES_MAC8_ISO9797_M1` generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.

Note:

- *This algorithm must not be implemented if export restrictions apply.*

ALG_DES_MAC4_ISO9797_M2

```
public static final byte ALG_DES_MAC4_ISO9797_M2
```

Signature algorithm ALG_DES_MAC4_ISO9797_M2 generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

ALG_DES_MAC8_ISO9797_M2

```
public static final byte ALG_DES_MAC8_ISO9797_M2
```

Signature algorithm ALG_DES_MAC8_ISO9797_M2 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

Note:

- *This algorithm must not be implemented if export restrictions apply.*

ALG_DES_MAC4_PKCS5

```
public static final byte ALG_DES_MAC4_PKCS5
```

Signature algorithm ALG_DES_MAC4_PKCS5 generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.

ALG_DES_MAC8_PKCS5

```
public static final byte ALG_DES_MAC8_PKCS5
```

Signature algorithm ALG_DES_MAC8_PKCS5 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.

Note:

- *This algorithm must not be implemented if export restrictions apply.*

ALG_RSA_SHA_ISO9796

```
public static final byte ALG_RSA_SHA_ISO9796
```

Signature algorithm ALG_RSA_SHA_ISO9796 encrypts the 20 byte SHA digest using RSA. The digest is padded according to the ISO 9796 (EMV'96) scheme.

ALG_RSA_SHA_PKCS1

```
public static final byte ALG_RSA_SHA_PKCS1
```

Signature algorithm ALG_RSA_SHA_PKCS1 encrypts the 20 byte SHA digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.

ALG_RSA_MD5_PKCS1

```
public static final byte ALG_RSA_MD5_PKCS1
```

Signature algorithm ALG_RSA_MD5_PKCS1 encrypts the 16 byte MD5 digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.

ALG_RSA_RIPEMD160_ISO9796

```
public static final byte ALG_RSA_RIPEMD160_ISO9796
```

Signature algorithm ALG_RSA_RIPEMD160_ISO9796 encrypts the 20 byte RIPE MD-160 digest using RSA. The digest is padded according to the ISO 9796 scheme.

ALG_RSA_RIPEMD160_PKCS1

```
public static final byte ALG_RSA_RIPEMD160_PKCS1
```

Signature algorithm ALG_RSA_RIPEMD160_PKCS1 encrypts the 20 byte RIPE MD-160 digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.

ALG_DSA_SHA

```
public static final byte ALG_DSA_SHA
```

Signature algorithm ALG_DSA_SHA signs/verifies the 20 byte SHA digest using DSA.

ALG_RSA_SHA_RFC2409

```
public static final byte ALG_RSA_SHA_RFC2409
```

Signature algorithm ALG_RSA_SHA_RFC2409 encrypts the 20 byte SHA digest using RSA. The digest is padded according to the RFC2409 scheme.

ALG_RSA_MD5_RFC2409

```
public static final byte ALG_RSA_MD5_RFC2409
```

Signature algorithm ALG_RSA_MD5_RFC2409 encrypts the 16 byte MD5 digest using RSA. The digest is padded according to the RFC2409 scheme.

MODE_SIGN

```
public static final byte MODE_SIGN
```

Used in init() methods to indicate signature sign mode.

MODE_VERIFY

```
public static final byte MODE_VERIFY
```

Used in init() methods to indicate signature verify mode.

Constructor Detail

Signature

```
protected Signature()
```

Protected Constructor

Method Detail

getInstance

```
public static final Signature getInstance(byte algorithm,
                                          boolean externalAccess)
                                   throws CryptoException
```

Creates a `Signature` object instance of the selected algorithm.

Parameters:

`algorithm` - the desired Signature algorithm. See above.

`externalAccess` - if `true` indicates that the instance will be shared among multiple applet instances and that the `Signature` instance will also be accessed (via a `Shareable` interface) when the owner of the `Signature` instance is not the currently selected applet.

Returns:

the `Signature` object instance of the requested algorithm.

Throws:

CryptoException - with the following reason codes:

- `CryptoException.NO_SUCH_ALGORITHM` if the requested algorithm is not supported.

---

init

```
public abstract void init(Key theKey,
                          byte theMode)
                   throws CryptoException
```

Initializes the `Signature` object with the appropriate `Key`. This method should be used for algorithms which do not need initialization parameters or use default parameter values.

Note:

- *DES and triple DES algorithms in CBC mode will use 0 for initial vector(IV) if this method is used.*

Parameters:

`theKey` - the key object to use for signing or verifying

`theMode` - one of `MODE_SIGN` or `MODE_VERIFY`

Throws:

CryptoException - with the following reason codes:

- `CryptoException.ILLEGAL_VALUE` if `theMode` option is an undefined value or if the `Key` is inconsistent with `theMode` or with the `Signature` implementation.

---

init

```
public abstract void init(Key theKey,
                          byte theMode,
                          byte[] bArray,
                          short bOff,
                          short bLen)
                   throws CryptoException
```

Initializes the `Signature` object with the appropriate `Key` and algorithm specific parameters.

Note:

- *DES and triple DES algorithms in outer CBC mode expect an 8 byte parameter value for the initial vector(IV) in* `bArray`.

- *RSA and DSA algorithms throw* `CryptoException.ILLEGAL_VALUE`.

Parameters:

`theKey` - the key object to use for signing
`theMode` - one of `MODE_SIGN` or `MODE_VERIFY`
`bArray` - byte array containing algorithm specific initialization info.
`bOff` - offset withing `bArray` where the algorithm specific data begins.
`bLen` - byte length of algorithm specific parameter data

Throws:

CryptoException - with the following reason codes:

- `CryptoException.ILLEGAL_VALUE` if `theMode` option is an undefined value or if a byte array parameter option is not supported by the algorithm or if the `bLen` is an incorrect byte length for the algorithm specific data or if the `Key` is inconsistent with `theMode` or with the `Signature` implementation.

---

getAlgorithm

```
public abstract byte getAlgorithm()
```

Gets the Signature algorithm.

Returns:

the algorithm code defined above.

---

getLength

```
public abstract short getLength()
```

Returns the byte length of the signature data.

Returns:

the byte length of the signature data.

---

update

```
public abstract void update(byte[] inBuff,
                            short inOffset,
                            short inLength)
                     throws CryptoException
```

Accumulates a signature of the input data. When this method is used temporary storage of intermediate results is required. This method should only be used if all the input data required for the signature is not available in one byte array. The `sign()` or `verify()` method is recommended whenever possible.

Parameters:

`inBuff` - the input buffer of data to be signed
`inOffset` - the offset into the input buffer at which to begin signature generation
`inLength` - the byte length to sign

Throws:

CryptoException - with the following reason codes:

- `CryptoException.UNINITIALIZED_KEY` if key not initialized.

See Also:

`sign(byte[], short, short, byte[], short)`, `verify(byte[], short, short, byte[], short, short)`

---

sign

```
public abstract short sign(byte[] inBuff,
                           short inOffset,
                           short inLength,
                           byte[] sigBuff,
                           short sigOffset)
                    throws CryptoException
```

Generates the signature of all/last input data. A call to this method also resets this `Signature` object to the state it was in when previously initialized via a call to `init()`. That is, the object is reset and available to sign another message.

The input and output buffer data may overlap.

Parameters:

`inBuff` - the input buffer of data to be signed

`inOffset` - the offset into the input buffer at which to begin signature generation

`inLength` - the byte length to sign

`sigBuff` - the output buffer to store signature data

`sigOffset` - the offset into sigBuff at which to begin signature data

Returns:

number of bytes of signature output in sigBuff

Throws:

CryptoException - with the following reason codes:

- `CryptoException.UNINITIALIZED_KEY` if key not initialized.
- `CryptoException.INVALID_INIT` if this `Signature` object is not initialized or initialized for signature verify mode.
- `CryptoException.ILLEGAL_USE` if this `Signature` algorithm does not pad the message and the message is not block aligned.

---

verify

```
public abstract boolean verify(byte[] inBuff,
                               short inOffset,
                               short inLength,
                               byte[] sigBuff,
                               short sigOffset,
                               short sigLength)
                        throws CryptoException
```

Verifies the signature of all/last input data against the passed in signature. A call to this method also resets this `Signature` object to the state it was in when previously initialized via a call to `init()`. That is, the object is reset and available to verify another message.

Parameters:

`inBuff` - the input buffer of data to be verified
`inOffset` - the offset into the input buffer at which to begin signature generation
`inLength` - the byte length to sign
`sigBuff` - the input buffer containing signature data
`sigOffset` - the offset into sigBuff where signature data begins.
`sigLength` - the byte length of the signature data

Returns:

`true` if signature verifies `false` otherwise.

Throws:

CryptoException - with the following reason codes:

- `CryptoException.UNINITIALIZED_KEY` if key not initialized.
- `CryptoException.INVALID_INIT` if this `Signature` object is not initialized or initialized for signature sign mode.
- `CryptoException.ILLEGAL_USE` if this `Signature` algorithm does not pad the message and the message is not block aligned.

Package javacardx.crypto

Extension package containing security classes and interfaces for export-controlled functionality.

See:

Description

Interface Summary	
KeyEncryption	`KeyEncryption` interface defines the methods used to enable encrypted key data access to a key implementation.

Class Summary	
Cipher	The `Cipher` class is the abstract base class for Cipher algorthims.

Package javacardx.crypto Description

Extension package containing security classes and interfaces for export-controlled functionality.

javacardx.crypto
Class Cipher

```
java.lang.Object
  |
  +--javacardx.crypto.Cipher
```

public abstract class **Cipher**
extends Object

The `Cipher` class is the abstract base class for Cipher algorthims. Implementations of Cipher algorithms must extend this class and implement all the abstract methods.

The term "pad" is used in the public key cipher algorithms below to refer to all the operations specified in the referenced scheme to transform the message block into the cipher block size.

Field Summary

`static byte`	**`ALG_DES_CBC_ISO9797_M1`** Cipher algorithm `ALG_DES_CBC_ISO9797_M1` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.
`static byte`	**`ALG_DES_CBC_ISO9797_M2`** Cipher algorithm `ALG_DES_CBC_ISO9797_M2` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.
`static byte`	**`ALG_DES_CBC_NOPAD`** Cipher algorithm `ALG_DES_CBC_NOPAD` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data.
`static byte`	**`ALG_DES_CBC_PKCS5`** Cipher algorithm ALG_DES_CBC_PKCS5 provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.
`static byte`	**`ALG_DES_ECB_ISO9797_M1`** Cipher algorithm `ALG_DES_ECB_ISO9797_M1` provides a cipher using DES in ECB mode. Input data is padded according to the ISO 9797 method 1 scheme.
`static byte`	**`ALG_DES_ECB_ISO9797_M2`** Cipher algorithm `ALG_DES_ECB_ISO9797_M2` provides a cipher using DES in ECB mode. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

static byte	**ALG_DES_ECB_NOPAD** Cipher algorithm ALG_DES_ECB_NOPAD provides a cipher using DES in ECB mode. This algorithm does not pad input data.
static byte	**ALG_DES_ECB_PKCS5** Cipher algorithm ALG_DES_ECB_PKCS5 provides a cipher using DES in ECB mode. Input data is padded according to the PKCS#5 scheme.
static byte	**ALG_RSA_ISO14888** Cipher algorithm ALG_RSA_ISO14888 provides a cipher using RSA. Input data is padded according to the ISO 14888 scheme.
static byte	**ALG_RSA_ISO9796** Cipher algorithm ALG_RSA_ISO9796 provides a cipher using RSA. Input data is padded according to the ISO 9796 (EMV'96) scheme.
static byte	**ALG_RSA_PKCS1** Cipher algorithm ALG_RSA_PKCS1 provides a cipher using RSA. Input data is padded according to the PKCS#1 (v1.5) scheme.
static byte	**MODE_DECRYPT** Used in init() methods to indicate decryption mode.
static byte	**MODE_ENCRYPT** Used in init() methods to indicate encryption mode.

Constructor Summary

protected	**Cipher()** Protected Constructor

Method Summary

abstract short	**doFinal**(byte[] inBuff, short inOffset, short inLength, byte[] outBuff, short outOffset) Generates encrypted/decrypted output from all/last input data.
abstract byte	**getAlgorithm**() Gets the Cipher algorithm.
static Cipher	**getInstance**(byte algorithm, boolean externalAccess) Creates a Cipher object instance of the selected algorithm.
abstract void	**init**(Key theKey, byte theMode) Initializes the Cipher object with the appropriate Key.
abstract void	**init**(Key theKey, byte theMode, byte[] bArray, short bOff, short bLen) Initializes the Cipher object with the appropriate Key and algorithm specific parameters.
abstract short	**update**(byte[] inBuff, short inOffset, short inLength, byte[] outBuff, short outOffset) Generates encrypted/decrypted output from input data.

Methods inherited from class java.lang.Object
equals

Field Detail

ALG_DES_CBC_NOPAD

```
public static final byte ALG_DES_CBC_NOPAD
```

Cipher algorithm ALG_DES_CBC_NOPAD provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data. If the input data is not (8 byte) block aligned it throws CryptoExeption with the reason code ILLEGAL_USE.

ALG_DES_CBC_ISO9797_M1

```
public static final byte ALG_DES_CBC_ISO9797_M1
```

Cipher algorithm `ALG_DES_CBC_ISO9797_M1` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.

ALG_DES_CBC_ISO9797_M2

`public static final byte` **`ALG_DES_CBC_ISO9797_M2`**

Cipher algorithm `ALG_DES_CBC_ISO9797_M2` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

ALG_DES_CBC_PKCS5

`public static final byte` **`ALG_DES_CBC_PKCS5`**

Cipher algorithm ALG_DES_CBC_PKCS5 provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.

ALG_DES_ECB_NOPAD

`public static final byte` **`ALG_DES_ECB_NOPAD`**

Cipher algorithm `ALG_DES_ECB_NOPAD` provides a cipher using DES in ECB mode. This algorithm does not pad input data. If the input data is not (8 byte) block aligned it throws `CryptoExeption` with the reason code `ILLEGAL_USE`.

ALG_DES_ECB_ISO9797_M1

`public static final byte` **`ALG_DES_ECB_ISO9797_M1`**

Cipher algorithm `ALG_DES_ECB_ISO9797_M1` provides a cipher using DES in ECB mode. Input data is padded according to the ISO 9797 method 1 scheme.

ALG_DES_ECB_ISO9797_M2

`public static final byte` **`ALG_DES_ECB_ISO9797_M2`**

Cipher algorithm `ALG_DES_ECB_ISO9797_M2` provides a cipher using DES in ECB mode. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

ALG_DES_ECB_PKCS5

```
public static final byte ALG_DES_ECB_PKCS5
```

Cipher algorithm `ALG_DES_ECB_PKCS5` provides a cipher using DES in ECB mode. Input data is padded according to the PKCS#5 scheme.

ALG_RSA_ISO14888

```
public static final byte ALG_RSA_ISO14888
```

Cipher algorithm `ALG_RSA_ISO14888` provides a cipher using RSA. Input data is padded according to the ISO 14888 scheme.

ALG_RSA_PKCS1

```
public static final byte ALG_RSA_PKCS1
```

Cipher algorithm `ALG_RSA_PKCS1` provides a cipher using RSA. Input data is padded according to the PKCS#1 (v1.5) scheme.

Note:

- *This algorithm is only suitable for messages of limited length. The total number of input bytes processed may not be more than k-11, where k is the RSA key's modulus size in bytes.*

ALG_RSA_ISO9796

```
public static final byte ALG_RSA_ISO9796
```

Cipher algorithm `ALG_RSA_ISO9796` provides a cipher using RSA. Input data is padded according to the ISO 9796 (EMV'96) scheme.

Note:

- *This algorithm is only suitable for messages of limited length. The total number of input bytes processed may not be more than k/2, where k is the RSA key's modulus size in bytes.*

MODE_DECRYPT

```
public static final byte MODE_DECRYPT
```

Used in `init()` methods to indicate decryption mode.

MODE_ENCRYPT

```
public static final byte MODE_ENCRYPT
```

Used in `init()` methods to indicate encryption mode.

Constructor Detail

Cipher

```
protected Cipher()
```

Protected Constructor

Method Detail

getInstance

```
public static final Cipher getInstance(byte algorithm,
                                       boolean externalAccess)
                                throws CryptoException
```

Creates a `Cipher` object instance of the selected algorithm.

Parameters:

`algorithm` - the desired Cipher algorithm. See above.

`externalAccess` - if `true` indicates that the instance will be shared among multiple applet instances and that the `Cipher` instance will also be accessed (via a `Shareable` interface) when the owner of the `Cipher` instance is not the currently selected applet.

Returns:

the `Cipher` object instance of the requested algorithm.

Throws:

CryptoException - with the following reason codes:

- `CryptoException.NO_SUCH_ALGORITHM` if the requested algorithm is not supported.

---

init

```
public abstract void init(Key theKey,
                          byte theMode)
                   throws CryptoException
```

Initializes the `Cipher` object with the appropriate `Key`. This method should be used for algorithms which do not need initialization parameters or use default parameter values.

Note:

- *DES and triple DES algorithms in CBC mode will use 0 for initial vector(IV) if this method is used.*

Parameters:

`theKey` - the key object to use for signing or verifying
`theMode` - one of `MODE_DECRYPT` or `MODE_ENCRYPT`

Throws:

CryptoException - with the following reason codes:

- `CryptoException.ILLEGAL_VALUE` if `theMode` option is an undefined value or if the `Key` is inconsistent with the `Cipher` implementation.

---

init

```
public abstract void init(Key theKey,
                          byte theMode,
                          byte[] bArray,
                          short bOff,
                          short bLen)
                   throws CryptoException
```

Initializes the `Cipher` object with the appropriate Key and algorithm specific parameters.

Note:

- *DES and triple DES algorithms in outer CBC mode expect an 8 byte parameter value for the initial vector(IV) in* `bArray`.
- *RSA and DSA algorithms throw* `CryptoException.ILLEGAL_VALUE`.

Parameters:

`theKey` - the key object to use for signing
`theMode` - one of `MODE_DECRYPT` or `MODE_ENCRYPT`
`bArray` - byte array containing algorithm specific initialization info.
`bOff` - offset withing bArray where the algorithm specific data begins.
`bLen` - byte length of algorithm specific parameter data

Throws:

CryptoException - with the following reason codes:

- `CryptoException.ILLEGAL_VALUE` if `theMode` option is an undefined value or if a byte array parameter option is not supported by the algorithm or if the `bLen` is an incorrect byte length for the algorithm specific data or if the `Key` is inconsistent with the `Cipher` implementation.

---

getAlgorithm

```
public abstract byte getAlgorithm()
```

Gets the Cipher algorithm.

Returns:
the algorithm code defined above.

---

doFinal

```
public abstract short doFinal(byte[] inBuff,
                              short inOffset,
                              short inLength,
                              byte[] outBuff,
                              short outOffset)
                       throws CryptoException
```

Generates encrypted/decrypted output from all/last input data. A call to this method also resets this `Cipher` object to the state it was in when previously initialized via a call to `init()`. That is, the object is reset and available to encrypt or decrypt (depending on the operation mode that was specified in the call to `init()`) more data.

The input and output buffer data may overlap.

Notes:

- *On decryption operations (except when ISO 9797 method 1 padding is used), the padding bytes are not written to* `outBuff`.
- *On encryption operations, the number of bytes output into* `outBuff` *may be larger than* `inLength`.

Parameters:
`inBuff` - the input buffer of data to be encrypted/decrypted.
`inOffset` - the offset into the input buffer at which to begin encryption/decryption.
`inLength` - the byte length to be encrypted/decrypted.
`outBuff` - the output buffer, may be the same as the input buffer
`outOffset` - the offset into the output buffer where the resulting hash value begins

Returns:
number of bytes output in `outBuff`

Throws:
CryptoException - with the following reason codes:

- `CryptoException.UNINITIALIZED_KEY` if key not initialized.
- `CryptoException.INVALID_INIT` if this `Cipher` object is not initialized.
- `CryptoException.ILLEGAL_USE` if this `Cipher` algorithm does not pad the message and the message is not block aligned or if the input message length is not supported.

---

update

```
public abstract short update(byte[] inBuff,
                             short inOffset,
```

```
                short inLength,
                byte[] outBuff,
                short outOffset)
        throws CryptoException
```

Generates encrypted/decrypted output from input data. When this method is used temporary storage of intermediate results is required. This method should only be used if all the input data required for the cipher is not available in one byte array. The `doFinal()` method is recommended whenever possible.

The input and output buffer data may overlap.

Notes:

- *On decryption operations(except when ISO 9797 method 1 padding is used), the padding bytes are not written to* `outBuff`.
- *On encryption operations, the number of bytes output into* `outBuff` *may be larger than* `inLength`.
- *On encryption and decryption operations(except when ISO 9797 method 1 padding is used), block alignment considerations may require that the number of bytes output into* `outBuff` *be smaller than* `inLength` *or even 0.*

Parameters:

`inBuff` - the input buffer of data to be encrypted/decrypted.
`inOffset` - the offset into the input buffer at which to begin encryption/decryption.
`inLength` - the byte length to be encrypted/decrypted.
`outBuff` - the output buffer, may be the same as the input buffer
`outOffset` - the offset into the output buffer where the resulting hash value begins

Returns:

number of bytes output in `outBuff`

Throws:

CryptoException - with the following reason codes:

- `CryptoException.UNINITIALIZED_KEY` if key not initialized.
- `CryptoException.INVALID_INIT` if this `Cipher` object is not initialized.
- `CryptoException.ILLEGAL_USE` if the input message length is not supported.

---

javacardx.crypto
Interface KeyEncryption

public abstract interface **KeyEncryption**

`KeyEncryption` interface defines the methods used to enable encrypted key data access to a key implementation.

See Also:
`KeyBuilder`, `Cipher`

Method Summary

`Cipher`	**`getKeyCipher`**`()` Returns the `Cipher` object to be used to decrypt the input key data and key parameters in the set methods. Default is `null` - no decryption performed.
`void`	**`setKeyCipher`**`(Cipher keyCipher)` Sets the `Cipher` object to be used to decrypt the input key data and key parameters in the set methods. Default `Cipher` object is `null` - no decryption performed.

Method Detail

setKeyCipher

```
public void setKeyCipher(Cipher keyCipher)
```

Sets the `Cipher` object to be used to decrypt the input key data and key parameters in the set methods.

Default `Cipher` object is `null` - no decryption performed.

Parameters:

`keyCipher` - the decryption `Cipher` object to decrypt the input key data. `null` parameter indicates that no decryption is required.

getKeyCipher

```
public Cipher getKeyCipher()
```

Returns the `Cipher` object to be used to decrypt the input key data and key parameters in the set methods.

Default is `null` - no decryption performed.

Returns:

keyCipher the decryption `Cipher` object to decrypt the input key data. `null` return indicates that no decryption is performed.

A

abortTransaction() - Static method in class javacard.framework.JCSystem
Aborts the atomic transaction.
AID - class javacard.framework.AID.
This class encapsulates the Application Identifier(AID) associated with an applet.
AID(byte[], short, byte) - Constructor for class javacard.framework.AID
The JCRE uses this constructor to create a new `AID` instance encapsulating the specified AID bytes.
ALG_DES_CBC_ISO9797_M1 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_CBC_ISO9797_M1` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.
ALG_DES_CBC_ISO9797_M2 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_CBC_ISO9797_M2` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.
ALG_DES_CBC_NOPAD - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_CBC_NOPAD` provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data.
ALG_DES_CBC_PKCS5 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm ALG_DES_CBC_PKCS5 provides a cipher using DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.
ALG_DES_ECB_ISO9797_M1 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_ECB_ISO9797_M1` provides a cipher using DES in ECB mode. Input data is padded according to the ISO 9797 method 1 scheme.
ALG_DES_ECB_ISO9797_M2 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_ECB_ISO9797_M2` provides a cipher using DES in ECB mode. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.
ALG_DES_ECB_NOPAD - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_ECB_NOPAD` provides a cipher using DES in ECB mode. This algorithm does not pad input data.
ALG_DES_ECB_PKCS5 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm `ALG_DES_ECB_PKCS5` provides a cipher using DES in ECB mode. Input data is padded according to the PKCS#5 scheme.
ALG_DES_MAC4_ISO9797_M1 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_DES_MAC4_ISO9797_M1` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.
ALG_DES_MAC4_ISO9797_M2 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_DES_MAC4_ISO9797_M2` generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

ALG_DES_MAC4_NOPAD - Static variable in class javacard.security.Signature
Signature algorithm ALG_DES_MAC4_NOPAD generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data.

ALG_DES_MAC4_PKCS5 - Static variable in class javacard.security.Signature
Signature algorithm ALG_DES_MAC4_PKCS5 generates a 4 byte MAC (most significant 4 bytes of encrypted block) using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.

ALG_DES_MAC8_ISO9797_M1 - Static variable in class javacard.security.Signature
Signature algorithm ALG_DES_MAC8_ISO9797_M1 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 1 scheme.

ALG_DES_MAC8_ISO9797_M2 - Static variable in class javacard.security.Signature
Signature algorithm ALG_DES_MAC8_ISO9797_M2 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the ISO 9797 method 2 (ISO 7816-4, EMV'96) scheme.

ALG_DES_MAC8_NOPAD - Static variable in class javacard.security.Signature
Signature algorithm ALG_DES_MAC_8_NOPAD generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. This algorithm does not pad input data.

ALG_DES_MAC8_PKCS5 - Static variable in class javacard.security.Signature
Signature algorithm ALG_DES_MAC8_PKCS5 generates a 8 byte MAC using DES or triple DES in CBC mode. This algorithm uses outer CBC for triple DES. Input data is padded according to the PKCS#5 scheme.

ALG_DSA_SHA - Static variable in class javacard.security.Signature
Signature algorithm ALG_DSA_SHA signs/verifies the 20 byte SHA digest using DSA.

ALG_MD5 - Static variable in class javacard.security.MessageDigest
Message Digest algorithm MD5.

ALG_PSEUDO_RANDOM - Static variable in class javacard.security.RandomData
Utility pseudo random number generation algorithms.

ALG_RIPEMD160 - Static variable in class javacard.security.MessageDigest
Message Digest algorithm RIPE MD-160.

ALG_RSA_ISO14888 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm ALG_RSA_ISO14888 provides a cipher using RSA. Input data is padded according to the ISO 14888 scheme.

ALG_RSA_ISO9796 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm ALG_RSA_ISO9796 provides a cipher using RSA. Input data is padded according to the ISO 9796 (EMV'96) scheme.

ALG_RSA_MD5_PKCS1 - Static variable in class javacard.security.Signature
Signature algorithm ALG_RSA_MD5_PKCS1 encrypts the 16 byte MD5 digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.

ALG_RSA_MD5_RFC2409 - Static variable in class javacard.security.Signature
Signature algorithm ALG_RSA_MD5_RFC2409 encrypts the 16 byte MD5 digest using RSA. The digest is padded according to the RFC2409 scheme.

ALG_RSA_PKCS1 - Static variable in class javacardx.crypto.Cipher
Cipher algorithm ALG_RSA_PKCS1 provides a cipher using RSA. Input data is padded according to

the PKCS#1 (v1.5) scheme.

ALG_RSA_RIPEMD160_ISO9796 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_RSA_RIPEMD160_ISO9796` encrypts the 20 byte RIPE MD-160 digest using RSA. The digest is padded according to the ISO 9796 scheme.

ALG_RSA_RIPEMD160_PKCS1 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_RSA_RIPEMD160_PKCS1` encrypts the 20 byte RIPE MD-160 digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.

ALG_RSA_SHA_ISO9796 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_RSA_SHA_ISO9796` encrypts the 20 byte SHA digest using RSA. The digest is padded according to the ISO 9796 (EMV'96) scheme.

ALG_RSA_SHA_PKCS1 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_RSA_SHA_PKCS1` encrypts the 20 byte SHA digest using RSA. The digest is padded according to the PKCS#1 (v1.5) scheme.

ALG_RSA_SHA_RFC2409 - Static variable in class javacard.security.Signature
Signature algorithm `ALG_RSA_SHA_RFC2409` encrypts the 20 byte SHA digest using RSA. The digest is padded according to the RFC2409 scheme.

ALG_SECURE_RANDOM - Static variable in class javacard.security.RandomData
Cryptographically secure random number generation algorithms.

ALG_SHA - Static variable in class javacard.security.MessageDigest
Message Digest algorithm SHA.

APDU - class javacard.framework.APDU.
Application Protocol Data Unit (APDU) is the communication format between the card and the off-card applications.

APDUException - exception javacard.framework.APDUException.
`APDUException` represents an `APDU` related exception.

APDUException(short) - Constructor for class javacard.framework.APDUException
Constructs an APDUException.

Applet - class javacard.framework.Applet.
This abstract class defines an applet in Java Card.

Applet() - Constructor for class javacard.framework.Applet
Only this class's `install()` method should create the applet object.

ArithmeticException - exception java.lang.ArithmeticException.
A JCRE owned instance of `ArithmethicException` is thrown when an exceptional arithmetic condition has occurred.

ArithmeticException() - Constructor for class java.lang.ArithmeticException
Constructs an `ArithmeticException`.

arrayCompare(byte[], short, byte[], short, short) - Static method in class javacard.framework.Util
Compares an array from the specified source array, beginning at the specified position, with the specified position of the destination array from left to right.

arrayCopy(byte[], short, byte[], short, short) - Static method in class javacard.framework.Util
Copies an array from the specified source array, beginning at the specified position, to the specified position of the destination array.

arrayCopyNonAtomic(byte[], short, byte[], short, short) - Static method in class javacard.framework.Util
Copies an array from the specified source array, beginning at the specified position, to the specified position of the destination array (non-atomically).

arrayFillNonAtomic(byte[], short, short, byte) - Static method in class javacard.framework.Util
Fills the byte array (non-atomically) beginning at the specified position, for the specified length with the specified byte value.
ArrayIndexOutOfBoundsException - exception java.lang.ArrayIndexOutOfBoundsException.
A JCRE owned instance of `IndexOutOfBoundsException` is thrown to indicate that an array has been accessed with an illegal index.
ArrayIndexOutOfBoundsException() - Constructor for class java.lang.ArrayIndexOutOfBoundsException
Constructs an `ArrayIndexOutOfBoundsException`.
ArrayStoreException - exception java.lang.ArrayStoreException.
A JCRE owned instance of `ArrayStoreException` is thrown to indicate that an attempt has been made to store the wrong type of object into an array of objects.
ArrayStoreException() - Constructor for class java.lang.ArrayStoreException
Constructs an `ArrayStoreException`.

B

BAD_LENGTH - Static variable in class javacard.framework.APDUException
This reason code is used by the `APDU.setOutgoingLength()` method to indicate that the length parameter is greater that 256 or if non BLOCK CHAINED data transfer is requested and `len` is greater than (IFSD-2), where IFSD is the Outgoing Block Size.
beginTransaction() - Static method in class javacard.framework.JCSystem
Begins an atomic transaction.
BUFFER_BOUNDS - Static variable in class javacard.framework.APDUException
This reason code is used by the `APDU.sendBytes()` method to indicate that the sum of buffer offset parameter and the byte length parameter exceeds the APDU buffer size.
BUFFER_FULL - Static variable in class javacard.framework.TransactionException
This reason code is used during a transaction to indicate that the commit buffer is full.
buildKey(byte, short, boolean) - Static method in class javacard.security.KeyBuilder
Creates cryptographic keys for signature and cipher algorithms.

C

CardException - exception javacard.framework.CardException.
The `CardException` class defines a field `reason` and two accessor methods `getReason()` and `setReason()`.
CardException(short) - Constructor for class javacard.framework.CardException
Construct a CardException instance with the specified reason.
CardRuntimeException - exception javacard.framework.CardRuntimeException.
The `CardRuntimeException` class defines a field `reason` and two accessor methods `getReason()` and `setReason()`.
CardRuntimeException(short) - Constructor for class javacard.framework.CardRuntimeException
Construct a CardRuntimeException instance with the specified reason.

check(byte[], short, byte) - Method in class javacard.framework.OwnerPIN
Compares `pin` against the PIN value.
check(byte[], short, byte) - Method in interface javacard.framework.PIN
Compares `pin` against the PIN value.
Cipher - class javacardx.crypto.Cipher.
The `Cipher` class is the abstract base class for Cipher algorthims.
Cipher() - Constructor for class javacardx.crypto.Cipher
Protected Constructor
CLA_ISO7816 - Static variable in interface javacard.framework.ISO7816
APDU command CLA : ISO 7816 = 0x00
ClassCastException - exception java.lang.ClassCastException.
A JCRE owned instance of `ClassCastException` is thrown to indicate that the code has attempted to cast an object to a subclass of which it is not an instance.
ClassCastException() - Constructor for class java.lang.ClassCastException
Constructs a `ClassCastException`.
CLEAR_ON_DESELECT - Static variable in class javacard.framework.JCSystem
This event code indicates that the contents of the transient object are cleared to the default value on applet deselection event or in `CLEAR_ON_RESET` cases.
CLEAR_ON_RESET - Static variable in class javacard.framework.JCSystem
This event code indicates that the contents of the transient object are cleared to the default value on card reset (or power on) event.
clearKey() - Method in interface javacard.security.Key
Clears the key and sets its initialized state to false.
commitTransaction() - Static method in class javacard.framework.JCSystem
Commits an atomic transaction.
CryptoException - exception javacard.security.CryptoException.
`CryptoException` represents a cryptography-related exception.
CryptoException(short) - Constructor for class javacard.security.CryptoException
Constructs a `CryptoException` with the specified reason.

D

deselect() - Method in class javacard.framework.Applet
Called by the JCRE to inform this currently selected applet that another (or the same) applet will be selected.
DESKey - interface javacard.security.DESKey.
`DESKey` contains an 8/16/24 byte key for single/2 key triple DES/3 key triple DES operations.
doFinal(byte[], short, short, byte[], short) - Method in class javacard.security.MessageDigest
Generates a hash of all/last input data.
doFinal(byte[], short, short, byte[], short) - Method in class javacardx.crypto.Cipher
Generates encrypted/decrypted output from all/last input data.
DSAKey - interface javacard.security.DSAKey.
The DSAKey interface is the base interface for the DSA algorithms private and public key implementaions.

DSAPrivateKey - interface javacard.security.DSAPrivateKey.
The `DSAPrivateKey` interface is used to sign data using the DSA algorithm.
DSAPublicKey - interface javacard.security.DSAPublicKey.
The `DSAPublicKey` interface is used to verify signatures on signed data using the DSA algorithm.

E

equals(byte[], short, byte) - Method in class javacard.framework.AID
Checks if the specified AID bytes in `bArray` are the same as those encapsulated in `this AID` object.
equals(Object) - Method in class java.lang.Object
Compares two Objects for equality.
equals(Object) - Method in class javacard.framework.AID
Compares the AID bytes in `this AID` instance to the AID bytes in the specified object.
Exception - exception java.lang.Exception.
The class `Exception` and its subclasses are a form of `Throwable` that indicates conditions that a reasonable applet might want to catch.
Exception() - Constructor for class java.lang.Exception
Constructs an `Exception` instance.

G

generateData(byte[], short, short) - Method in class javacard.security.RandomData
Generates random data.
getAID() - Static method in class javacard.framework.JCSystem
Returns the JCRE owned instance of the `AID` object associated with the current applet context.
getAlgorithm() - Method in class javacard.security.MessageDigest
Gets the Message digest algorithm.
getAlgorithm() - Method in class javacard.security.Signature
Gets the Signature algorithm.
getAlgorithm() - Method in class javacardx.crypto.Cipher
Gets the Cipher algorithm.
getAppletShareableInterfaceObject(AID, byte) - Static method in class javacard.framework.JCSystem
This method is called by a client applet to get a server applet’s shareable interface object.
getBuffer() - Method in class javacard.framework.APDU
Returns the APDU buffer byte array.
getBytes(byte[], short) - Method in class javacard.framework.AID
Called to get the AID bytes encapsulated within `AID` object.
getDP1(byte[], short) - Method in interface javacard.security.RSAPrivateCrtKey
Returns the value of the DP1 parameter in plain text.
getDQ1(byte[], short) - Method in interface javacard.security.RSAPrivateCrtKey
Returns the value of the DQ1 parameter in plain text.

getExponent(byte[], short) - Method in interface javacard.security.RSAPrivateKey
Returns the private exponent value of the key in plain text.
getExponent(byte[], short) - Method in interface javacard.security.RSAPublicKey
Returns the private exponent value of the key in plain text.
getG(byte[], short) - Method in interface javacard.security.DSAKey
Returns the subprime parameter value of the key in plain text.
getInBlockSize() - Static method in class javacard.framework.APDU
Returns the configured incoming block size. In T=1 protocol, this corresponds to IFSC (information field size for ICC), the maximum size of incoming data blocks into the card. In T=0 protocol, this method returns 1.
getInstance(byte) - Static method in class javacard.security.RandomData
Creates a `RandomData` instance of the selected algorithm.
getInstance(byte, boolean) - Static method in class javacard.security.MessageDigest
Creates a `MessageDigest` object instance of the selected algorithm.
getInstance(byte, boolean) - Static method in class javacard.security.Signature
Creates a `Signature` object instance of the selected algorithm.
getInstance(byte, boolean) - Static method in class javacardx.crypto.Cipher
Creates a `Cipher` object instance of the selected algorithm.
getKey(byte[], short) - Method in interface javacard.security.DESKey
Returns the `Key` data in plain text.
getKeyCipher() - Method in interface javacardx.crypto.KeyEncryption
Returns the `Cipher` object to be used to decrypt the input key data and key parameters in the set methods. Default is `null` - no decryption performed.
getLength() - Method in class javacard.security.MessageDigest
Returns the byte length of the hash.
getLength() - Method in class javacard.security.Signature
Returns the byte length of the signature data.
getMaxCommitCapacity() - Static method in class javacard.framework.JCSystem
Returns the total number of bytes in the commit buffer.
getModulus(byte[], short) - Method in interface javacard.security.RSAPrivateKey
Returns the modulus value of the key in plain text.
getModulus(byte[], short) - Method in interface javacard.security.RSAPublicKey
Returns the modulus value of the key in plain text.
getNAD() - Method in class javacard.framework.APDU
In T=1 protocol, this method returns the Node Address byte, NAD. In T=0 protocol, this method returns 0.
getOutBlockSize() - Static method in class javacard.framework.APDU
Returns the configured outgoing block size. In T=1 protocol, this corresponds to IFSD (information field size for interface device), the maximum size of outgoing data blocks to the CAD. In T=0 protocol, this method returns 258 (accounts for 2 status bytes).
getP(byte[], short) - Method in interface javacard.security.DSAKey
Returns the base parameter value of the key in plain text.
getP(byte[], short) - Method in interface javacard.security.RSAPrivateCrtKey
Returns the value of the P parameter in plain text.
getPQ(byte[], short) - Method in interface javacard.security.RSAPrivateCrtKey
Returns the value of the PQ parameter in plain text.

getPreviousContextAID() - Static method in class javacard.framework.JCSystem
This method is called to obtain the JCRE owned instance of the `AID` object associated with the previously active applet context.
getProtocol() - Static method in class javacard.framework.APDU
Returns the ISO 7816 transport protocol type, T=1 or T=0 in progress.
getQ(byte[], short) - Method in interface javacard.security.DSAKey
Returns the prime parameter value of the key in plain text.
getQ(byte[], short) - Method in interface javacard.security.RSAPrivateCrtKey
Returns the value of the Q parameter in plain text.
getReason() - Method in class javacard.framework.CardRuntimeException
Get reason code
getReason() - Method in class javacard.framework.CardException
Get reason code
getShareableInterfaceObject(AID, byte) - Method in class javacard.framework.Applet
Called by the JCRE to obtain a shareable interface object from this server applet, on behalf of a request from a client applet.
getShort(byte[], short) - Static method in class javacard.framework.Util
Concatenates two bytes in a byte array to form a short value.
getSize() - Method in interface javacard.security.Key
Returns the key size in number of bits.
getTransactionDepth() - Static method in class javacard.framework.JCSystem
Returns the current transaction nesting depth level.
getTriesRemaining() - Method in class javacard.framework.OwnerPIN
Returns the number of times remaining that an incorrect PIN can be presented before the `PIN` is blocked.
getTriesRemaining() - Method in interface javacard.framework.PIN
Returns the number of times remaining that an incorrect PIN can be presented before the `PIN` is blocked.
getType() - Method in interface javacard.security.Key
Returns the key interface type.
getUnusedCommitCapacity() - Static method in class javacard.framework.JCSystem
Returns the number of bytes left in the commit buffer.
getValidatedFlag() - Method in class javacard.framework.OwnerPIN
This protected method returns the validated flag.
getVersion() - Static method in class javacard.framework.JCSystem
Returns the current major and minor version of the Java Card API.
getX(byte[], short) - Method in interface javacard.security.DSAPrivateKey
Returns the value of the key in plain text.
getY(byte[], short) - Method in interface javacard.security.DSAPublicKey
Returns the value of the key in plain text.

I

ILLEGAL_AID - Static variable in class javacard.framework.SystemException
This reason code is used by the `javacard.framework.Applet.register()` method to indicate that the input AID parameter is not a legal AID value.

ILLEGAL_TRANSIENT - Static variable in class javacard.framework.SystemException
This reason code is used to indicate that the request to create a transient object is not allowed in the current applet context.

ILLEGAL_USE - Static variable in class javacard.framework.APDUException
This APDUException reason code indicates that the method should not be invoked based on the current state of the APDU.

ILLEGAL_USE - Static variable in class javacard.security.CryptoException
This reason code is used to indicate that the signature or cipher algorithm does not pad the incoming message and the input message is not block aligned.

ILLEGAL_VALUE - Static variable in class javacard.framework.PINException
This reason code is used to indicate that one or more input parameters is out of allowed bounds.

ILLEGAL_VALUE - Static variable in class javacard.framework.SystemException
This reason code is used to indicate that one or more input parameters is out of allowed bounds.

ILLEGAL_VALUE - Static variable in class javacard.security.CryptoException
This reason code is used to indicate that one or more input parameters is out of allowed bounds.

IN_PROGRESS - Static variable in class javacard.framework.TransactionException
This reason code is used by the `beginTransaction` method to indicate a transaction is already in progress.

IndexOutOfBoundsException - exception java.lang.IndexOutOfBoundsException.
A JCRE owned instance of `IndexOutOfBoundsException` is thrown to indicate that an index of some sort (such as to an array) is out of range.

IndexOutOfBoundsException() - Constructor for class java.lang.IndexOutOfBoundsException
Constructs an `IndexOutOfBoundsException`.

init(Key, byte) - Method in class javacard.security.Signature
Initializes the `Signature` object with the appropriate `Key`.

init(Key, byte) - Method in class javacardx.crypto.Cipher
Initializes the `Cipher` object with the appropriate `Key`.

init(Key, byte, byte[], short, short) - Method in class javacard.security.Signature
Initializes the `Signature` object with the appropriate `Key` and algorithm specific parameters.

init(Key, byte, byte[], short, short) - Method in class javacardx.crypto.Cipher
Initializes the `Cipher` object with the appropriate Key and algorithm specific parameters.

INS_EXTERNAL_AUTHENTICATE - Static variable in interface javacard.framework.ISO7816
APDU command INS : EXTERNAL AUTHENTICATE = 0x82

INS_SELECT - Static variable in interface javacard.framework.ISO7816
APDU command INS : SELECT = 0xA4

install(byte[], short, byte) - Static method in class javacard.framework.Applet
To create an instance of the `Applet` subclass, the JCRE will call this static method first.

INTERNAL_FAILURE - Static variable in class javacard.framework.TransactionException
This reason code is used during a transaction to indicate an internal JCRE problem (fatal error).

INVALID_INIT - Static variable in class javacard.security.CryptoException
This reason code is used to indicate that the signature or cipher object has not been correctly initialized for the requested operation.
IO_ERROR - Static variable in class javacard.framework.APDUException
This reason code indicates that an unrecoverable error occurred in the I/O transmission layer.
isInitialized() - Method in interface javacard.security.Key
Reports the initialized state of the key.
ISO7816 - interface javacard.framework.ISO7816.
`ISO7816` encapsulates constants related to ISO 7816-3 and ISO 7816-4.
ISOException - exception javacard.framework.ISOException.
`ISOException` class encapsulates an ISO 7816-4 response status word as its `reason` code.
ISOException(short) - Constructor for class javacard.framework.ISOException
Constructs an ISOException instance with the specified status word.
isTransient(Object) - Static method in class javacard.framework.JCSystem
Used to check if the specified object is transient.
isValidated() - Method in class javacard.framework.OwnerPIN
Returns `true` if a valid PIN has been presented since the last card reset or last call to `reset()`.
isValidated() - Method in interface javacard.framework.PIN
Returns `true` if a valid PIN value has been presented since the last card reset or last call to `reset()`.

J

java.lang - package java.lang
Provides classes that are fundamental to the design of the Java Card technology subset of the Java programming language.
javacard.framework - package javacard.framework
Provides framework of classes and interfaces for the core functionality of a Java Card applet.
javacard.security - package javacard.security
Provides the classes and interfaces for the Java Card security framework.
javacardx.crypto - package javacardx.crypto
Extension package containing security classes and interfaces for export-controlled functionality.
JCSystem - class javacard.framework.JCSystem.
The `JCSystem` class includes a collection of methods to control applet execution, resource management, atomic transaction management and inter-applet object sharing in Java Card.

K

Key - interface javacard.security.Key.
The `Key` interface is the base interface for all keys.
KeyBuilder - class javacard.security.KeyBuilder.
The `KeyBuilder` class is a key object factory.

KeyEncryption - interface javacardx.crypto.KeyEncryption.
`KeyEncryption` interface defines the methods used to enable encrypted key data access to a key implementation.

L

LENGTH_DES - Static variable in class javacard.security.KeyBuilder
DES Key Length `LENGTH_DES` = 64.
LENGTH_DES3_2KEY - Static variable in class javacard.security.KeyBuilder
DES Key Length `LENGTH_DES3_2KEY` = 128.
LENGTH_DES3_3KEY - Static variable in class javacard.security.KeyBuilder
DES Key Length `LENGTH_DES3_3KEY` = 192.
LENGTH_DSA_1024 - Static variable in class javacard.security.KeyBuilder
DSA Key Length `LENGTH_DSA_1024` = 1024.
LENGTH_DSA_512 - Static variable in class javacard.security.KeyBuilder
DSA Key Length `LENGTH_DSA_512` = 512.
LENGTH_DSA_768 - Static variable in class javacard.security.KeyBuilder
DSA Key Length `LENGTH_DSA_768` = 768.
LENGTH_RSA_1024 - Static variable in class javacard.security.KeyBuilder
RSA Key Length `LENGTH_RSA_1024` = 1024.
LENGTH_RSA_2048 - Static variable in class javacard.security.KeyBuilder
RSA Key Length `LENGTH_RSA_2048` = 2048.
LENGTH_RSA_512 - Static variable in class javacard.security.KeyBuilder
RSA Key Length `LENGTH_RSA_512` = 512.
LENGTH_RSA_768 - Static variable in class javacard.security.KeyBuilder
RSA Key Length `LENGTH_RSA_768` = 768.
lookupAID(byte[], short, byte) - Static method in class javacard.framework.JCSystem
Returns the JCRE owned instance of the `AID` object, if any, encapsulating the specified AID bytes in the `buffer` parameter if there exists a successfully installed applet on the card whose instance AID exactly matches that of the specified AID bytes.

M

makeShort(byte, byte) - Static method in class javacard.framework.Util
Concatenates the two parameter bytes to form a short value.
makeTransientBooleanArray(short, byte) - Static method in class javacard.framework.JCSystem
Create a transient boolean array with the specified array length.
makeTransientByteArray(short, byte) - Static method in class javacard.framework.JCSystem
Create a transient byte array with the specified array length.
makeTransientObjectArray(short, byte) - Static method in class javacard.framework.JCSystem
Create a transient array of `Object` with the specified array length.
makeTransientShortArray(short, byte) - Static method in class javacard.framework.JCSystem
Create a transient short array with the specified array length.

MessageDigest - class javacard.security.MessageDigest.
The `MessageDigest` class is the base class for hashing algorthims.
MessageDigest() - Constructor for class javacard.security.MessageDigest
Protected Constructor
MODE_DECRYPT - Static variable in class javacardx.crypto.Cipher
Used in `init()` methods to indicate decryption mode.
MODE_ENCRYPT - Static variable in class javacardx.crypto.Cipher
Used in `init()` methods to indicate encryption mode.
MODE_SIGN - Static variable in class javacard.security.Signature
Used in `init()` methods to indicate signature sign mode.
MODE_VERIFY - Static variable in class javacard.security.Signature
Used in `init()` methods to indicate signature verify mode.

N

NegativeArraySizeException - exception java.lang.NegativeArraySizeException.
A JCRE owned instance of `NegativeArraySizeException` is thrown if an applet tries to create an array with negative size.
NegativeArraySizeException() - Constructor for class java.lang.NegativeArraySizeException
Constructs a `NegativeArraySizeException`.
NO_RESOURCE - Static variable in class javacard.framework.SystemException
This reason code is used to indicate that there is insufficient resource in the Card for the request.
NO_SUCH_ALGORITHM - Static variable in class javacard.security.CryptoException
This reason code is used to indicate that the requested algorithm or key type is not supported.
NO_T0_GETRESPONSE - Static variable in class javacard.framework.APDUException
This reason code indicates that during T=0 protocol, the CAD did not return a GET RESPONSE command in response to a <61xx> response status to send additional data.
NO_TRANSIENT_SPACE - Static variable in class javacard.framework.SystemException
This reason code is used by the `makeTransient..()` methods to indicate that no room is available in volatile memory for the requested object.
NOT_A_TRANSIENT_OBJECT - Static variable in class javacard.framework.JCSystem
This event code indicates that the object is not transient.
NOT_IN_PROGRESS - Static variable in class javacard.framework.TransactionException
This reason code is used by the `abortTransaction` and `commintTransaction` methods when a transaction is not in progress.
NullPointerException - exception java.lang.NullPointerException.
A JCRE owned instance of `NullPointerException`is thrown when an applet attempts to use `null` in a case where an object is required.
NullPointerException() - Constructor for class java.lang.NullPointerException
Constructs a `NullPointerException`.

O

Object - class java.lang.Object.
Class `Object` is the root of the Java Card class hierarchy.
Object() - Constructor for class java.lang.Object

OFFSET_CDATA - Static variable in interface javacard.framework.ISO7816
APDU command data offset : CDATA = 5
OFFSET_CLA - Static variable in interface javacard.framework.ISO7816
APDU header offset : CLA = 0
OFFSET_INS - Static variable in interface javacard.framework.ISO7816
APDU header offset : INS = 1
OFFSET_LC - Static variable in interface javacard.framework.ISO7816
APDU header offset : LC = 4
OFFSET_P1 - Static variable in interface javacard.framework.ISO7816
APDU header offset : P1 = 2
OFFSET_P2 - Static variable in interface javacard.framework.ISO7816
APDU header offset : P2 = 3
OwnerPIN - class javacard.framework.OwnerPIN.
This class represents an Owner PIN.
OwnerPIN(byte, byte) - Constructor for class javacard.framework.OwnerPIN
Constructor.

P

partialEquals(byte[], short, byte) - Method in class javacard.framework.AID
Checks if the specified partial AID byte sequence matches the first `length` bytes of the encapsulated AID bytes within `this AID` object.
PIN - interface javacard.framework.PIN.
This interface represents a PIN.
PINException - exception javacard.framework.PINException.
`PINException` represents a `OwnerPIN` class access-related exception.
PINException(short) - Constructor for class javacard.framework.PINException
Constructs a PINException.
PrivateKey - interface javacard.security.PrivateKey.
The `PrivateKey` class is the base class for private keys used in asymmetric algorithms.
process(APDU) - Method in class javacard.framework.Applet
Called by the JCRE to process an incoming APDU command.
PROTOCOL_T0 - Static variable in class javacard.framework.APDU
ISO 7816 transport protocol type T=0
PROTOCOL_T1 - Static variable in class javacard.framework.APDU
ISO 7816 transport protocol type T=1
PublicKey - interface javacard.security.PublicKey.
The `PublicKey` class is the base class for public keys used in asymmetric algorithms.

R

RandomData - class javacard.security.RandomData.
The `RandomData` abstract class is the base class for random number generation.
RandomData() - Constructor for class javacard.security.RandomData
Protected constructor for subclassing.
receiveBytes(short) - Method in class javacard.framework.APDU
Gets as many data bytes as will fit without APDU buffer overflow, at the specified offset `bOff`. Gets all the remaining bytes if they fit.
register() - Method in class javacard.framework.Applet
This method is used by the applet to register `this` applet instance with the JCRE and to assign the `Applet` subclass AID bytes as its instance AID bytes.
register(byte[], short, byte) - Method in class javacard.framework.Applet
This method is used by the applet to register `this` applet instance with the JCRE and assign the specified AID bytes as its instance AID bytes.
reset() - Method in class javacard.framework.OwnerPIN
If the validated flag is set, this method resets it.
reset() - Method in interface javacard.framework.PIN
If the validated flag is set, this method resets it.
reset() - Method in class javacard.security.MessageDigest
Resets the `MessageDigest` object to the initial state for further use.
resetAndUnblock() - Method in class javacard.framework.OwnerPIN
This method resets the validated flag and resets the `PIN` try counter to the value of the `PIN` try limit.
RIDEquals(AID) - Method in class javacard.framework.AID
Checks if the RID (National Registered Application provider identifier) portion of the encapsulated AID bytes within the `otherAID` object matches that of `this AID` object.
RSAPrivateCrtKey - interface javacard.security.RSAPrivateCrtKey.
The `RSAPrivateCrtKey` interface is used to sign data using the RSA algorithm in its Chinese Remainder Theorem form.
RSAPrivateKey - interface javacard.security.RSAPrivateKey.
The `RSAPrivateKey` class is used to sign data using the RSA algorithm in its modulus/exponent form.
RSAPublicKey - interface javacard.security.RSAPublicKey.
The `RSAPublicKey` is used to verify signatures on signed data using the RSA algorithm.
RuntimeException - exception java.lang.RuntimeException.
`RuntimeException` is the superclass of those exceptions that can be thrown during the normal operation of the Java Card Virtual Machine. A method is not required to declare in its throws clause any subclasses of `RuntimeException` that might be thrown during the execution of the method but not caught.
RuntimeException() - Constructor for class java.lang.RuntimeException
Constructs a `RuntimeException` instance.

S

SecretKey - interface javacard.security.SecretKey.
The `SecretKey` class is the base interface for keys used in symmetric alogrightms (e.g. DES).

SecurityException - exception java.lang.SecurityException.
A JCRE owned instance of `SecurityException` is thrown by the Java Card Virtual Machine to indicate a security violation. This exception is thrown when an attempt is made to illegally access an object belonging to a another applet.

SecurityException() - Constructor for class java.lang.SecurityException
Constructs a `SecurityException`.

select() - Method in class javacard.framework.Applet
Called by the JCRE to inform this applet that it has been selected.

selectingApplet() - Method in class javacard.framework.Applet
This method is used by the applet `process()` method to distinguish the SELECT APDU command which selected `this` applet, from all other other SELECT APDU commands which may relate to file or internal applet state selection.

sendBytes(short, short) - Method in class javacard.framework.APDU
Sends `len` more bytes from APDU buffer at specified offset `bOff`.

sendBytesLong(byte[], short, short) - Method in class javacard.framework.APDU
Sends `len` more bytes from `outData` byte array starting at specified offset `bOff`.

setDP1(byte[], short, short) - Method in interface javacard.security.RSAPrivateCrtKey
Sets the value of the DP1 parameter.

setDQ1(byte[], short, short) - Method in interface javacard.security.RSAPrivateCrtKey
Sets the value of the DQ1 parameter.

setExponent(byte[], short, short) - Method in interface javacard.security.RSAPrivateKey
Sets the private exponent value of the key.

setExponent(byte[], short, short) - Method in interface javacard.security.RSAPublicKey
Sets the public exponent value of the key.

setG(byte[], short, short) - Method in interface javacard.security.DSAKey
Sets the subprime parameter value of the key.

setIncomingAndReceive() - Method in class javacard.framework.APDU
This is the primary receive method.

setKey(byte[], short) - Method in interface javacard.security.DESKey
Sets the `Key` data.

setKeyCipher(Cipher) - Method in interface javacardx.crypto.KeyEncryption
Sets the `Cipher` object to be used to decrypt the input key data and key parameters in the set methods. Default `Cipher` object is `null` - no decryption performed.

setModulus(byte[], short, short) - Method in interface javacard.security.RSAPrivateKey
Sets the modulus value of the key.

setModulus(byte[], short, short) - Method in interface javacard.security.RSAPublicKey
Sets the modulus value of the key.

setOutgoing() - Method in class javacard.framework.APDU
This method is used to set the data transfer direction to outbound and to obtain the expected length of response (Le).

setOutgoingAndSend(short, short) - Method in class javacard.framework.APDU
This is the "convenience" send method.
setOutgoingLength(short) - Method in class javacard.framework.APDU
Sets the actual length of response data.
setOutgoingNoChaining() - Method in class javacard.framework.APDU
This method is used to set the data transfer direction to outbound without using BLOCK CHAINING(See ISO 7816-3/4) and to obtain the expected length of response (Le).
setP(byte[], short, short) - Method in interface javacard.security.DSAKey
Sets the base parameter value of the key.
setP(byte[], short, short) - Method in interface javacard.security.RSAPrivateCrtKey
Sets the value of the P parameter.
setPQ(byte[], short, short) - Method in interface javacard.security.RSAPrivateCrtKey
Sets the value of the PQ parameter.
setQ(byte[], short, short) - Method in interface javacard.security.DSAKey
Sets the prime parameter value of the key.
setQ(byte[], short, short) - Method in interface javacard.security.RSAPrivateCrtKey
Sets the value of the Q parameter.
setReason(short) - Method in class javacard.framework.CardRuntimeException
Set reason code
setReason(short) - Method in class javacard.framework.CardException
Set reason code
setSeed(byte[], short, short) - Method in class javacard.security.RandomData
Seeds the random data generator.
setShort(byte[], short, short) - Static method in class javacard.framework.Util
Deposits the short value as two successive bytes at the specified offset in the byte array.
setValidatedFlag(boolean) - Method in class javacard.framework.OwnerPIN
This protected method sets the value of the validated flag.
setX(byte[], short, short) - Method in interface javacard.security.DSAPrivateKey
Sets the value of the key.
setY(byte[], short, short) - Method in interface javacard.security.DSAPublicKey
Sets the value of the key.
Shareable - interface javacard.framework.Shareable.
The Shareable interface serves to identify all shared objects.
sign(byte[], short, short, byte[], short) - Method in class javacard.security.Signature
Generates the signature of all/last input data.
Signature - class javacard.security.Signature.
The `Signature` class is the base class for Signature algorthims.
Signature() - Constructor for class javacard.security.Signature
Protected Constructor
SW_APPLET_SELECT_FAILED - Static variable in interface javacard.framework.ISO7816
Response status : Applet selection failed = 0x6999;
SW_BYTES_REMAINING_00 - Static variable in interface javacard.framework.ISO7816
Response status : Response bytes remaining = 0x6100
SW_CLA_NOT_SUPPORTED - Static variable in interface javacard.framework.ISO7816
Response status : CLA value not supported = 0x6E00

SW_COMMAND_NOT_ALLOWED - Static variable in interface javacard.framework.ISO7816
Response status : Command not allowed (no current EF) = 0x6986
SW_CONDITIONS_NOT_SATISFIED - Static variable in interface javacard.framework.ISO7816
Response status : Conditions of use not satisfied = 0x6985
SW_CORRECT_LENGTH_00 - Static variable in interface javacard.framework.ISO7816
Response status : Correct Expected Length (Le) = 0x6C00
SW_DATA_INVALID - Static variable in interface javacard.framework.ISO7816
Response status : Data invalid = 0x6984
SW_FILE_FULL - Static variable in interface javacard.framework.ISO7816
Response status : Not enough memory space in the file = 0x6A84
SW_FILE_INVALID - Static variable in interface javacard.framework.ISO7816
Response status : File invalid = 0x6983
SW_FILE_NOT_FOUND - Static variable in interface javacard.framework.ISO7816
Response status : File not found = 0x6A82
SW_FUNC_NOT_SUPPORTED - Static variable in interface javacard.framework.ISO7816
Response status : Function not supported = 0x6A81
SW_INCORRECT_P1P2 - Static variable in interface javacard.framework.ISO7816
Response status : Incorrect parameters (P1,P2) = 0x6A86
SW_INS_NOT_SUPPORTED - Static variable in interface javacard.framework.ISO7816
Response status : INS value not supported = 0x6D00
SW_NO_ERROR - Static variable in interface javacard.framework.ISO7816
Response status : No Error = (short)0x9000
SW_RECORD_NOT_FOUND - Static variable in interface javacard.framework.ISO7816
Response status : Record not found = 0x6A83
SW_SECURITY_STATUS_NOT_SATISFIED - Static variable in interface javacard.framework.ISO7816
Response status : Security condition not satisfied = 0x6982
SW_UNKNOWN - Static variable in interface javacard.framework.ISO7816
Response status : No precise diagnosis = 0x6F00
SW_WRONG_DATA - Static variable in interface javacard.framework.ISO7816
Response status : Wrong data = 0x6A80
SW_WRONG_LENGTH - Static variable in interface javacard.framework.ISO7816
Response status : Wrong length = 0x6700
SW_WRONG_P1P2 - Static variable in interface javacard.framework.ISO7816
Response status : Incorrect parameters (P1,P2) = 0x6B00
SystemException - exception javacard.framework.SystemException.
`SystemException` represents a `JCSystem` class related exception.
SystemException(short) - Constructor for class javacard.framework.SystemException
Constructs a SystemException.

T

T1_IFD_ABORT - Static variable in class javacard.framework.APDUException
This reason code indicates that during T=1 protocol, the CAD returned an ABORT S-Block command and aborted the data transfer.

Throwable - class java.lang.Throwable.
The Throwable class is the superclass of all errors and exceptions in the Java Card subset of the Java language.

Throwable() - Constructor for class java.lang.Throwable
Constructs a new `Throwable`.

throwIt(short) - Static method in class javacard.framework.CardRuntimeException
Throw the JCRE owned instance of the `CardRuntimeException` class with the specified reason.

throwIt(short) - Static method in class javacard.framework.PINException
Throws the JCRE owned instance of `PINException` with the specified reason.

throwIt(short) - Static method in class javacard.framework.ISOException
Throws the JCRE owned instance of the ISOException class with the specified status word.

throwIt(short) - Static method in class javacard.framework.CardException
Throw the JCRE owned instance of `CardException` class with the specified reason.

throwIt(short) - Static method in class javacard.framework.UserException
Throws the JCRE owned instance of `UserException` with the specified reason.

throwIt(short) - Static method in class javacard.framework.SystemException
Throws the JCRE owned instance of `SystemException` with the specified reason.

throwIt(short) - Static method in class javacard.framework.TransactionException
Throws the JCRE owned instance of `TransactionException` with the specified reason.

throwIt(short) - Static method in class javacard.framework.APDUException
Throws the JCRE owned instance of `APDUException` with the specified reason.

throwIt(short) - Static method in class javacard.security.CryptoException
Throws the JCRE owned instance of `CryptoException` with the specified reason.

TransactionException - exception javacard.framework.TransactionException.
`TransactionException` represents an exception in the transaction subsystem.

TransactionException(short) - Constructor for class javacard.framework.TransactionException
Constructs a TransactionException with the specified reason.

TYPE_DES - Static variable in class javacard.security.KeyBuilder
`Key` object which implements interface type `DESKey` with persistent key data.

TYPE_DES_TRANSIENT_DESELECT - Static variable in class javacard.security.KeyBuilder
`Key` object which implements interface type `DESKey` with CLEAR_ON_DESELECT transient key data.

TYPE_DES_TRANSIENT_RESET - Static variable in class javacard.security.KeyBuilder
`Key` object which implements interface type `DESKey` with CLEAR_ON_RESET transient key data.

TYPE_DSA_PRIVATE - Static variable in class javacard.security.KeyBuilder
`Key` object which implements the interface type `DSAPrivateKey` for the DSA algorithm.

TYPE_DSA_PUBLIC - Static variable in class javacard.security.KeyBuilder
`Key` object which implements the interface type `DSAPublicKey` for the DSA algorithm.

TYPE_RSA_CRT_PRIVATE - Static variable in class javacard.security.KeyBuilder
`Key` object which implements interface type `RSAPrivateCrtKey` which uses Chinese Remainder Theorem.

TYPE_RSA_PRIVATE - Static variable in class javacard.security.KeyBuilder
`Key` object which implements interface type `RSAPrivateKey` which uses modulus/exponent form.

TYPE_RSA_PUBLIC - Static variable in class javacard.security.KeyBuilder
`Key` object which implements interface type `RSAPublicKey`.

U

UNINITIALIZED_KEY - Static variable in class javacard.security.CryptoException
This reason code is used to indicate that the key is uninitialized.
update(byte[], short, byte) - Method in class javacard.framework.OwnerPIN
This method sets a new value for the PIN and resets the `PIN` try counter to the value of the `PIN` try limit.
update(byte[], short, short) - Method in class javacard.security.MessageDigest
Accumulates a hash of the input data.
update(byte[], short, short) - Method in class javacard.security.Signature
Accumulates a signature of the input data.
update(byte[], short, short, byte[], short) - Method in class javacardx.crypto.Cipher
Generates encrypted/decrypted output from input data.
UserException - exception javacard.framework.UserException.
`UserException` represents a User exception.
UserException() - Constructor for class javacard.framework.UserException
Constructs a `UserException` with reason = 0.
UserException(short) - Constructor for class javacard.framework.UserException
Constructs a `UserException` with the specified reason.
Util - class javacard.framework.Util.
The `Util` class contains common utility functions.

V

verify(byte[], short, short, byte[], short, short) - Method in class javacard.security.Signature
Verifies the signature of all/last input data against the passed in signature.

W

waitExtension() - Static method in class javacard.framework.APDU
Requests additional processsing time from CAD.

A B C D E G I J K L M N O P R S T U V W